Appendix D

The Pacific Northwest Maritime Region (PNMR) BCG panel approach to defining BCG level 1

Susan Jackson/USEPA Project Coordinator

The Pacific NW maritime region has undergone extensive ecological changes over past 200 years due to impacts of human settlement and land use. These changes have altered the landscape in such a way that no undisturbed conditions remain that the PNMR expert panel could use in development of a numeric BCG model. However, describing BCG level 1 conditions is a critical component of a BCG model since the upper end of the gradient is defined as natural, or conditions undisturbed by human activity. In absence of BCG level 1 conditions, the panel thought it possible to hypothesize a narrative, if not quantitative, descriptions of BCG level 1 and use that information to help define the upper level of the gradient. At a minimum, a narrative description would provide context for understanding current conditions in context of what could be there absent human disturbance. To begin to address this challenge, at their first meeting the expert panel developed a broad, conceptual description of BCG level 1 condition characterized by high biological and habitat diversity though influenced by elevation and gradient.

The panel readily acknowledged that human impacts on the natural environment have occurred in different ways and to different degrees over thousands of years. So, where to set the boundary for "no disturbance" posed an additional challenge.

The PNMR BCG expert panel proceeded to address these challenges by conducting a literature search and synthesis of historic records and observations and then using this information to:

1) Inform our understanding of how natural physical factors and human disturbance may combine to influence what we observe in PNMR streams.
2) Refine the initial narrative definition of BCG level 1 and provide documentation.
3) Develop an approach to detect biological indicators of ecological condition that may indicate stream conditions close to, or at least containing some elements of, a natural, undisturbed stream. The proposed numeric index is not a part of the BCG model but can be used to flag streams that support in whole or in part BCG level 1 conditions. The numeric index is proposed as a hypothesis for testing and further refinement.

The technical report, Calibration of the Biological Condition Gradient (BCG) for Macroinvertebrate Assemblages in Freshwater Wadeable Streams in the Pacific Northwest Maritime Region of Oregon and Washington, has incorporated the above findings and recommendation.[1]

[1] Calibration of the Biological Condition Gradient (BCG) for Macroinvertebrate Assemblages in Freshwater Wadeable Streams in the Pacific Northwest Maritime Region of Oregon and Washington. Technical Report to USEPA prepared by Jennifer Stamp, Tetra Tech, September 29, 2022.

Appendix D

Disturbance History and Natural Condition Assessment for the Pacific Northwest Maritime Region

Prepared for U. S. Environmental Protection Agency

by

Rick Hafele, Rob Plotnikoff, and Bob Wisseman

Rick Hafele: Apolysis, LLC, former manager of biomonitoring program at Oregon Dept. of Environmental Quality (retired).

Rob Plotnikoff: Senior aquatic biologist, Snohomish County

Bob Wisseman: Aquatic Biology Associates.

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1.0 BACKGROUND

1.1 Defining "Natural Conditions" for the Pacific Northwest Maritime Region

The objective of this report is to provide as much information as possible about historic freshwater stream habitats and biological communities within the Pacific Northwest Maritime Region (PNMR), and to describe how conditions have changed over time in order to establish an understanding of BCG Level 1, natural or native conditions. While such "natural" conditions no longer exist, and it is not practical nor expected to return to conditions unimpaired by 150+ years of anthropogenic disturbance, understanding and defining natural conditions is critical for determining where the other BCG biological conditions (levels 2-6) fall along the human disturbance gradient. Understanding natural conditions also provides information for making better informed decisions for restoration projects, and provides the public a clear understanding of how much change has actually occurred to the land and streams, and thus helps avoid the sliding baseline phenomenon where each generation accepts as "natural" the current "best available" environmental conditions.

The PNMR includes five Omernik Level 3 ecoregions (Figure 1). Ecoregions as described by Omernik (1987) have been used as an effective tool for delineating regions of similar ecological characteristics and biological expectations since the late 1980s. Omernik describes ecoregions as ecological regions that can be identified through the analysis of the patterns and composition of biotic and abiotic conditions. Regional traits used to define ecoregions include geology, physiography, vegetation, climate, soils, land use, wildlife, and hydrology (Omernik 1995). Ecoregions have been defined at multiple scales across the United States and Canada, with Level 1 being the largest (15 ecological regions across North America) to Level 4 defining 967 ecological regions in North America. Level 3 ecoregions represent 98 regions across continental US and is considered an effective scale for assessing and managing natural resources (US EPA 1996). In addition, as shown in Figure 2, the ecoregions within the PNMR have been the subject of numerous water quality studies, including widespread bioassessment programs, that provide a rich data base for developing a Biotic Condition Gradient (BCG) model for the region.

Initial BCG model development included just the Willamette Valley and Puget Lowlands ecoregions (Stamp 2022). When that model was finished, however, there was agreement among the expert panelists that expanding the model to include the entire Pacific Northwest Maritime Region (PNMR) would help agencies and others interpret biological stream assessments for a much larger portion of the Pacific Northwest region. Therefore, beginning in 2020 the workgroup began working on expanding that model to cover the entire PNMR. As a result, this report describes the historical changes from the early 1800's to present day of the physical and biological conditions of the Willamette Valley, Puget Lowlands, and Coast Range ecoregions. We describe what sources of information and data we uncovered (from historical records spanning to more recent data), what the landscape used to be like, and how the physical template/stressor backdrop has changed over time with respect to land use and land cover, hydrology and water quality. Of particular interest were historical accounts of aquatic communities in

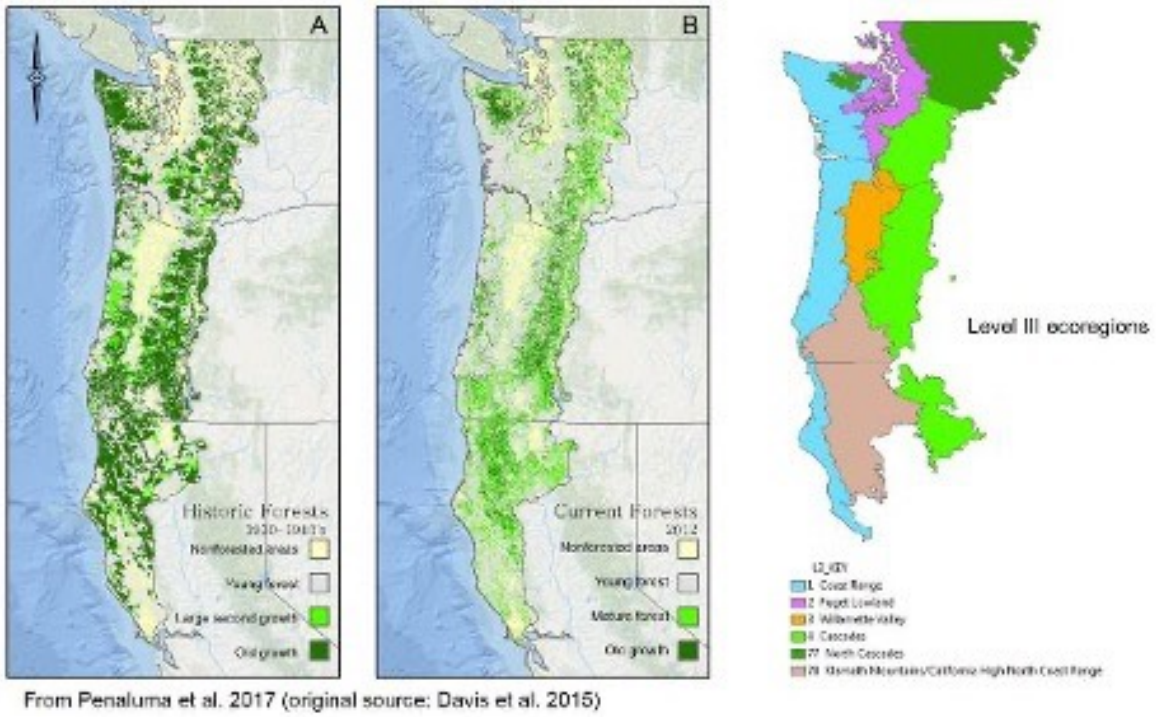


Figure 1. The maritime Pacific Northwest refers to the humid temperate areas extending from western Washington into northern California (Cascade Mountain crest to the Pacific Ocean), as depicted in the two maps on the left, which show historic forest cover (1930-1940's) and more current (2012) conditions (from Penaluma et al. 2017; originally from Davis et al. 2015). The crest of the Cascades generally divides the Humid Temperate Domain from the Dry Domain of the Pacific Northwest (Kimerling & Jackson 1985). The forest boundaries in the maps on the left correspond closely with the Level III Ecoregions shown on the right (Omernik 1987, 1995).

completely undisturbed stream habitat ("as naturally occurs"), but such historical data are rare, especially for macroinvertebrates.

We used the 1830s-1850s as the baseline for describing "natural conditions" in the region because it was around 1830-1850 that land use changed from subsistence hunting and fishing with minimal agriculture as practiced by Native tribes to widespread agriculture, mining, forestry, and permanent settlements introduced by European and Early American settlers. This transformation includes the elimination of beaver populations due to trapping and commercial interests that profoundly transformed the landscape from a complex, multi-habitat, floodplain system to a more simplified river channel network to allow for the transportation of goods and the reduction of flood risk to protect permanent ports and towns and foster European style farming practices. To a large degree, the mid-1830s marks the end of the beaver-dominated landscape, but is before the landscape had been unalterably transformed by agriculture, mining and forestry. The most robust historical data available of landscape and stream conditions generally begins in the early 1850s when the United States General Land Office (GLO) was established in the Oregon Territory and charged with surveying the region (Vaughan 2018). The GLO surveys provide detailed descriptions of vegetation, land use, and stream channel characteristics that can be compared to current landscape conditions.

1.2 Stream classes

The focus for the BCG is on smaller, wadeable streams (drainage area <100 mi²), spanning from low gradient-low elevation valley bottom streams to high gradient-high elevation streams in the Cascade Mountains. While the BCG model is based on the response of macroinvertebrates to the human stressor gradient, for this report, we also included information about fish - primarily salmonids - as they play such a critical role in aquatic ecosystems of the Pacific Northwest and have a long history of commercial and recreational interests in the region.

Both Washington and Oregon have extensive macroinvertebrate data sets from the PNMR region (Figure 2), which provided a robust data set for developing and calibrating the BCG model. The current model has been developed for three subclasses of stream types: 1) low elevation (< 750 meters), low gradient (< 1% slope); 2) high elevation (> 750 meters), low gradient (< 1% slope); and 3) high elevation (> 750 meters), high gradient (> 1% slope) streams. For an in-depth comparison of these stream classes see Appendix A

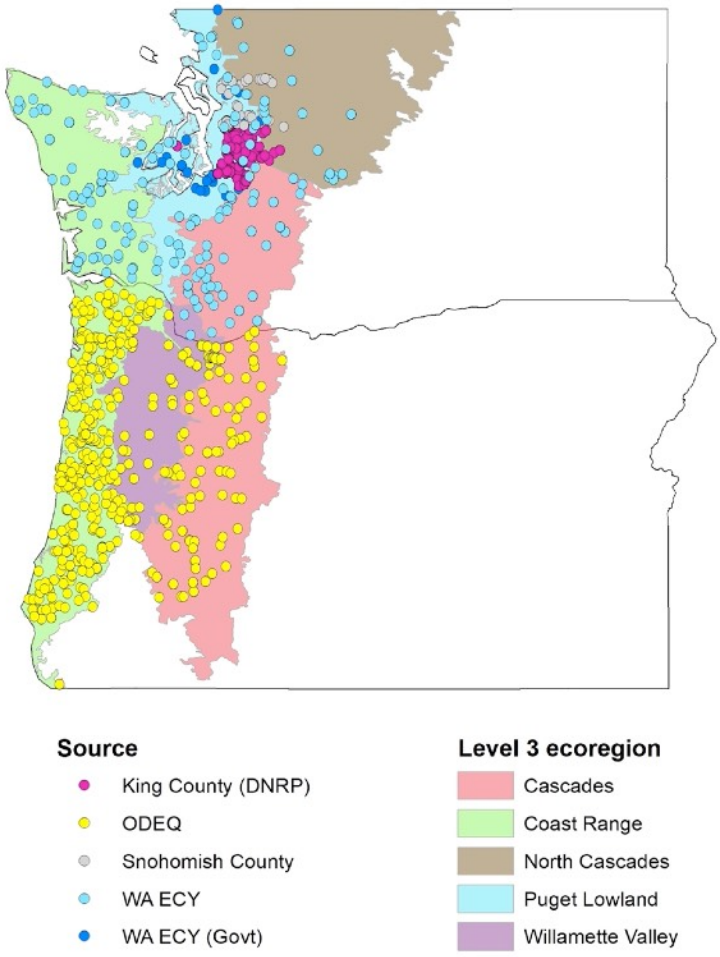


Figure 2. Macroinvertebrate samples collected across the PNMR and used to develop the BCG model.

2 Willamette Valley

2.1 Physical characteristics

The Willamette River is approximately 180 miles long from its headwaters in the Cascade Mountains southeast of Eugene to where it empties into the Columbia River by Portland. The Willamette River Basin (WRB) drains an area of 11,478 square miles, bounded by the Coast Range to the west and the Cascade Mountains to the east. The Willamette Valley ecoregion (level 3) lies within the WRB and is about half the size of the entire basin (Figure 3 & 4).



Figure 3. Topographic map of western Oregon. The Willamette Valley Level 3 ecoregion is outlined in red and the WRB is shaded in blue. HUC=Hydrologic Unit code. HUC 6 is the hydrologic unit code used by USGS to identify the Willamette River Basin.

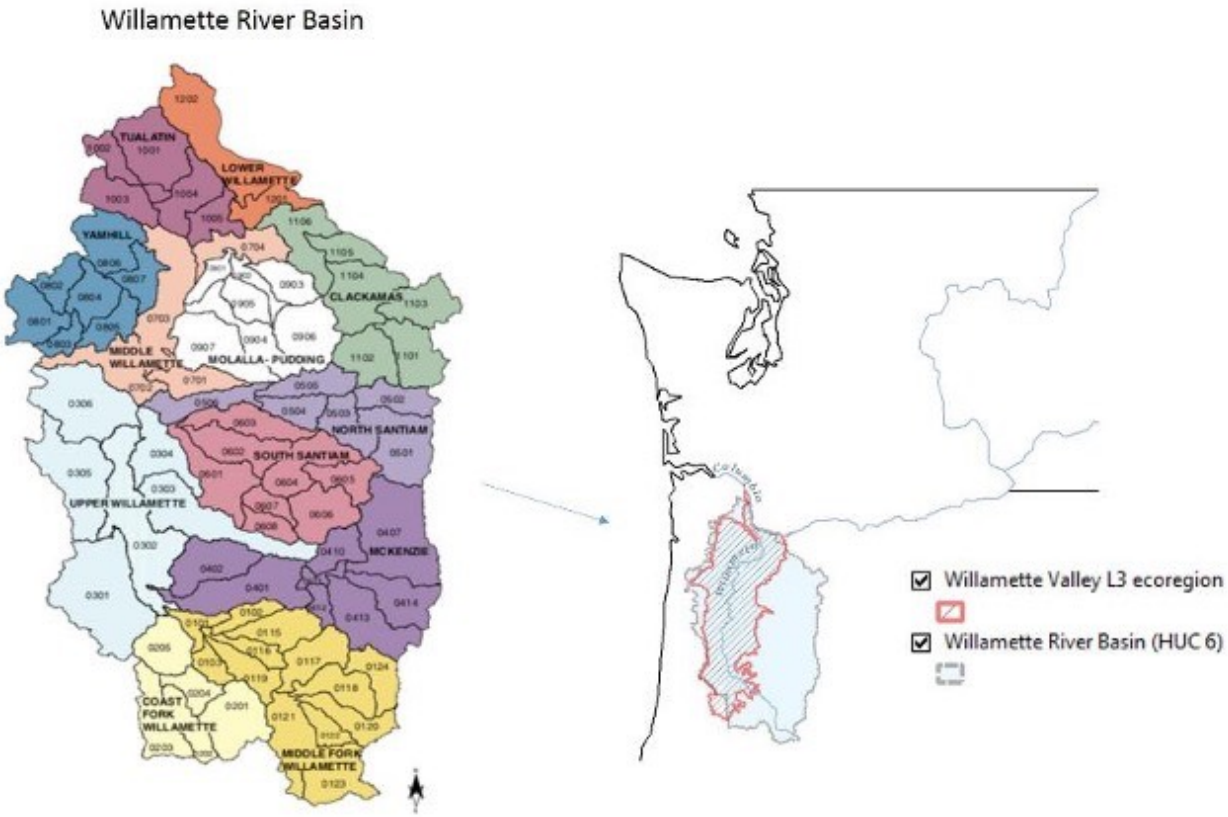


Figure 4. The Willamette Valley level 3 ecoregion (study area) is located within the Willamette River Basin (right). The map on the left shows the 12 major river sub-basins within the Willamette Valley ecoregion (from the Willamette River Basin Atlas). HUC=Hydrologic Unit Code.

2.1.1 Topography

From west to east the WRB ranges from the top of the Coast Range to the crest of the Cascade Range, a distance of up to 100 miles from crest to crest. In the middle lies the valley floor of the Willamette River. The valley floor drops from an elevation of 450 feet (137 m) at the south end to approximately 10 ft (3 m) above sea level at the confluence with the Columbia River. The highest point in the Coast Range is Marys Peak at 4097 ft. (1,249 m), while the highest point in the Cascades still within the WRB is Mt. Jefferson at 10,495 ft. (3,199 m).

Streams traversing the slopes and foothills of the Coast Range and Cascades have steep to moderate gradients(2 to >4%) This produces streams with a regular mix of fast water riffles interspersed with pools, with a substrate ranging from small gravel to boulders that is mixed with sand and fines along stream margins and within low gradient reaches. Stream segments in the valley floor by contrast have very low gradients, often less than one percent. This results in slow moving water with stream substrates dominated by fine sediments. Stream channels have also had greater modification due to agriculture and urban development in the valley floor compared to the reaches of streams in the Coast Range or Cascades.

2.1.2 Climate

The climate of the WRB is greatly influenced by the Coast Range to the west and the higher Cascade mountains to the east. The western, or ocean side of the Coast Range, receives large amounts of rainfall with average annual rainfall greater than 100 inches (254 cm) common. The eastern slopes of the Coast Range in the WRB, however, receive much less rain with a total annual rainfall typically between 35-50 inches per year. Rainfall increases once again along the western slopes of the Cascades. Average annual rainfall in Detroit, OR, near the crest of the Cascades for example, is 90.7 inches. Annual rainfall in the WRB from north to south ranges from 35 inches per year in Portland to 46 inches per year in Eugene (U.S. Climate data). Overall the WRB has cool, wet winters and warm, dry summers. From July through September, the warmest and driest part of the year, rainfall comprises just 5% of the average annual total (Hulse et al. 2002). Summer streamflows rely heavily on winter snowfall accumulation in the Cascades. Rainfall dominates in the Coast Range with winter snowfall occurring at irregular intervals and any accumulation is typically restricted to the highest peaks. As a result streams on the west side of the basin tend to be more sluggish, with lower base and minimum discharges and higher water temperatures than streams on the east side of the basin where geology and snowpack create more uniform temperature and flow rates.

2.1.3 Soils, geology and glaciation

Two great forces, volcanic activity and glaciation, shaped the landscape we see today in the WRB. The formation of the Willamette Valley as we now see it was completed about 15 million years ago (mya). By this time the northern end of the Coast Range had completed its development. Five million years earlier (about 20 mya) the floor of the Willamette Valley had uplifted sufficiently to drain a shallow inland sea. At the same time, volcanic activity finished creating the Cascade Mountains along the eastern border of the WRB over a period from 30 to 10 mya (Branscomb 2002).

After volcanic activity created the basic landscape of the WRB, the effects of glaciation during the last great ice age (40 to 13 thousand years ago) finished forming what the first Europeans saw when they arrived in the Willamette Basin in the late 1700s and early 1800s, first as fur trappers and later as settlers. While the continental ice sheet during the last ice age did not reach as far south as the

Willamette River Basin, glaciation still played a major role in the geology and soils of the WRB by creating conditions for massive flooding and transportation of nutrient rich sediments into the basin. From 15 to 13 thousand years ago as many as several dozen great floods (known as the Bretz floods for J. Harlan Bretz, the geologist who first postulated their existence) occurred when ice periodically dammed the Clark Fork River east of Spokane, Washington. The ice dams created a lake called Glacial Lake Missoula with a volume half that of present day Lake Michigan (Alt 2001). When the ice dam broke, a volume of water greater than the annual volume of all the current rivers on earth came roaring across Eastern Washington and down into the Columbia River Gorge (Branscomb 2002). When the flood, with a wall of water 500 feet high, reached what is today Portland its waters continued into the Willamette Valley to a depth of 400 feet. The sediments left by these multiple floods now cover the floor of the Willamette Valley. Additional sediments have been carried into the valley floor from the major rivers draining into it, all which have created excellent soil conditions for agriculture (Enright et al. 2002).

2.2 Brief History of Human Settlement

When the first Europeans arrived by ship in late 1770s along Pacific Northwest coast and sailed up into the Columbia River (Barbour 2016), a thriving and diverse community of Native tribes had existed throughout the region for more than 10,000 years (Atlas of Oregon). While the exact number has been difficult to determine, perhaps as many as twenty different groups of Native Americans lived in the Willamette Valley (Atlas of Oregon). The names of some of the Native groups – e.g. Kalapuya, Santiam, Molala, Tualatin, and Yamhill – account for the names of many of the major tributaries to the Willamette River. In 1806 Lewis and Clark estimated the Native population in the Portland Basin alone at between 4,840 and 10,940 depending on the season with the higher number present during spring fishing activity (Boyd & Zenik 2016). For example, in the Tualatin Plains (todays Beaverton, Hillsboro, Mountaindale, Forest Grove) sixteen villages were known by name.

The impact of the Native population on the habitat and ecological resources of the WRB consisted primarily of using fire to maintain habitat for key plants gathered for food such as nuts and berries and especially camas. They also used fire in hunting drives to concentrate deer or to help gather insects such as grasshoppers (Warre & Richardson 2003). It is believed that the use of fire was critical for maintaining Oak Savanah habitat in the Willamette Valley for thousands of years prior to the arrival of EuroAmericans. In addition, Native tribes maintained extensive fisheries for the large runs of anadromous fish in the region for millennia. The massive runs of fish made the Native communities along the lower Columbia, including major tributaries like the Willamette River, some of the richest in trade goods and overall resources in North America (Lang 2019). While thousands of Native Americans gathered annually at key locations, such as Celilo Falls on the Columbia and Willamette Falls on the Willamette, to harvest fish and trade goods, their methods of harvest (spear, dip net, and seine net) maintained sustainable runs of fish. Harvest methods later implemented by EuroAmericans in the early to mid 1800s quickly depleted the seemingly endless supply of salmon (see section 2.6.1 for details).

As early as the 1780's diseases contracted from the early European explorers and settlers began decimating the Native populations throughout the Pacific Northwest including the Willamette Valley. Outbreaks of smallpox, influenza, dysentery, measles, and malaria all took their toll. By the end of a malaria outbreak between 1830-1833 the Native population in the Willamette Valley had been reduced by 90% (Robbins 2016). A series of treaties with Willamette Valley tribes negotiated between 1851-1855 culminated in January-March 1856 when tribes from throughout western Oregon were marched to a reservation called the Grande Ronde Agency or Yamhill River Agency along the banks for the South Yamhill River (Lewis 2019). This allowed unrestrained settlement by EuroAmericans throughout the Willamette Valley.

European Explorers and Settlers:
The first Europeans to explore, and later settle the Willamette Valley, were fur traders primarily from the Hudson's Bay Company (HBC) (Barton Barbour 2016). From 1818-1830, HBC trappers from Fort Vancouver, supervised by Peter Skene Ogden, implemented a policy of creating a "fur desert" throughout the Northwest to prevent the influx of competing fur traders (Barbour 2016). By the late 1800's beaver populations throughout Oregon had plummeted to the point that the Oregon legislature prohibited beaver trapping from 1899-1919 and again between 1937-1951. The near extermination of beavers had a profound effect on all aspects of the aquatic system, and these impacts will be discussed in further detail in the sections on land use, hydrology, and aquatic life.

Once the beaver were eliminated early fur traders began settling in the Willamette Valley. In the 1830's, they created one of the first colonies in the Willamette Valley known as French Prairie, which was located near present day Champoeg State Park on the Willamette River. The population of French Prairie increased from 26 families in 1836 to a population of approximately 1,200 by 1850 (Atlas of Oregon). The European population in the Willamette Valley grew quickly beginning in the 1850's, increasing from about 10,000 people in 1860, to over 115,000 by 1900 (Atlas of Oregon). The General Land Office (GLO) surveys in 1851-1854 provide the first detailed measurements and description of the Willamette River and the surrounding landscape. In 2015 the estimated population of the Willamette Valley was just over 2.8 million, which accounts for nearly 70% of Oregon's total population of 4.1 million.

2.3 Land use and land cover

A radical shift in land use/land cover in the WRB began when fur trading companies started extensive beaver trapping in the basin beginning around 1818. Unrestrained trapping continued with the expressed goal of trapping every beaver in the region, and by 1833 beaver populations had been reduced to near zero (Barbour 2016). Besides beaver, other animals such as otter, muskrat, and martins were also sought after, but beaver had by far the greatest economic value. The international demand for beaver products (primarily fur for clothing and beaver felt hats) drove both the trade and cultural relationship between Native tribes and Europeans starting as early as the late 1500s on the East Coast (Naiman et al. 1988). The uncontrolled trapping of beaver in North America continued for nearly 300

years (~1580-1880), and only ended when beavers were nearly exterminated from across the continent. The end result was major landscape and watershed changes throughout North America (Goldfarb 2018).

Numerous studies have shown how the removal of beaver ripples through all aspects of watershed ecology. Ben Goldfarb in his book Eager Beavers (2018), describes in some detail how beavers created much of the landscape across North America and how their removal radically changed plant and animal communities across the continent. Goldfarb describes beavers' effect on land as, "…a beaver complex is one of the few opportunities we mortals have to watch geological processes unfold within the duration of a single human life span. Often the progression runs from water to earth, as beaver ponds slow flows, trap sediment, and gradually fill with silt and pioneering plants, whereupon their creators move upstream to begin the cycle again. Left behind are open, grass-filled meadows - their surfaces flat, treeless, and flan-like underfoot, bermed with the overgrown contours of long-ago dams." He then explains that, "When trappers de-beavered North America, they threw that natural evolution into fast forward. As dams deteriorated and ponds drained, they left behind some of the finest soil a farmer could till, a newly exposed mélange of leaves, bark, rotten wood and other manure." Just as in other regions of North America, the abandoned and dried-up beaver ponds that the first EuroAmerican settlers found in the Willamette Valley became the first areas to be farmed.

Today the Willamette Valley ecoregion is home to 70 percent of Oregon's entire human population. The land outside of dense urban areas is used extensively for agriculture (Figure 5). Historically, agricultural land use increased significantly as European settlement quickly expanded from the 1850's to today. Because Native tribes relied primarily on gathering native plants and hunting (Atlas of Oregon), prior to European settlement there was no intensive agriculture in the Willamette Valley. By 2002, approximately 43% of the land area in the Willamette Valley was being intensively farmed (Baker et al. 2004). In addition, as the population in the Willamette Valley increased from under 10,000 in 1850 to more than 2.8 million in 2015, cities and urban areas also expanded significantly. Table 1 lists changes in land use and land cover from approximately 1850 with projections to 2050 under different levels of development.

Christy et al. (2011) provides good descriptions of the typical vegetation in the Willamette Valley based on 1850 Government Land Office (GLO) surveys (overall, as well as the dominant vegetation in riparian areas). The dominant land cover across the Willamette Valley in 1850 was upland prairie and wet prairie, followed by upland forest, savanna, and woodland (Table 2). The primary vegetation classes along streams and in the floodplain of Willamette Valley rivers was wet prairie and wetland forest. Prairie was characterized as "open grassland with at least some bearing trees [trees used as survey markers] absent or too distant for convenient use." Vegetation type varied within the prairie class between upland prairie and wet prairie. Two-thirds of prairie fell into the upland prairie category. Upland prairie was found on well-drained soils and likely included a combination of grass species such as Roemer's fescue (*Festuca idahoensis ssp. roemeri*), Sandberg bluegrass (*Poa secunda*), prairie June grass (*Koeleria macrantha*), Lemmon's needlegrass (*Achnatherum lemmonii*), wheatgrass (*Elymus trachycaulus*), and California oatgrass (*Danthonia californica*). Wet prairie, while covering only about one third of the total prairie area, was a significant vegetation type in riparian areas occurring where heavy clay soils dominated and

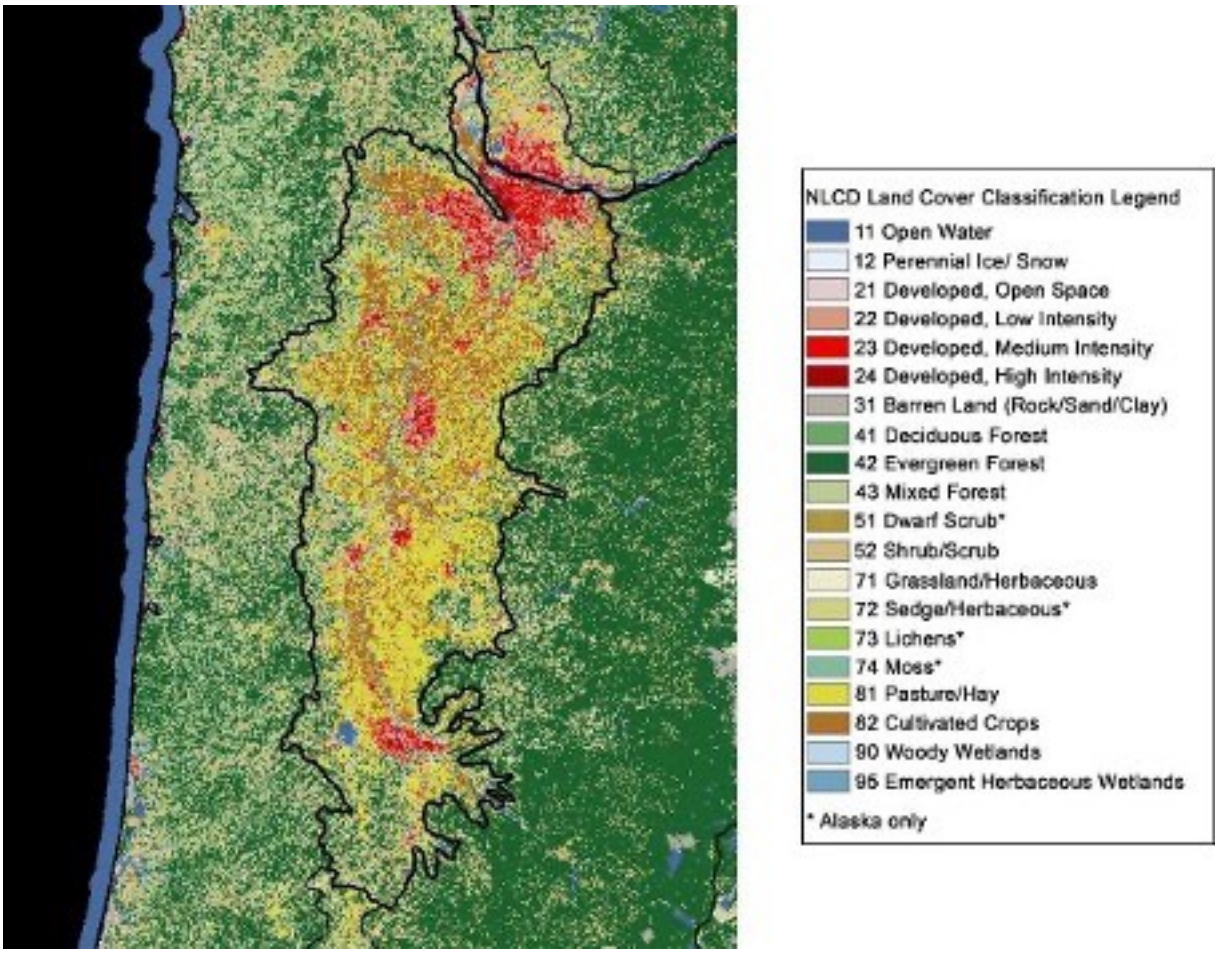


Figure 5. Land cover in the Willamette Valley and neighboring ecoregions, based on the National Land Cover Database (2011 Edition, amended 2014; USGS 2014).

standing water was evident during the winter. Wet prairie was also called “camas prairie” because of the high abundance of camas (*Camassia quamash*). Other common species included tufted hairgrass (*Deschampsia cespitosa*), rushes (*Juncus spp*.), and sedges (*Carex spp*.). Less than two percent of the prairie habitat present in 1850 is estimated to exist in the Willamette Valley today (Christy et al. 2011).

Table 1. Riparian vegetation comparison 1850-2050 for 1-4th order streams (from Table 1 in Hulse et al. 2002).

LOWLAND ECOREGION - 120 METER RIPARIAN AREA: 1- 4th ORDER STREAMS

		Percent composition							
	Acres	Devlp	Ag	NatGrass	Natshrub	Hdwd	Mix For	Conif For	Wetlands
PESVEG	467693	0.0	0.0	16.8	13.9	24.7	5.1	25.4	14.2
LULC90	470071	10.3	41.6	1.0	12.7	9.9	13.2	9.6	1.1
PT2050	470072	12.7	40.5	0.9	12.1	9.7	12.5	9.9	1.1
CON2050	469072	11.7	26.7	5.7	15.4	11.9	14.9	9.9	3.5
DEV2050	470089	15.8	36.8	1.0	13.5	9.3	12.1	9.8	1.1

UPLAND ECOREGION - 120 METER RIPARIAN AREA: 1 - 4th ORDER STREAMS

		Percent composition							
	Acres	Devlp	Ag	NatGrass	Natshrub	Hdwd	Mix For	Conif For	Wetlands
PESVEG	590522	0.0	0.0	0.4	0.7	0.1	0.3	98.2	0.1
LULC90	587954	0.9	0.7	0.0	0.6	9.0	36.6	52.1	0.0
PT2050	587959	0.9	0.7	0.0	0.6	9.0	36.3	52.4	0.0
CON2050	588012	0.9	0.5	0.0	0.6	9.1	36.6	52.2	0.0
DEV2050	587912	1.0	0.6	0.0	0.6	9.0	36.2	52.5	0.0

PESVEG = PreEuropean Settlement; LULC90 = Land use/Landcover circa 1990; PT2050 = Present Trend 2050; CON2050 = Conservation 2050; DEV2050 = Development 2050.
(From *Willamette River Basin Planning Atlas*, Edited by D. Hulse, S. Gregory, and J. Baker. 2002, OSU Press.

Table 2. Classes of historical vegetation in Willamette Valley, ca. 1850s, based on General Land Office survey data (from Christy et al. 2011).

Vegetation	***Percent Cover***
Upland Prairie and Wet Prairie	31.4
Upland Forest	26.3
Savanna	18.2
Woodland	14.4
Riparian and Wetland Forest	7.2

Riparian and wetland forest, though not a dominant vegetation type overall (7.2% of valley area), was a common habitat in the floodplains of the mainstem Willamette River and smaller valley streams, extending up to 10 km wide along the Willamette River (Figure 6). Willow, red alder, white alder, and black cottonwood were common along active channels, while Oregon ash, bigleaf maple, and white oak occurred on higher terraces. In addition, white oak and Oregon ash were common along smaller streams. Christy et al. (2011) gives the following description of vegetation along small valley streams:

“Stands along smaller streams were described as ash or ash and willow swales. Low-gradient sites often contained ash, alder and willow swamps with beaver dams that were subsequently drained and cleared for their coveted beaver dam soils.” Of note is that GLO surveys recorded emergent wetlands (also called marshes and swamps) as covering 3,5592 hectares (ha), or just under 0.3% of the valley area. Swamps were frequently associated with small streams and often contained beaver dams, though by the 1850s only a small number of beaver remained in the valley. Extensive human development, plus the invasion of shrubs and trees following fire suppression, has led to nearly 100% loss of some pre-settlement habitats today, especially wet and dry prairie and oak savanna (USEPA 2002a).

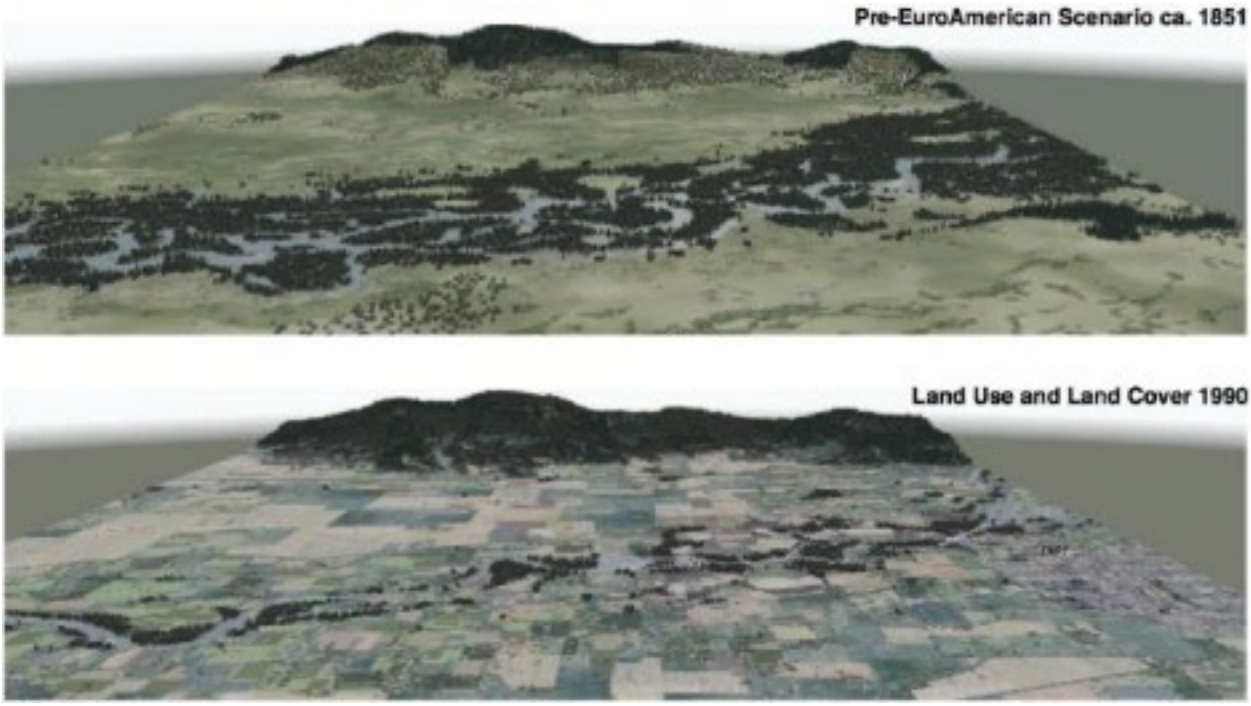


Figure 6. Computer simulation of the upper Willamette River and floodplain between Harrisburg and Eugene-Springfield, ca. 1850 and ca. 1990 (from USEPA 2002a).

2.4 Hydrology

Since the time of European settlement (early to mid-1800’s), streams have been extensively altered through channelization and dam construction (Hulse et al. 2002). Flood control and more intensive management of the river channel became a high priority following the rapid increase in agricultural and urban development after the late 1800’s. Hydrologic changes, however, first occurred with the removal of beaver in the early 1800s. Beaver dams store water and moderate annual stream discharge, reduce current velocity, lower stream gradient and retain sediment and organic matter (Naiman 1988). As a

result of the near complete removal of beaver by the late 1830s, we can infer that the hydrology of streams in the WRB was already significantly altered by 1851 when the GLO began surveying stream habitat and channel conditions in the WRB. The landuse changes since 1850, including extensive channelization and dam construction, have only further altered the hydrology of the basin. The overall effect has been a dramatic decline in wetlands, loss of natural water storage that maintained streamflows during dry summer months, and changes in the timing and magnitude of peak streamflows throughout the basin (Taft & Haig 2003; Gregory et al. 2002).

Compared to pre-European conditions, present day peak flows in the mainstem Willamette and its major tributaries are lower due to flow control at dams, while small streams have greater and more rapid fluctuations in flow. This, plus the systematic removal of wood debris and other impediments to boat traffic from both large and small stream channels, has led to a significant simplification of channel conditions and stream habitat for aquatic life (details below) (Gregory et al. 2002).

Damaging floods occurred in the Willamette Valley at somewhat regular intervals, with major flood disasters noted in 1861, 1890, 1943, 1945, 1964, and 1996 (Figure 7). Dam construction became the primary means to control floods and also provide irrigation, navigation, power, recreation, and pollution abatement (Willamette River Basin Commission 1945). As of 2002, 371 dams (federal, state, and private) had been constructed throughout the WRB (Payne 2002) (Figure 8). The first dam was built in 1894 for water supply by the City of Portland. Dam construction increased significantly in the 1940's with the greatest number of federal and private dams constructed between 1950 and 1980 (Payne 2002). While the large federal dams on the mainstem tributaries of the Willamette and McKenzie Rivers are best known, the majority of dams are small private dams on small streams in the basin. The Tualatin watershed, for example, has the highest number and concentration of dams (82) in the WRB.

At the same time that dams were being built and altering river hydrology, stream channels were also being simplified through channelization and constructing revetments to minimize channel movement and to protect property along the river. This resulted in a significant loss of channel complexity. Gregory et al. (2002) compared channel complexity of the Willamette River between 1850 and 1995 (Figure 9). They found that the total area of river channels and islands decreased from 41,000 acres to less than 23,000 acres, and the total length of all channels dropped from 355 miles to 264 miles. In addition, more than half of the area of small floodplain tributaries and more than one-third of all alcoves and sloughs were gone by 1995. In the reach between Albany and Eugene, from 1850 to 1995, roughly 80% of the area of islands, 41% of the area of side channels, 80% of the area of riparian forest, and 74% of alcoves and sloughs had been lost (Landers et al. 2002). With increasing population, water use also increased dramatically. By the beginning of the 21st century, water use in the WRB for irrigation, municipal, industrial and other out-of-stream uses consumed over 37,000 cubic feet of surface water per day, causing approximately 80 miles of 2nd to 4th order streams to go dry in moderately dry summers (USEPA 2002b).

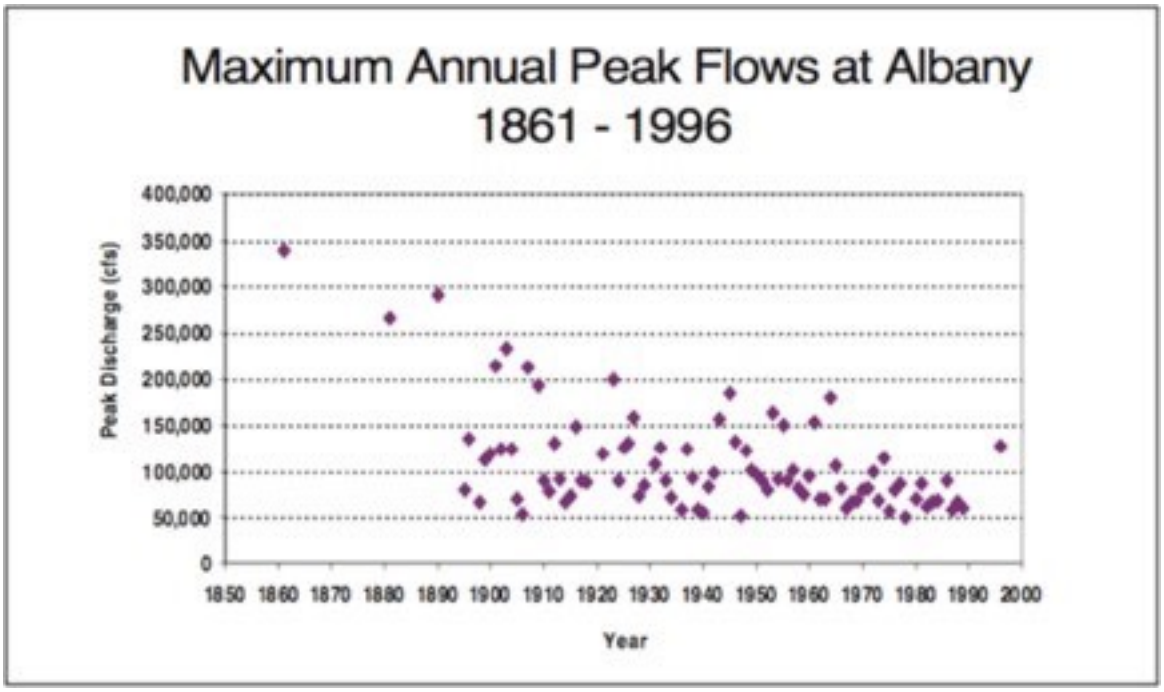


Figure 7. Historic Willamette River high-water flood events from 1850 to 2000. (From Willamette River Basin Atlas 2002).

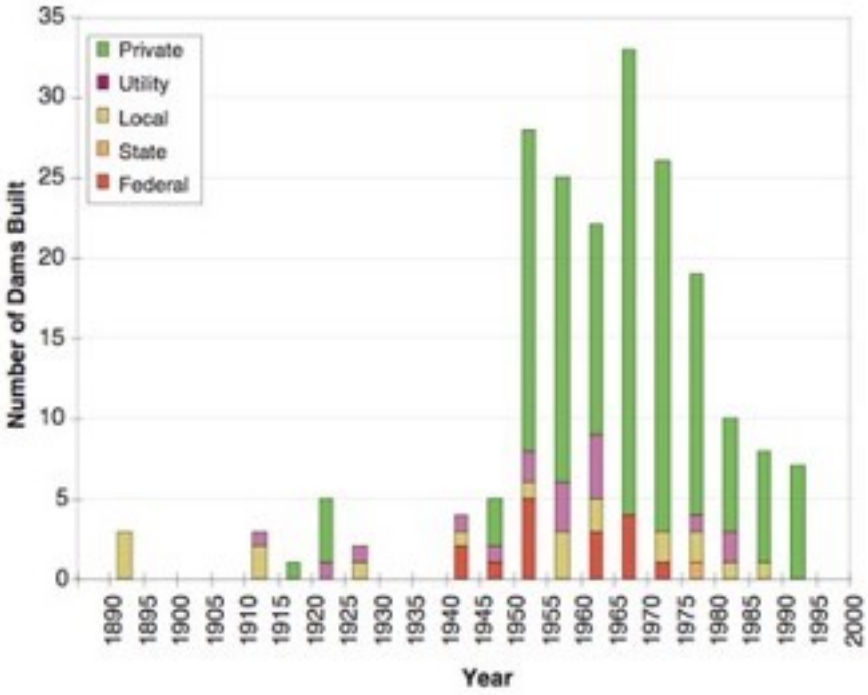


Figure 8. History of dam construction in the Willamette Basin (from Willamette River Planning Atlas 2002).

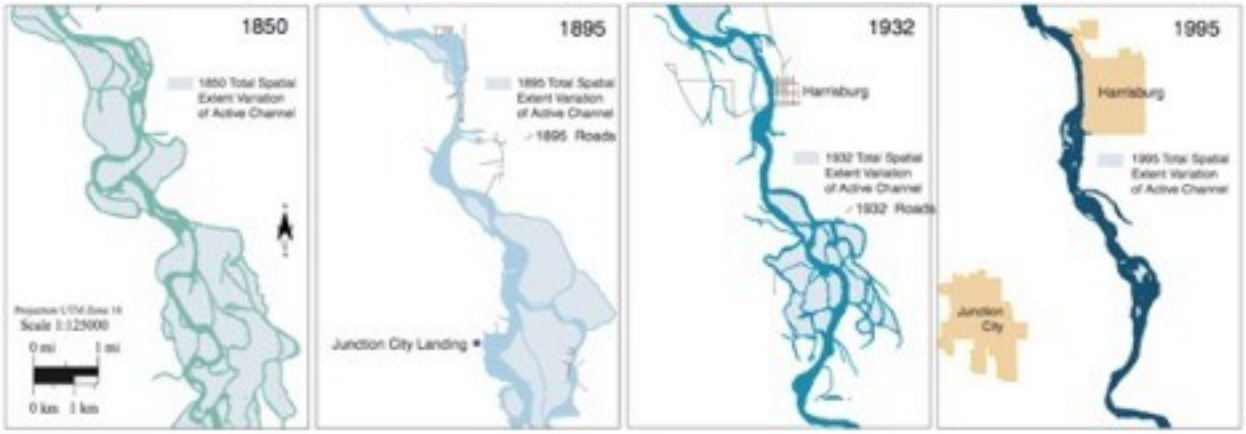


Figure 9. Comparison of Willamette River channel complexity from 1850 to 1995 near Harrisburg, Oregon (from Willamette River Planning Atlas 2002)

2.5 Water quality

Changes in land use, urban development, and alterations to stream channels and hydrologic conditions in the WRB led to serious declines in water quality (Britton 1965). Part of the rationale for building dams was to improve water quality by storing water in reservoirs for release during the lowest flows of summer and fall, and thereby improve water quality by increasing streamflows. By the 1920's, water quality in the Willamette River had declined to very poor conditions. In 1926 the Oregon State Board of Health convened a meeting in Salem to outline the problem and organize an "Anti-Stream Pollution League" to develop a plan to address water pollution (Britton 1965). The result was a water quality study at 28 sites along the mainstem Willamette River from Cottage Grove to Portland in August 1929. Results of the survey found a continuous decline in water quality from the upper river, downstream to Portland. Dissolved oxygen, for example, remained above 8 mg/L over a 24-hour period (samples collected every 6 hours) at the two uppermost sites, but never rose above 4 mg/L at the lowest downstream site in Portland, and did not rise above 8 mg/L at the five lowest sites. Extremely high bacteria levels were also collected at numerous locations where untreated sewage and animal waste entered the river.

As a result of this and other water quality studies in the early 1930's, Oregon voters passed a ballot measure in 1938 to create the Oregon State Sanitary Authority, which was established in 1939 (Britton 1965). Conditions, however, did not improve. A follow-up study in 1944 to the study conducted in 1929 found that dissolved oxygen conditions were worse at nearly every site on the river, with oxygen concentrations in the Portland harbor near 1 mg/L. Healthy cutthroat trout fry placed in a live box in the South Santiam River below Lebanon, Oregon died within two minutes due to low oxygen and other toxic materials. Federal dollars became available in the 1950's for pollution abatement and sewage treatment. Between 1957 and 1961, federal grants were used to construct waste treatment facilities for cities along

the Willamette River and its major tributaries (Britton 1965). In 1969 the Oregon legislature established the Oregon Department of Environmental Quality (ODEQ) to replace the Oregon Sanitary Authority.

Long-term water quality monitoring in the Willamette River was established by ODEQ. A WRB report in 2015 showed an improving trend from 1980 to 2015 throughout the mainstem Willamette based on the Oregon Water Quality Index, which integrates eight water quality variables (pH, dissolved oxygen, Biochemical Oxygen Demand (BOD), temperature, ammonia+nitrate+nitrite, total phosphorus, total solids, and fecal bacteria) into a single index (Figure 10) (Mulvey 2015a).

While water quality in the WRB has improved from critical conditions in the 1930s through 1940s, many water quality problems still exist. For example, the exposure of streams to numerous toxic chemicals is largely unknown, though several more recent studies have looked at the concentration of toxic chemicals in Willamette Valley streams and their potential effects on aquatic life (Anderson et al. 1997, Hope 2012, Carpenter et al. 2016). For example, Carpenter et al. (2016) found impaired benthic invertebrate communities in urban streams exposed to pyrethroid insecticides. An ODEQ report on toxics in the Willamette River found over 25 toxic chemicals though most were at levels below criterion set to protect human and aquatic life (Mulvey 2015b). While regulations for point sources of pollution have resulted in significant improvement in water quality, impacts on aquatic life from non-point source pollution from agricultural and urban runoff persist (USEPA 2016).

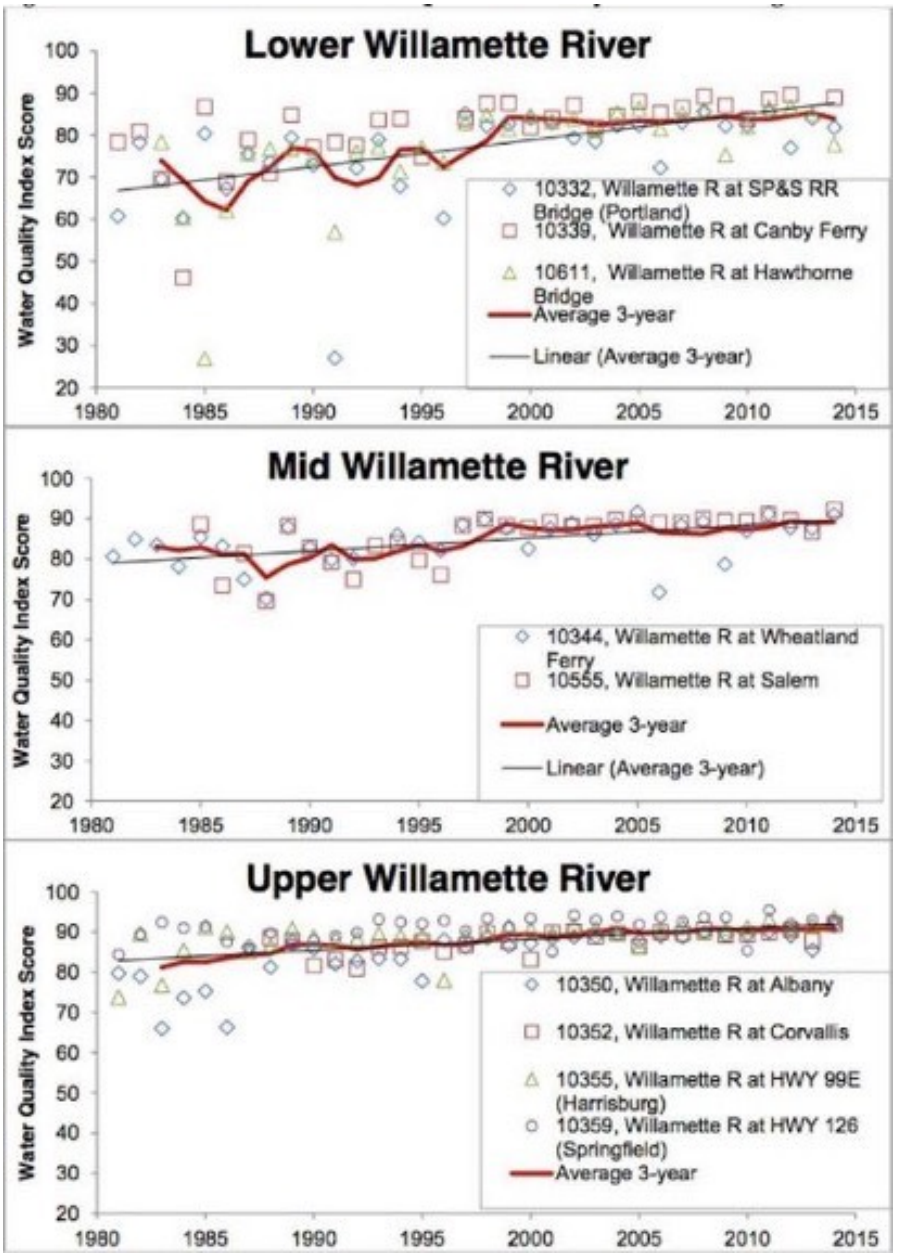


Figure 10. Willamette River annual OWQI scores with 3-year reach average and reach trend.

2.6 Historic information on aquatic communities

2.6.1 Anadromous Fish

Unlike other Columbia River tributaries, the Willamette River has a major natural barrier to upstream salmon migration - Willamette Falls. Willamette Falls is located at Oregon City, 26 miles upstream from the confluence of the Willamette with the Columbia River. At 1500 feet long and 42 feet high, it is the largest waterfall by volume in the Pacific Northwest (Northwest Waterfall Survey: https://www.waterfallsnorthwest.com/index.php). Flow over the falls typically ranges from around 6,000 cubic feet per second (cfs) in the summer to over 60,000 cfs in the winter. During flood events flows may exceed 100,000 cfs.

While Willamette Falls is a natural obstruction to upstream migrating salmon, to the first white settlers it created both opportunity and challenges. The first city in Oregon, Oregon City, was founded at the base of the falls in 1845. The power generated by the water was first put to use by settlers with the construction of a flour mill in 1844. A paper mill was added in 1867, and in 1889 the first major hydroelectric facility in the country was added at the base of the falls to power electric lights and cable cars in Portland.

Because of flow conditions at the falls, anadromous fish passage was only possible during high flows in the winter and spring months. As a result the only natural runs of anadromous fish into the upper Willamette Basin were spring Chinook and winter steelhead. Due to the development at the falls, the only natural routes available for salmon and steelhead to pass above the falls were blocked. Therefore, in order to prevent the complete elimination of anadromous fish into the upper Willamette Basin, in 1882 a fish ladder was blasted out of the basalt on the west side of the falls. Improvements were made to this rough ladder in the mid-1890's (Frazier 1988). Anadromous fish passage continued up the historical ladder until 1971, when the Oregon Department of Fish and Wildlife (ODFW) had a new, more effective fish ladder constructed.

Accurate counts of the number of returning salmon and steelhead in the Columbia River and its tributaries in the early years of settlement are generally not available, primarily because there was no direct way to count fish. Later, with the construction of dams and fish ladders through which returning fish had to pass, adults could be observed and counted with a high degree of accuracy. As a result, the only early estimates of the number of anadromous fish often come from the number of fish caught by commercial fishermen in the lower Columbia River, or from records of how much canned salmon was produced at canneries along the lower Columbia River. Mattson (1948) estimated the run of spring Chinook salmon in the upper Willamette Basin in the 1920s was up to five times greater than the run of 55,000 fish in 1947, or 275,000 fish. Some historical accounts suggest that the Native American fishery at Willamette Falls when settlers first arrived in Oregon took as many as 450,000 fish (Myers 2003).

Since 1946, ODFW has maintained annual fish counts at Willamette Falls. From 1946 until 1971, fish counts focused on spring Chinook salmon. Beginning in 1971, counts have been made monthly for spring Chinook, fall Chinook, coho, and winter and summer steelhead. The 50-year average for spring Chinook is 41,478, with a minimum of slightly more than 14,000 occurring in 1950, 1960, and 2008. From 1946 onward, maximum counts for spring Chinook occurred in 1953 (76,446), 1990 (71,273), and 2004 (96,725). In 2017, total spring Chinook count was 36,628. Figure 11 shows the historical and current range of spring Chinook salmon in the Willamette Basin (Kostow 1995).

Winter steelhead counts at Willamette Falls shows a 46-year average of 9,486 fish. The lowest count on record occurred in 2017 (822 fish). Other low counts occurred in 1996 (1,801), and 2009 (2,813). The highest counts occurred in 1971 (26,647) and 1988 (23,378). Figure 12 shows the historical and current distribution of winter steelhead in the Willamette Basin.

With the addition of fish ladders at Willamette Falls, other anadromous fish runs have been introduced and are able to migrate above the falls at times of the year that were historically not possible. As a result fall Chinook (50-year average run of 8,831 fish), coho salmon (50-year average run of 7,822 fish), and summer steelhead (50-year average run of 13,909 fish) now use the upper Willamette Basin for spawning and rearing (Table 3).

Table 3. Summary of fish counts at Willamette Falls.

Fish Run	1850 (historical accounts)	50-year average	Maximum (Year)	Minimum (Year)	2017
Spring Chinook	~450,000 *~275,000 (1920)*	41,478	96,725 (2004)	14,000 (2008)	36,628
Winter Steelhead	N.A.	9,486	23,378 (1988)	1,801 (1996	822
Summer Steelhead	N.A.	13,909	40,719 (1986)	690 (1972)	2,182
Fall Chinook	N.A.	8,831	34,189 (1974)	352 (1999)	3,462
Coho	N.A.	7,822	27,392 (2009)	683 (1993)	5,721

Though accounts are limited, the Willamette River and its tributaries were also known to harbor healthy populations of resident trout (rainbow, cutthroat, and Dolly Varden, i.e. bull trout). The following quotes from the Capital Journal newspaper (Salem, Oregon) provide some interesting insights about the abundance of resident trout and the impacts of water pollution (Allen 2013).

> *The Willamette River was a fine trout stream; it held cutthroat, redside (rainbow) trout, and Dolly Varden - - before pollution. The last trout I caught in the Willamette were two 22-inch rainbow about where Gerth Street would hit the river, sometime around 1922.*

> *The Rickreal provided excellent fishing from its mouth near Eola to its source. In the fall, the Dolly Varden ran up this stream to spawn. During hop-picking time, the hop pickers at the Horst Brothers' hop ranch added to the larder with Dollies that ran five to ten pounds and more. Some of locals did likewise (circa 1923).*

From the Oregonian April 25, 1916:

> *Two catch 21 Trout In Hour - While fishing on the North Santiam River last week, Rocky Willis and a Detroit man caught 21 trout averaging 20 inches in length, in a little more than an hour. Nineteen of the fish were Dolly Varden and the other two were redsides.*

These historical accounts indicate that resident salmonid populations in the mainstem and tributaries of the Willamette River were abundant and diverse. Bull trout (called Dolly Varden in historical accounts) are known to require colder water than other salmonid species (Oregon's maximum water temperature standard for bull trout is 10°C versus 18°C for other salmonids). Populations of bull trout today are restricted to headwater reaches of a few streams where water temperatures remain adequately cold. Early accounts of bull trout populations over a wide area of the Willamette Basin provide evidence that historical water temperatures were notably colder than current conditions.



Figure 11. Current and historical distribution of spring Chinook in the Willamette Basin. Green shows extinct populations and red shows existing populations. The Clackamas population is shown in yellow (from Kostow 1995).

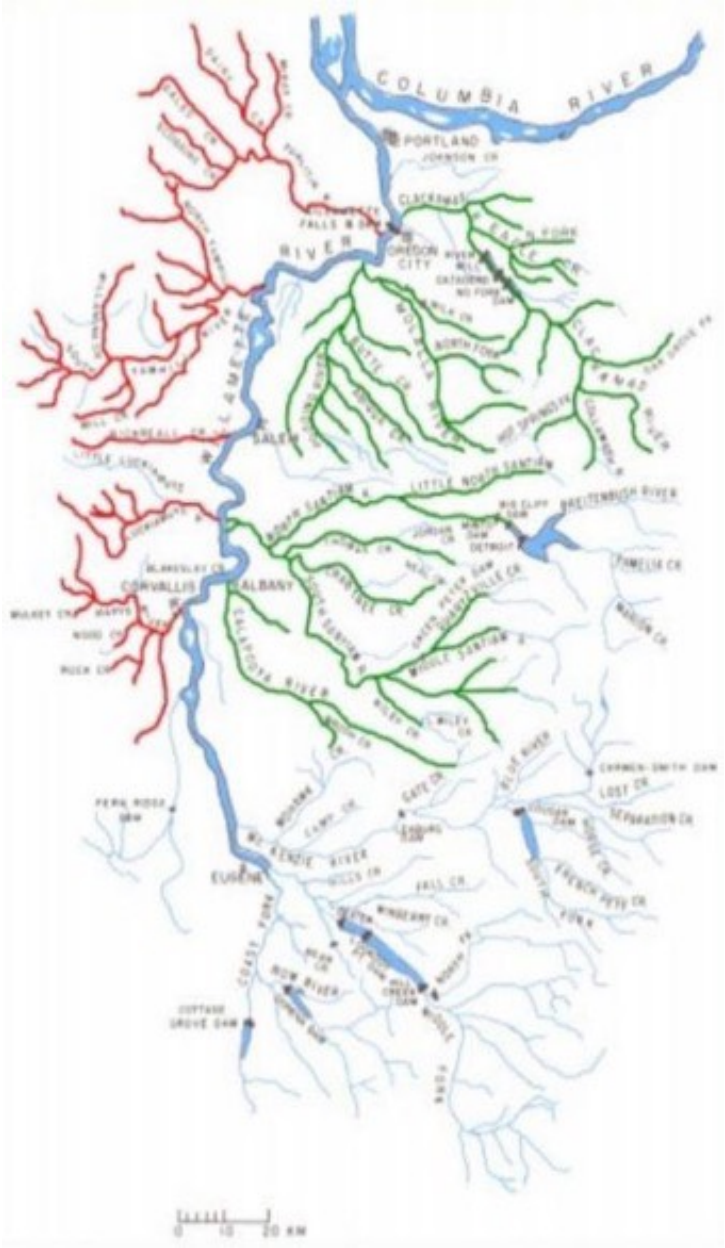


Figure 12. Current and historical distribution of winter steelhead in the Willamette Basin. Green indicates historical range and red indicates probable recent introduction (from Kostow 1995).

2.6.2 Resident fish and aquatic macroinvertebrates

Because of their commercial value, the most complete information about changes in aquatic life over time is for anadromous fish. However, a model has been developed that compares aquatic life indicators of fish and aquatic macroinvertebrates between pre-EuroAmerican conditions and 1990, as well as for future conditions in 2050 (Baker et al. 2002, VanSickle et. al. 2004). This model showed that lowland streams in the WRB had been impacted more severely than upland streams in the basin. For example, the median Ephemeroptera-Plecoptera-Trichoptera (EPT) richness across all lowland streams was estimated to have been 90% higher during the pre-EuroAmerican period than in 1990 (a loss of 5.2 EPT species) (Baker et al. 2002). While EPT richness could not be compared in upland streams between historical (pre-EuroAmerican) conditions and 1990 because of limited macroinvertebrate data, in upland streams a habitat suitability index (HSI) for cutthroat trout declined half as much in upland streams compared to lowland streams. Though limited in scope, these models suggest that lowland streams have been severely degraded over time while the upland streams (with minimal agriculture and urban development) show moderate degradation. These findings are consistent with other water quality, habitat, and aquatic life data collected in the WRB that have compared impacts from agricultural and urban land use to other land uses in the basin (Mulvey et al. 2009).

We infer that changes in habitat and food resources have caused shifts in macroinvertebrate communities over time. For example, because macroinvertebrates are known to feed on salmonid eggs and decaying salmon carcasses (Cederholm et al. 2000), the drastic reduction in fish spawning in Willamette Valley streams over time is likely to have affected the composition of macroinvertebrate communities. Shifts in macroinvertebrate taxa have also likely occurred in response to the near-complete loss of beavers in Willamette Valley streams. A study by Naiman et al. (1984) contrasted insect communities from beaver ponds with those from unmodified wadeable stream reaches. They found that the biomass and density of Dipteran taxa (particularly Chironomidae) increased dramatically compared to a modest representation of EPT taxa. These results suggest that Chironomidae would likely have been a dominant part of macroinvertebrate assemblages in beaver-impounded stream reaches (and likely in other slow-moving habitats as well, such as side channels or margin habitats). Besides shifts in species structure, studies in comparable aquatic systems in North American have reported total biomass and density of invertebrates in beaver ponds as two to five times greater than in stream riffle sites (Naiman et al. 1988). With the loss of salmon-derived nutrients and habitat changes from the loss of beaver, extensive agriculture, plus urban development, one can understand how present-day invertebrate communities, at even the best remaining stream sites, do not represent the invertebrate community that existed in WRB streams prior to European settlement. Appendix C describes a biodiversity index for identifying invertebrate communities with high diversity and low human disturbance.

Note: The model by Baker (2002) and VanSickle (2004) used relationships between aquatic life indicators and land cover/land use and stream water withdrawal data as surrogates for human disturbance. It did not include other factors such as changes in water quality due to chemical contamination.

3 Puget Lowlands

3.1 Physical characteristics

The Puget Sound Basin (PSB) is bordered by the Cascade Range to the east and the Coast Range (commonly referred to as the Olympic Mountains) to the west, and extends from the San Juan Islands in the north to past the southern end of the Puget Sound, covering a 13,700-square-mile area that drains to Puget Sound and adjacent marine waters (Figure 13). The PSB encompasses all or part of 13 counties in western Washington, as well as the headwaters of the Skagit River and part of the Nooksack River in British Columbia, Canada (Ebbert et al. 2000). The Puget Lowland EPA level 3 ecoregion covers the south, central portion of the PSB (Figure 14).

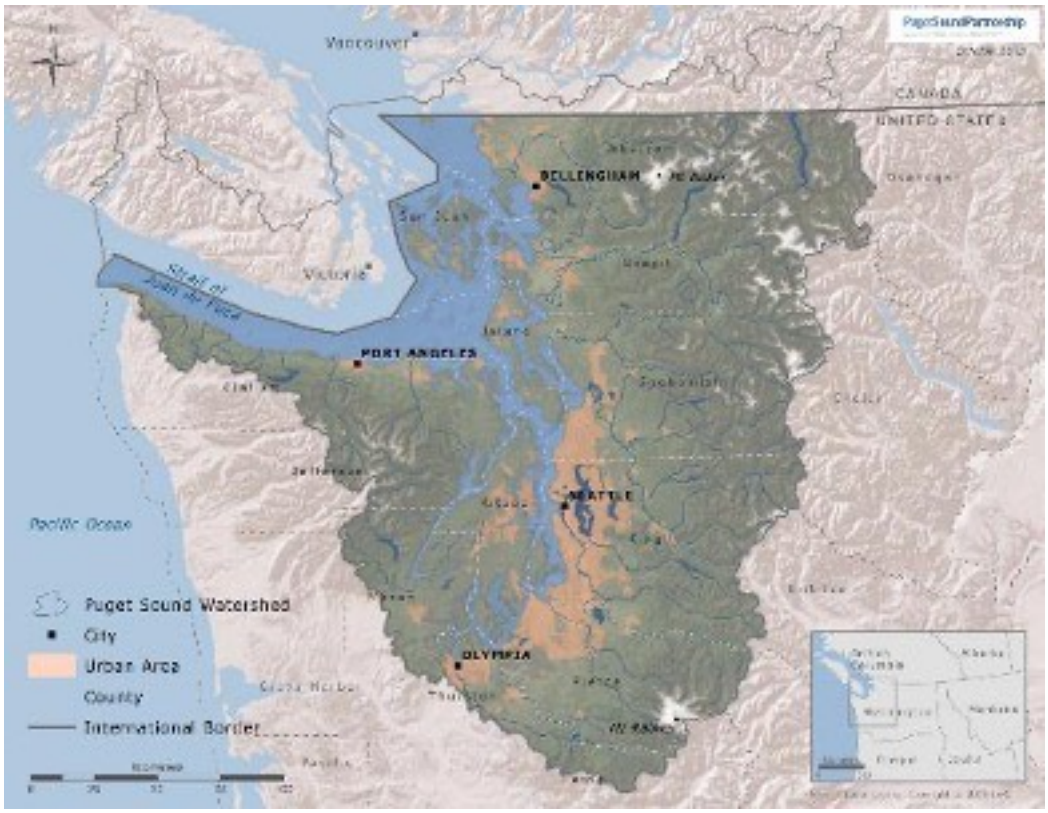


Figure 13. Map of the Puget Sound Basin. (Puget Sound Partnership 2013 - https://www.eopugetsound.org/maps/puget-sound-watershed-boundary).

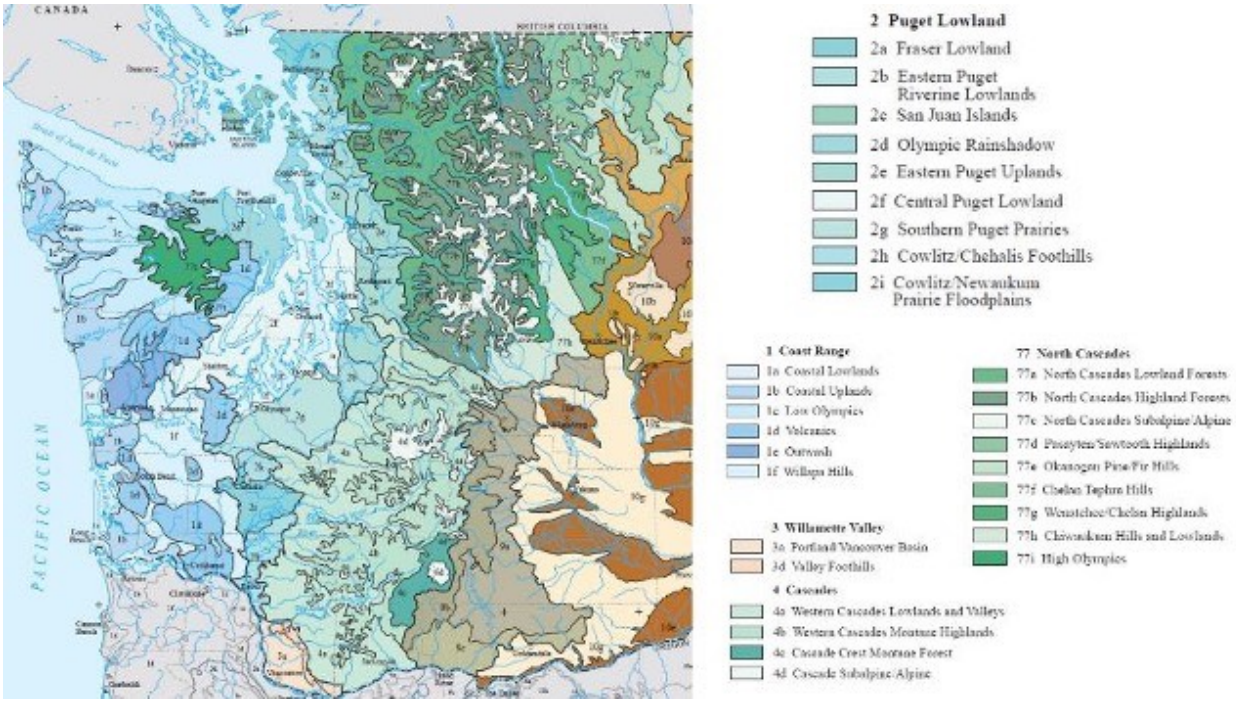


Figure 14. EPA level 3 & 4 ecoregions in western Washington (Pater et al. 2011). The Puget Lowland level 3 ecoregion (identified as “2 Puget Lowland”) is bounded by the Coast Range on the west and the Cascades on the east. Level 4 ecoregions are noted as “2a,” “2b,” “2c,” etc.

3.1.1 Topography

The larger basin (PSB) contains areas with dramatic changes in topographic relief (Figure 14), shaped largely by glaciation, volcanism, and plate tectonics. The relief within the low-lying Puget Lowlands is moderate. The area consists of moderately dissected tableland covered by glacial till, glacial outwash, and lacustrine deposits (Franklin and Dyrness 1973, Bailey 1995). Elevations range from sea level to 1500 feet (460 m) and average approximately 492 feet (150 m) (DellaSala et al. 2001). The southern portion of the ecoregion has more rugged relief than the north.

3.1.2 Climate

The sharp topographic relief creates highly variable local-scale climate in the PSB. The west (windward) slopes of the Cascade and Olympic mountains receive very large quantities of rain and snow during the wet season, exceeding 200 inches (5 meters) of water equivalent per year at some locations on the Olympic Peninsula. In contrast, the northeast corner of the Olympic Peninsula and the San Juan Island archipelago are relatively dry areas because they lie in “rain shadows” (the Olympic Mountains located to the west and southwest shield these areas from the direct impact of storms) (Sound Science Contributors 2012).

Overall, the Puget Lowlands ecoregion has warm, dry summers, and mild wet winters. Most of the moisture is deposited as rain and only a small percentage (~5% of total) appears as snow. Most of the rain and snowfall is concentrated between the months of October through March, with summer (June through August) rainfall events occurring at a lower frequency and smaller accumulations (Kruckeberg 1991). The large-scale drivers for these conditions include effects from the Pacific Ocean, interactions between large-scale wind and weather patterns, and the complex topography of the region. Of particular note is a semi-permanent low-pressure atmospheric circulation pattern commonly referred to as the Aleutian Low. The Aleutian Low works in combination with a semi-permanent cell of high pressure typically centered offshore of California. Together, these two systems bring a moist, mild onshore flow into the Puget Lowlands from October through early spring. Then, from late spring through summer, the large-scale wind patterns shift and the high-pressure system takes on a more dominant role, resulting in fewer land falling storms and a strong reduction in precipitation (Sound Science Contributors 2012). For example, average monthly rainfall in Seattle ranges from a high of 5.43 inches in December to a low of 0.63 inches in July (U.S. Climate Data).

3.1.3 Soils, geology and glaciation

The geologic history of the PSB is complex. Forces from volcanism and glaciation formed a complex mix of soils and sediment deposits in major river valleys and along nearshore marine areas. Glaciers covered the northern two-thirds of the Puget Lowland ecoregion in several thousand feet of ice, extending south to just beyond Olympia (Figure 15). Glacial till (sediment deposited directly by the ice) and outwash (sediment deposited by meltwater in front of the glacier) make up most of what is found at or near the surface. These glacial sediments were deposited during the last 2 million years by numerous glacial advances, the most recent of which was around 15,000 years ago (WA-DNR 2019). The southern area (Southern Puget Prairies (south of Olympia, WA); also known as Level IV ecoregion 2g (Pater et al. 2011) (Figure 14)) was not glaciated but was subject to glacial outwash which formed rolling to steep sloping hills. The southern portion of the Puget Lowland is also partly covered with ancient lava flows, similar to those found at Mount Rainier and Mount St. Helens. To the north, the San Juan islands are composed of metamorphic rocks accreted onto North America 160 million years ago (WA-DNR 2019). The dominant soil types in the Puget Lowlands were influenced by glacial processes and repeated volcanic activity and were formed largely from glacial drift (Staubitz et al. 1997, Pater et al. 2011).

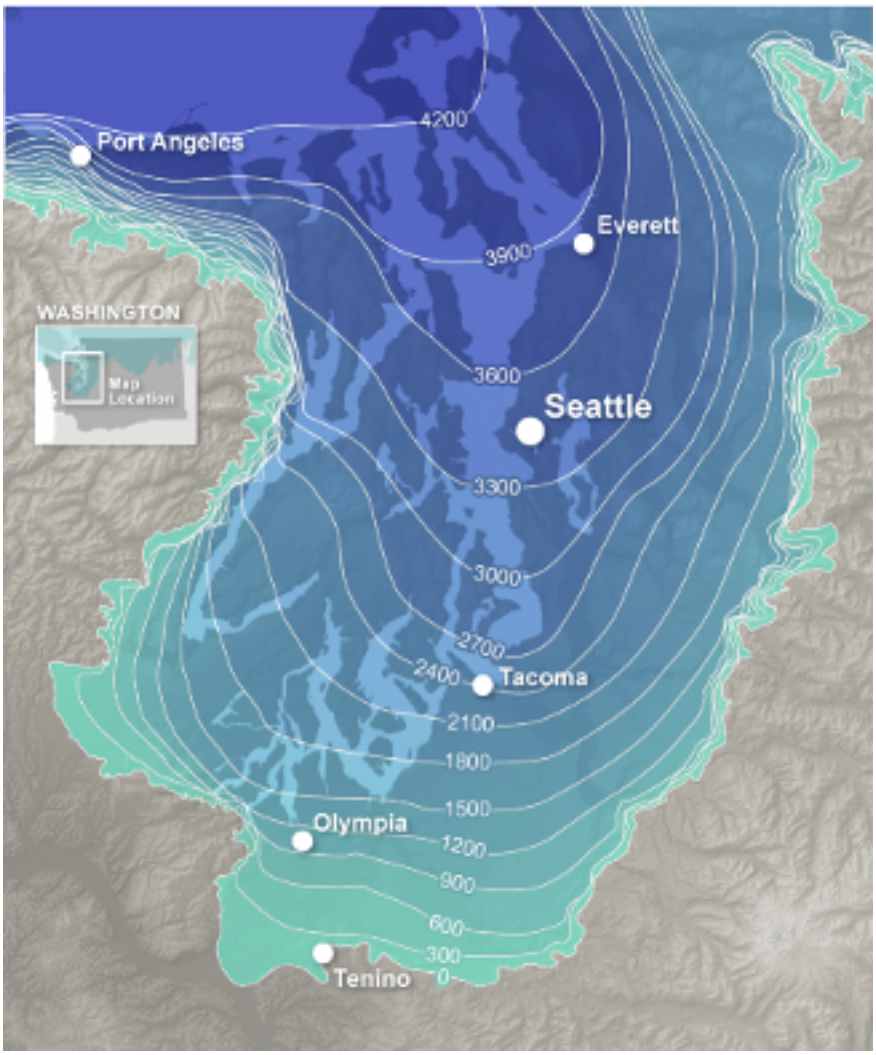


Figure 15. During the most recent period of glaciation (around 15,000 years ago), an ice sheet (referred to as the Puget Lobe) covered the northern two-thirds of the Puget Lowlands ecoregion. The ice sheet, which was several thousand feet thick in places (as shown by the contours), extended south of Olympia. Source of map: https://www.dnr.wa.gov/programs-and-services/geology/glaciers.

3.2 Brief History of Human Settlement

The first humans to settle the PSB were drawn to it by its rich ecological resources. The first inhabitants, who likely arrived as far back as four to six thousand years ago, were mobile and likely hunter-fisher gatherers. By the time Captain George Vancouver first visited Puget Sound in 1792, Native Americans had established permanent villages, where they spent the winters. The transition from a nomadic to a more sedentary life-style was likely driven by food resources, social complexity and organization, technological advances, and expansion of estuaries (resulting from remnant effects of the last glaciation). Food resources that sustained these early settlements included abundant salmon, smelt, eulachon, herring (roe), and shellfish. The wild and natural streams and rivers supported abundant populations of all five salmon species (Quinn 2010).

The first European settlers were also attracted by the rich natural resources of the region. The earliest explorers arrived in Washington via ship or the Oregon Trail in the late 18th and early 19th centuries. Many were fur trappers who came for the abundant supply of otter and beaver pelts. In 1846 the first settlement called New Market, or Tumwater, was established. The Washington Territory was established from part of the Oregon Territory in 1853 (Quinn 2010). European settlement occurred rapidly following the initial development of natural resource use (e.g. beaver trapping and fur trading in the early to mid-1800s). During the 79 or so years following the beginning of full-scale development, many of the towns and cities along the Interstate-5 corridor were established. They were connected primarily by railroads but major highways were also established (which follow routes similar to those used by travelers today - see red lines in Figure 16). With European settlement came industrialization and a different philosophy regarding resource use, which resulted in a measurable changes to the landscape as land was cleared for railroads, agriculture, and residential development, resulting in diminished salmon returns and poor water quality.

By the early 1850's logging became an important part of the economy. Timber harvest by hand axe and horse teams in the 1850's gave way to more mechanized timber harvest with railroads moving timber by the 1890's. The California Gold Rush in the 1870's created a huge market for timber from Puget Sound, which was shipped by boat to San Francisco. Commercial salmon catches also increased dramatically during this time, with landings increasing by 2,000 percent in the 1890's compared to the 1870's (Center for the Study of the Pacific Northwest, 2009). The result of this mechanization was a rapid increase in resource extraction. The level of impact of this industrial transformation, however, varied between Washington and Oregon. Oregon had been settled primarily by farmers and as a result had a different land-use philosophy. A quote by Ivan Doig (1982) brings this difference to light: "Even what I have been calling the Pacific Northwest is multiple. A basic division begins at the Columbian River; south of it, in Oregon, they have been the sounder citizens, we in Washington the sharper strivers. Transport fifty from each state as a colony on Mars and by nightfall the Oregonians will put up a school and a city hall, the Washingtonians will establish a bank and union."

By the mid-20th century much of the resources that supported the economy (large salmon runs, mature forests, coastal wetlands, and clean water) in the late 19th century had been badly degraded. For example, the peak in salmon harvest occurred from 1910 to 1919, and lumber production in the 12 counties surrounding Puget Sound declined from roughly 6 million board feet in 1926 to 2 million in 1951 (Chasen 1981). With continuing declines in natural resources, the economy gradually diversified, especially following the Second World War. As a result, interest in natural resources shifted from a purely extraction viewpoint to viewing it as important for recreational opportunities as well.



Figure 16. Puget Sound Country map produced by the Kroll Map Company in Seattle, 1919 (Scale 1:652,600). Major highways are depicted by red lines on the Puget Sound Country map and look like some of the same routes used by travelers today.

3.3 Land use and land cover

Urban/rural residential development and industrial activity are among the principal land use activities in the PSB (Figure 17). Currently, 67 percent of Washington State's entire population lives in the PSB, which includes the cities of Seattle, Tacoma and Olympia. In recent decades, the PSB has experienced rapid urban growth. In 2010, the U.S. Census Bureau reported approximately 4 million people within its counties (King, Kitsap, Pierce, Snohomish, and Thurston), a 38 percent increase from 1990 (Villarreal et al. 2017). Another 1.1 million people are expected to arrive to the urban and suburban regions of PSB by 2020. To date the trend has been an increase in urban sprawl across central Puget Sound basin (Ebbert et al. 2000).

Areas of agricultural land use surround the urban areas (both are concentrated in the lowlands) (Figure 17). The distribution of agricultural lands is largely driven by the locations of prime agricultural soils, most of which have been fully developed. More than half of the agricultural acreage in the PSB is located on the outwash deposits, alluvial valleys, and river deltas of Whatcom, Skagit, and Snohomish Counties (Staubitz et al. 1997). The Puget Lowlands wet climate coupled with low topographical relief and rich pastureland creates conditions highly favorable for dairy production. Other agricultural activities include pastureland, hay farming and cropland. Agricultural land use consists of about 60 percent cropland and 40 percent pasture (Ebbert et al. 2000). Some areas that once had large tracts of wetlands were drained before the 19th century and converted to pastureland, cropland, and forests. An example of wetland eradication can be found in the Eastern Puget Riverine Lowland (Level IV ecoregion; 2b) that includes the Skagit River floodplain (Figure 14). Here, through diking and draining, there has been an 80 percent decline in tidal marsh and estuarine ecosystems in the last 150 years (Quinn 2010).

Logging and cultivation of harvestable forests has also had a large effect on the landscape. In the last 100 years over 60 percent of Washington's old-growth forests have been cut. All the remaining old-growth is limited to relatively high elevation public lands, while 23 percent of forestland in Puget Sound has been converted to agriculture and urban land uses (Washington Department of Natural Resources 1988).

Before European settlement, the Puget Sound Lowland forests were dominated by dense coniferous forests most commonly made up of western red cedar, western hemlock (*Tsuga heterophylla*), and Douglas-fir (*Pseudotsuga menziesii*) (Bailey 1995). Mixed stands of Douglas-fir with some Garry oak (*Quercus garrayana*), Pacific dogwood (*Cornus nuttallii*) and arbutus (*Arbutus menziesii*) were common on drier sites. Periodic flooding and infrequent fires were once the predominant disturbance regimes in the region (McNab and Bailey 1994). Long intervals (centuries) between large-scale fire events were more typical of moister forest types (Agee 1993), with drier forests (*Quercus* sp., *P. ponderosa*) and prairies experiencing frequent fires.

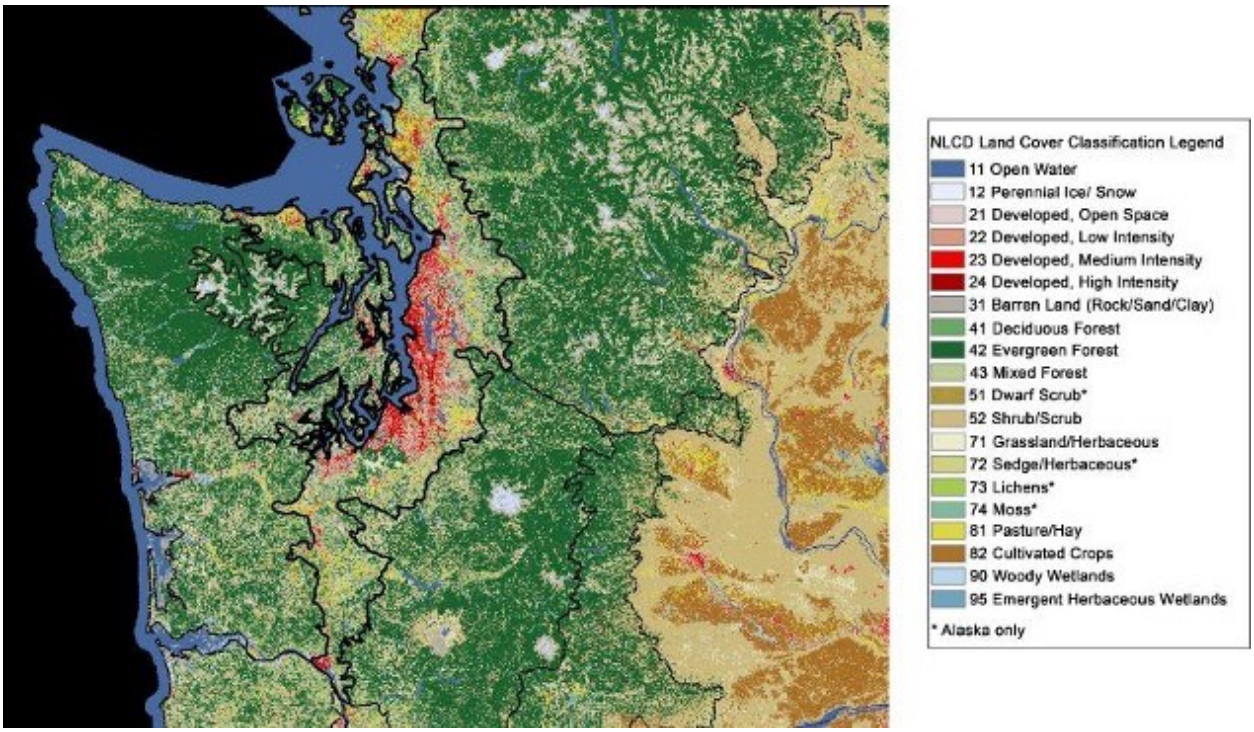


Figure 17. Land cover in the Puget Lowlands and neighboring ecoregions, based on the National Land Cover Database (2011 Edition, amended 2014) (USGS 2014).

3.4 Hydrology

Stream conditions in the Puget Lowlands prior to European settlement provided stable and plentiful habitat for salmonids and other cold-water species. The natural state of low to mid-order streams included ample shade, cold water, clean spawning gravels, abundant pools formed by woody debris, and abundant food. Higher order streams included side channels, wetlands, and sloughs that provided ample refuge habitat for developing juvenile fish as well as migration corridors unblocked by dams, dikes, or irrigation channels.

From the 1840's onward stream conditions have been steadily altered. Major changes first resulted from the loss of beaver and the ripple effects on riparian vegetation and stream channel complexity. Continued human population growth and resource use have altered the landscape and habitat from railroads, agriculture, and residential development, resulting in further loss of stream habitat complexity. A study of low-order streams in the PSB by the U.S. Forest Service (1993) reported the number of pools per mile declined from 25 to 60 per mile to only 5 to 25, with even greater loss of pools in low- and mid-order lowland streams due to removal of large woody debris for flood control and fisheries management (Staubitz et al. 1997). An increase in storm peak flows, stream-bank erosion and failures, and landslides have also increased due to clearcut logging and urbanization. One result of these changes has been a loss of spawning gravels from scouring and armoring of coarse substrate in high gradient streams and excessive deposition of fine sediments in low gradient streams. Altered habitat also includes major losses

of salt marshes with declines of 74 percent in the Snohomish River delta, 99 percent in the Duwamish River delta, and 100 percent in the Puyallup River delta (Staubitz et al. 1997).

A study by Haring (2002) estimates that the following habitat changes have taken place since 1895:

- Majority of streams have lost more than 20% of habitat historically accessible to fish
- Majority of streams have lost more than 66% of wetted area
- Loss of large woody debris
- Loss of pool habitat
- Degradation or loss of riparian habitat
- Less than 60% of a watershed with forest stands aged 25 years or more

Hydrology in the PSR has also been affected by the construction of dams for flood control or power generation. Flows in the Skagit, Nisqually, Green, Skokomish, and Cedar rivers are regulated by dams (Puget Sound Partnership 2009). In King County alone, there are 124 dams (WA ECY 2018). Water withdrawals and over-appropriation of water rights (not enough water for granted water rights plus support for fish and water quality) has been noted for up to one-fourth of the watersheds in the PSB (Washington Department of Natural Resources 2000). A limiting factors analysis of Puget Sound ecosystems by Smith (2005) found that 12 of 19 basins had a poor rating for multiple problems including: 303(d) listing for low flow, known salmon mortality due to low flow or other studies documenting low flow problems, and prohibition to additional water allocation due to over appropriation (Quinn 2010). As increasing demands for surface and ground water continues, the need to maintain instream flows for fish and other aquatic life becomes ever more critical (Ebbert 2000).

Natural factors such as climate, geology and beaver activity also have an important influence on hydrology. Most rivers in the Puget Lowlands are rain-dominant and exhibit peak flows during winter (when most of the rainfall occurs). During the summer, flows are typically at their lowest (Kruckeberg 1991).

Geology also has a strong influence on hydrology. The porous nature of the glacial till and glacial outwash in the Puget Lowlands creates the potential for strong connectivity between streams and groundwater. However, human activities have altered this connectivity in many places through conversion of large tracts of wetlands to agricultural or other land use - and accompanying consumptive water uses for irrigation, etc. – plus the reduction/eradication of beavers. Drainages with riparian wetlands that persist throughout the dry months are recharge areas when stream flows are at the lowest levels (Ward 1989). These areas also serve as important habitat for Coho salmon, both during the dry months (as thermal refugia) and wet months (to avoid high flows in the main channel) (Quinn 2005).The loss of wetlands and reductions in stream recharge also affects hydrology during the wet season; flows during storm events are typically flashier, and more sediment transport occurs (and is carried farther downstream).

3.5 Water quality

Since European settlement began in the mid-1800's impacts to water quality have accelerated. The abundant populations of aquatic species, including all five species of salmon, show that when Europeans arrived in the PSB water quality was high throughout the region. Over time, aquatic resources have been degraded due to human induced changes such as soil erosion, instream habitat disturbance, reduced streamflows, dams, and toxic runoff of pesticides and herbicides. Finally, in 1917, salmon runs and water quality had reached such high levels of degradation where the State of Washington decided to take action and the State Legislature passed the Water Code (Chapter 117) (Tanner 1917). This legislation established water use rights for agriculture, power generation, reservoirs, municipal water supply, and mining, all industries that had been growing for the past 50 years with no regulatory laws (Quinn 2010).

Water quality has been affected throughout the PSB by both point-source and nonpoint-source runoff. Approximately half of all contaminant loadings in Puget Sound are thought to come from industrial and municipal point sources, while the other half comes from nonpoint source pollution (Puget Sound Water Quality Authority 1992). Over time point sources of pollution have been controlled more effectively than nonpoint sources, but many point source pollution problems still exist. In 2008, for example, the state identified 115 sites covering 3,900 acres in Puget Sound with contaminated sediments (Washington Dept of Ecology 2008). Fifty percent of this contamination results from point sources including pulp, paper, and chemical production and petroleum refining, transport, and storage. Nonpoint sources of pollution are even harder to control because of their diffuse nature, and the many small sources of pollution that together produce a large cumulative effect. Nonpoint sources of pollution include urban runoff, seepage from on-site septic systems, infiltration or washoff of land applied manure, fertilizer, or pesticides, and deposition from atmospheric sources, plus new classes of chemicals such as endocrine disrupters and fire retardants. In 2005 the Center for Biological Diversity identified 1,000 species at risk in PSB including iconic species such as Puget Sound Chinook Salmon, Steelhead Salmon, and the Southern (Puget Sound) Resident Killer Whale, all of which are imperiled mostly by human related activities in the region (Quinn 2010).

The most common current water quality impairments in streams throughout the Puget Lowland ecoregion include temperature, dissolved oxygen concentrations, pH, bacteria (fecal coliform concentrations) and biological impairment (increasingly in the urban areas). A list of impaired stream segments can be found at:
https://fortress.wa.gov/ecy/waterqualityatlas/map.aspx?CustomMap=y&RT=0&Layers=23,29&Filters=n,n,n,n

Legacy contaminants that are less common but persistent in the environment, like PCBs, are also included in this list. This type of toxic chemical presents a challenge when implementing cleanup plans and expecting an improvement in biological communities including fisheries at all life stages. Even though water quality impairments are corrected, the persistent toxics in the environment often prevent important aquatic life communities from achieving viable and sustaining populations (Carpenter et al. 2016).

Legacy effects from extensive past logging still affect water quality and stream morphology. The logging exposed steep-sided slopes and riparian areas continue to introduce large quantities of sediments into streams. Impacts are particularly evident in the stream morphology in the southern foothills, where streams were historically deeply incised and heavily shaded. The historic logging practices resulted in filling of the original channels with sediment. As a result, streambeds in this area have wider wetted widths and are susceptible to erosion during higher flows. A study of flow records and suspended sediment samples by Glancy (1971) on the mainstem of the Chehalis River and its tributaries showed that the highest sediment yields occurred in tributaries located in this southern part of the ecoregion. Deforestation of select areas have also resulted in catastrophic debris flows that scoured streambeds to bedrock, leaving little or no habitat for biota to colonize. Landslide occurrence is episodic and depends on several factors, including hillslope gradient, quantity of water in the soil, composition of parent material, depth of soil, and degree of weathering (Swanston 1991, McHenry 1991).

3.6 Historic information on aquatic communities

3.6.1 Anadromous Fish

Five species of salmon are found in the rivers and streams throughout the PSB, and the native range of these salmon species extends throughout the Pacific Northwest. Salmon have been a staple in the diet of inhabitants of the Puget Sound region from the time tribal communities first arrived.

Salmon provide a direct linkage between the marine and freshwater ecosystems, as they use both habitats for growth, survival, and reproduction (Quinn 2005). As a result they play an important role in freshwater ecosystems affecting primary productivity, food-web dynamics, and development of community composition (Pess et al. 2002). Due to the reduction of densities and loss of salmon species in watersheds throughout the Puget Sound, there has been a dramatic impact on stream ecosystem functions in both aquatic and terrestrial communities. The return of adult salmon to freshwater ecosystems transport marine derived nutrients that support both aquatic and terrestrial communities, while out-migrating smolts represent the export of nutrients (particularly phosphorus) out of the freshwater ecosystem. The decline in adult salmon returns has resulted in less phosphorus input to freshwater ecosystems compared to the smolt export of phosphorus, leaving a net loss of nutrients in some drainages (Scheuerell et al. 2005, Kohler et al. 2013). This new perspective on flux of nutrients between marine and freshwater ecosystems highlights the dramatic change in productivity that has likely occurred in Puget Sound streams that are already, naturally nutrient-poor.

Numerous factors have contributed to the decline in salmon populations, most of which are the result of human induced disturbances such as urbanization, logging, agriculture, dams, habitat loss, and water quality impacts (Figure 18).

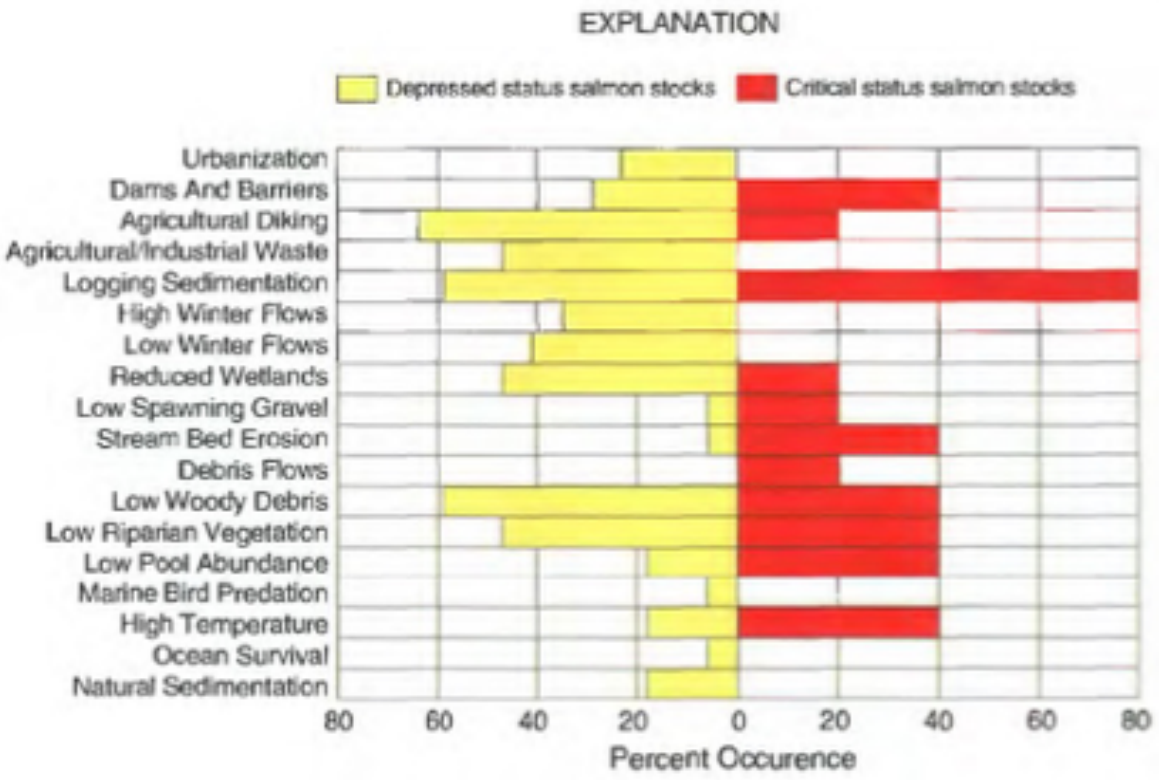


Figure 18. Percentage of times that each human or natural disturbance was cited as a potential cause of fish stock declines (data from Washington State Department of Fish and Wildlife and others, 1993; source: Staubitz et al. 1997). Depressed stock are those that are experiencing production levels below levels expected on the basis of available habitat and natural variations in survival rates, but are above the level at which permanent damage to the stock is likely. Critical stock are those that are experiencing production levels that are so low that permanent damage to the stock is likely or has already occurred.

Figures 19 through 22 show current and historic distributions of winter steelhead, spring and fall Chinook salmon, respectively. The maps show reaches where specific functions of the migratory run in freshwaters occur in the Puget Lowlands ecoregion. The distribution of winter steelhead is extensive throughout the Puget Lowland ecoregion, as they use all of the major drainages in the Puget Lowlands for different functions of their freshwater life cycle (e.g., spawning, rearing, or presence).



Figure 19. Current and historic presence of winter steelhead in Puget Sound streams and functions of the migratory run in stream reaches.

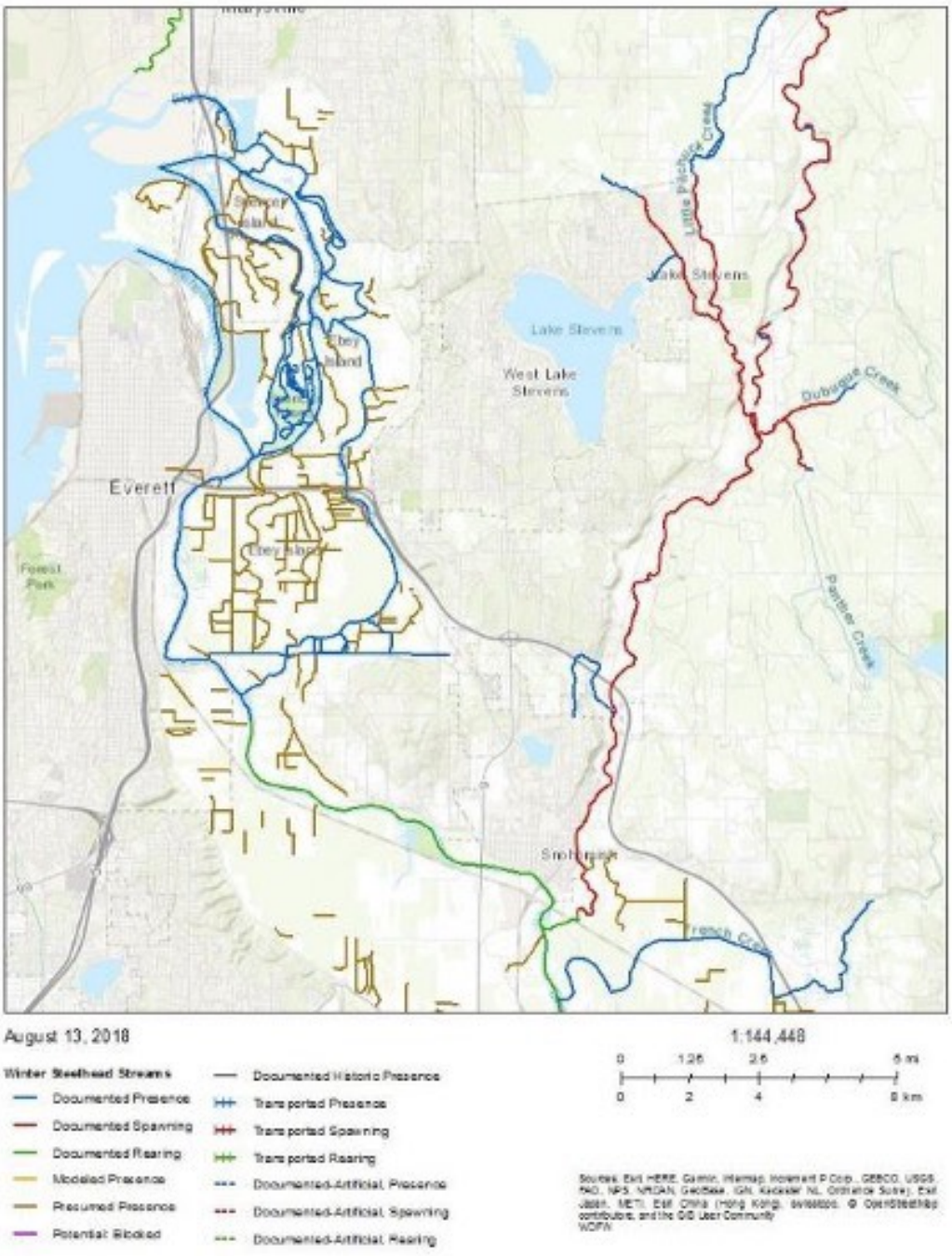


Figure 20. High resolution map of winter steelhead distribution and functions of the migratory run of the Lower Snohomish Basin mainstem and tributaries.



Figure 21. Current and historic presence of spring Chinook salmon in Puget Sound streams and functions of the migratory run in stream reaches.



Figure 22. Current and historic presence of fall Chinook salmon in Puget Sound streams and functions of the migratory run in stream reaches.

3.6.2 Resident fish and aquatic macroinvertebrates

As with the Willamette Valley, the most complete information about changes in aquatic life over time is limited to anadromous fish. While specifics about individual macroinvertebrate taxa are lacking, some assumptions can be made about changes to the overall assemblage based on changes that are known to have occurred in the food base and energy sources, habitat (both riparian and in-stream), and hydrology (flow and water quality). Table 4 summarizes the extent of changes for a range of factors starting from 1895. Among the changes is the drastic reduction in number of salmonids, which were once widespread in the Puget Sound Basin (Figures 19-22). Work by Haring (2002) suggests that there were 25 times more spawners in 1895 versus present day. Because macroinvertebrates are known to utilize salmonid eggs and decaying salmon carcasses as a food source (Cederholm et al. 2000), this has likely had an effect on composition of macroinvertebrate communities over time.

Examples of other changes that have likely caused shifts in macroinvertebrate communities over time include reductions in intact habitat areas (in particular in riparian zones) due to logging, development and other human activities. This has resulted in reductions in leaf litter, which is an important food source for some macroinvertebrates (Frady et al. 2007). Past research shows some examples where deciduous forests in the western Puget Lowlands contributed as much as 300 g C m^{-2} to streams (Fisher 2006). The loss of intact riparian areas has also resulted in a loss of shading and bank stability (Haring 2002), which indirectly affects macroinvertebrates via decreases in habitat and water quality. Reductions in woody debris inputs (due to logging, development, etc.) have also changed habitat in Puget Lowland streams over time. In particular, this has affected braided streams, which offer complex living spaces for a variety of benthic communities as well as rearing areas for sub-yearling life stages of juvenile salmon and smolts (Quinn 2010). Finally, the drastic reductions in beaver activity over time have also likely affected macroinvertebrate communities. Certain types of depositional macroinvertebrate taxa, particularly Chironomidae, are typically found in beaver-impounded streams (Naiman et al. 1984) (and likely in other slow-moving habitats as well, such as side channels or margin habitats).

Table 4. Changes in key factors affecting fish and aquatic life in Puget Lowland streams from 1895 to present (from Haring 2002).

Category	Key characteristics	Stream Condition in 1895 vs. today
Food Base	Allochthonous Input	> 25 times more return spawners
	Autochthonous Input	
	Marine-Derived Nutrients	
Riparian Condition	Leaf litter/Wood	> 2.5 times more intact area
	Shading (Water Quality)	
	Bank stability	
Hydrology	Water Quantity	> 3 times more tributary wetted area
		> 3m tributary stream width
	Flow Characteristics	< 0.5m/sec average water velocity
Habitat Diversity	Hard-bottomed (Cobble/Gravel)	> 1.25 times more useable habitat
	Soft-bottomed (Silt/Muck/Fines)	
	Woody debris	

4 The Pacific Northwest Coastal Region

The entire PNMR BCG model covers that portion of Oregon and Washington from the crest of the Cascade mountains to the Pacific Ocean (Figure 23). The coastal region primarily adds the region west of the Willamette Valley and Puget lowlands, from the crest of the Coast Range to the Pacific Ocean (Figure 24). This region will be referred to as the Coastal PNMR throughout the following discussion.

The Coastal PNMR covers the western most area of the two states. It is heavily influenced by the Pacific Ocean, particularly in terms of weather, but also from the ocean's impact on the ecology of the region due to the wet climate that supports extensive coniferous forests, large number of streams and rivers, and once significant runs of anadromous salmonids. The current condition of Coastal PNMR streams have been heavily influenced by forest practices and logging activity, which will be described in detail in this report.

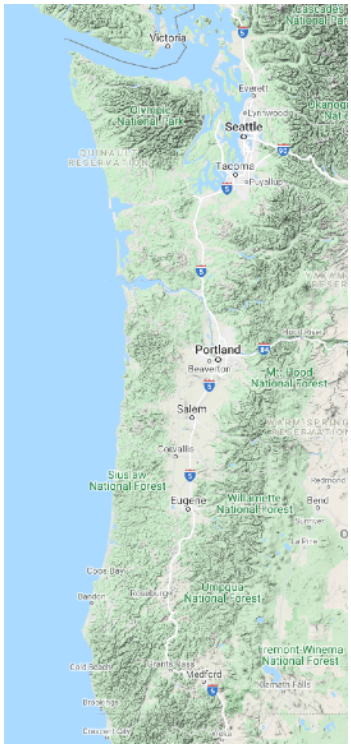


Figure 23. The Pacific Maritime Northwest Region (PNMR) extends from the crest of the Cascade Mountains on the east to the Pacific Ocean on the west.

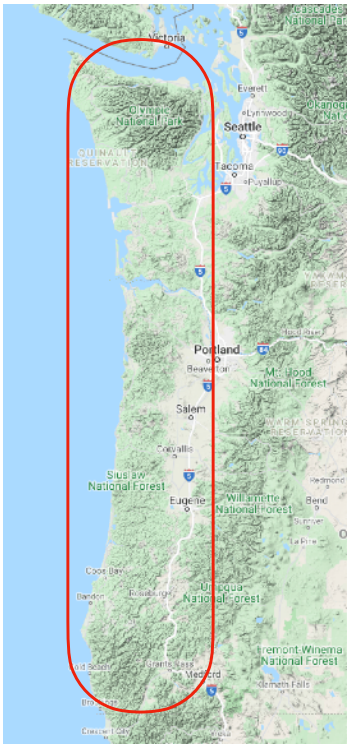


Figure 24. The coastal portion of the PNMR region (Coastal PNMR) highlighted above adds the area west of the Willamette Valley and Puget Lowlands ecoregions covered in the initial BCG model.

4.1 Physical Characteristics

4.1.1 Topography & Geologic History

The topography of the Coastal PNMR could best be described as "varied." The region covers both Washington and Oregon, and extends from the Canadian-US border to the California-Oregon state line. The settlement history and ecosystem traits of the two states share many characteristics, especially concerning land use and human disturbance patterns. Enough differences exist in topography and geologic history across the two states, however, that they are described separately below.

4.1.1.1 Washington

The Olympic Peninsula, at 3,600 square miles, comprises the majority of the Coastal PNMR in Washington (Figure 25). The Peninsula remained relatively unexplored by Europeans until Arthur Dodwell and Theodore Rixon mapped most of its topography and timber resources between 1898 and 1900 (Majors 1975).

Mount Olympus at 7,965 feet is the highest peak in Washington or Oregon west of the Cascade Mountains. In total a half dozen or more peaks from roughly 4,000 to 7,965feet in elevation create the heart of the Olympic Mountains, and most of the area is protected within the Olympic National Park. Comprising over 1400 square miles, Olympic National Park covers almost half of the Olympic Peninsula and is one of the largest contiguous areas in the Coastal PNMR with relatively low human disturbance (Figure 25). The park also contains 3,000 miles of rivers and streams and over 60 named glaciers (Olympic National Park website). However, given its relatively high elevation compared to the rest of the coast range in Oregon and Washington, it stands out as a unique region with its own weather and mix of habitat conditions.

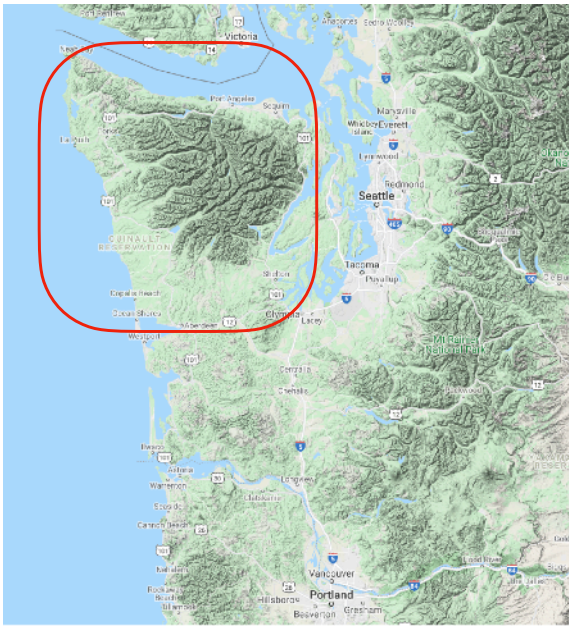


Figure 25. Olympic Peninsula circled in red and Olympic National Park (in dark green).

The topography of the Olympics and Puget Lowlands is partly the result of the last ice age that reached its maximum about 14,000 years ago (Mathews 1988). At its maximum, the largest ice sheet flowed around the northern and eastern flanks of the Olympic Mountains and down the Puget trough as far south as Olympia. While the ice sheet didn't cover the Olympic Mountains it did block river drainages and formed ancient Lake Elwha and other valley lakes thousands of feet deep. In addition this glacial period produced hundreds of alpine valley glaciers that shaped many of the river valleys in the northern portion of the Coastal PNMR (Mathews 1988).

The geology of the Olympic Peninsula includes extensive high peaks and ridges of erosion resistant basalt dating from more than 55 million years ago (mya) (Olympic National Park website). Mixed with the basalt are erosive marine sediments composed of shale, sandstone, and graywacke (type of sandstone) and their metamorphic byproducts of slate and schist. The collision of the oceanic and continental plates uplifted these marine sediments into their current position.

South of the Olympic Peninsula and north of the Columbia River lies a region of low coastal mountains known as the Willapa Hills (Figure 26). The Willapa Hills are the lowest elevation group of uplands along the entire Coastal PNMR. The highest point is Boistfort Peak at 3,087 feet. The core of the Willapa Hills region has been formed by submarine basalts uplifted from the sea floor approximately 53-55 mya. A variety of marine derived sandstones overlay and interbed the basalts (Moothart 1992).

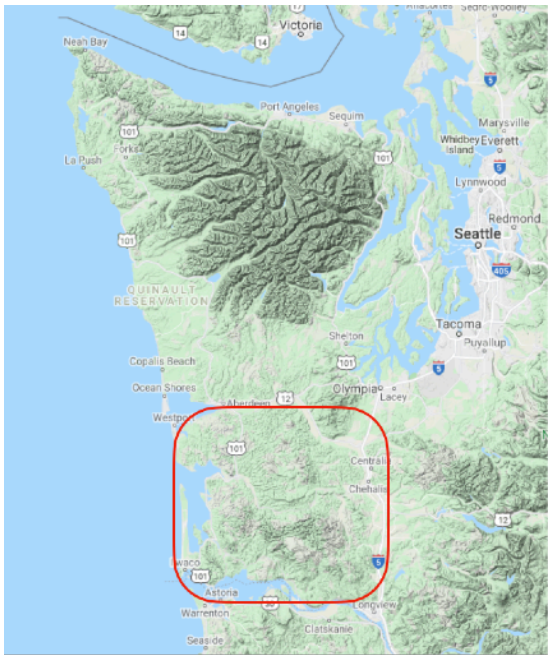


Figure 26. Willapa Hills region of Olympic Peninsula.

Over a dozen rivers originate here. The largest, the Chehalis River, flows northeast then west and empties into Grays Harbor, the northern boundary of the Willapa Hills. Other rivers like the Naselle and Willapa Rivers flow west and empty into Willapa Bay. Forestry is the dominant land use throughout the region.

4.1.1.2 Oregon

The Coast Range in Oregon forms a contiguous range of low elevation mountains that extend from the Columbia River in the north to the Siskiyou Mountains in the south (Figure 27). Marys Peak, just west of Corvallis, at 4,097 feet high is the highest peak. The Coast Range is approximately 200 miles long, and ranges from 30 to 60 miles wide from the Pacific Ocean to its eastern flanks. With the exception of Marys Peak the maximum elevation for most of the Coast Range ranges from 1400 to 2500 feet with an average elevation of 1500 feet (Atlas of Oregon 1976).

Like much of the coastal region in Washington, the geology of the Oregon Coast Range is largely composed of marine derived basalt and sedimentary rock originating along the sea floor approximately 50 mya. Basalt flows from 40-35 mya created notable headlands along the coast such as Heceta Head, Cascade Head, and the Tillamook highlands (Atlas of Oregon 1976). From 30 to 25 mya harder rock intruded into the earlier period sedimentary strata. Twenty million years of rainfall and weathering eroded the softer sedimentary rock leaving the harder intrusive rock to form the higher peaks. A number of major rivers, and many smaller streams, emanate from the coastal highlands and either flow

west into the Pacific Ocean (e.g. the Nehalem, Wilson, Nestucca, Siletz, Alsea, Siuslaw, and Umpqua rivers) or flow east into the Willamette Valley and Willamette River (e.g. the Yamhill, Tualatin, and Luckiamute). Two large rivers, the Umpqua and Rogue, have their headwaters in the Cascade Mountains and completely intersect the Coast Range as they flow west to the Pacific Ocean.

While the Siskiyou Mountains appear to be a continuation of the Coast Range, they are geologically distinct and part of the larger Klamath Mountain region that extends well into Northern California. The region is composed of much older rock, most notably sandstones, chert, and mudstones of the Dothan Formation of the Late Jurassic and Late Cretaceous periods (150-100 mya) mixed with volcanic pillow lava and granitic cobble (Beaulieu 1971). The middle fork of the Coquille River forms the Siskiyou's northern boundary, and the California-Oregon border roughly defines its southern extent. From west to east, the area extends from the Pacific Ocean eastward to the Cascade Range.

The complexity of the underlying geology of the Siskiyou Mountains results in exceptional biodiversity of plant and animal species, including 30 different conifer species (seven endemic to the region) making it one of the highest conifer diversity regions in the world (Norse 1990). The Rogue River, with its headwaters in the Cascades, is the largest river draining this region, but other important rivers include the Chetco, Pistol, Elk, and Sixes among others.

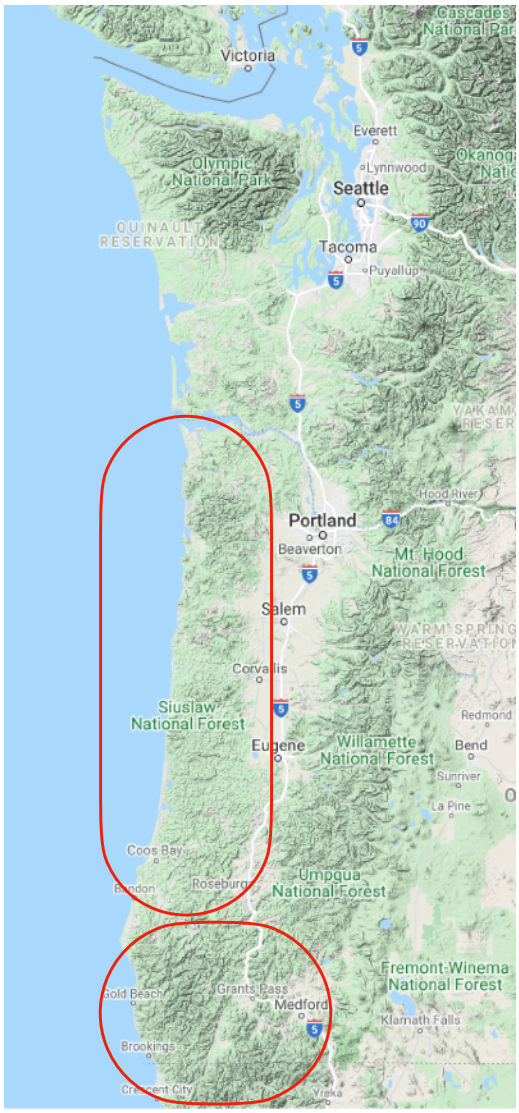


Figure 27. Oregon Coast Range (upper circled area) and Siskiyou Mountains (lower circled area) make up the two distinct regions of the Coastal PNMR in Oregon.

Within the Siskiyou Mountains lies the 280 square mile Kalmiopsis Wilderness created by Congress in 1964. The wilderness is noted as having deep, rough canyons, sharp rock ridges, and many streams and rivers. In 2002 the Biscuit Fire burned nearly 780 square miles of the Siskiyou Mountains including much of the Kalmiopsis Wilderness.

4.2 Climate

While distinct local weather conditions occur throughout the Coastal PNMR, overall the region is classified as a maritime climate or marine west coast climate (McKnight & Hess 2000). It is characterized by year round westerly air currents, mild winters and summers, moderate to abundant rainfall, except for dry summer months, and high humidity year round. Such climates occur between 40˚ and 65˚ latitude along the western sides of continents (McKnight & Hess 2000). The coastal climate of Oregon and Washinton is somewhat unique for the large difference between winter and summer rainfall. Between six and 20 times more rain falls during the wettest winter month compared to the driest summer month. This is a more pronounced difference than is found in any other marine west coast climate region on the globe (Mathews 1988).

Due to the high rainfall, with the exception of the extended dry period from July through September, much of the area along the western slopes of the Coast Range is classified as "rain forest." The Hoh River valley on the Olympic Peninsula for example, is famous for its lush rain forest habitat, and the nearby town for Forks, Washington averages upwards of 110 inches of rain a year with over 200 rainy days. Likewise the town of Siletz in the central portion of the Oregon Coast Range receives over 80 inches of rain annually with an average of 180 rainy days. In 1996 Laurel Mountain (3,592 ft) on Oregon's Central Coast Range set the one-year record for rainfall in the contiguous United States with 204.04 inches. The southern coast of Oregon from Bandon south, however, is considered to have the nicest weather along the entire Washington - Oregon coast line with more sunny days and less rain. Even so this area still gets plenty of rain and fog with cloudy overcast conditions on over 200 days a year.

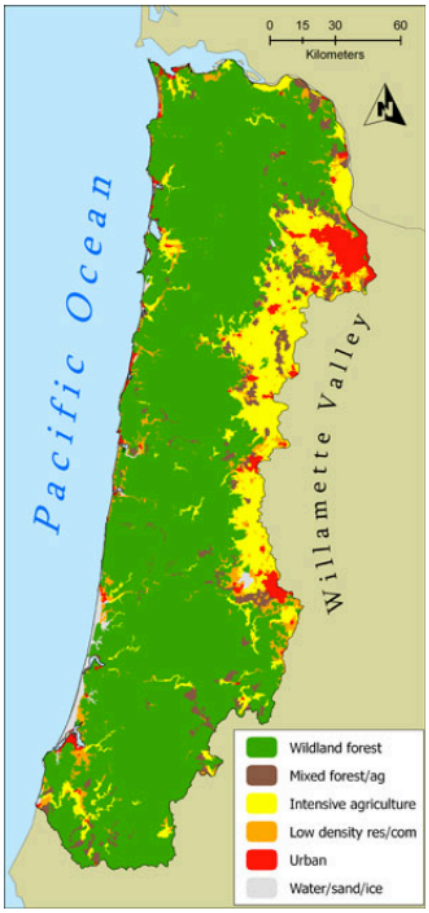


Figure 28. Current Land Use in Oregon's Coast Range.
(Cartography by Keith Olsen, CLAMS, OSU.)

The Pacific Ocean has a moderating effect on temperatures throughout the year. Average summer temperatures range between 60˚ and 70˚F, while average winter temperatures fall between 35˚ and 45˚F (McKnight & Hess 2000). In addition, the daily or diel range in temperature is often no more than 10-15˚F, a small difference especially when compared to interior regions such as Eastern Oregon and Washington, where the difference between the low nighttime temperature and high midday temperature may be as much as 40˚F (Atlas of Oregon 1976).

The overall climate of the region has a direct affect on the dominant habitats and stream conditions found in it. For example, the wet, cool conditions are ideal for growing large conifers (Figure 28). As a result, one activity has been the primary economic driver and land use across the entire Coastal PNMR - logging.

4.3 Land Use and Land Cover

The Coastal PNMR has the greatest diversity of conifers found anywhere on the planet, and the species of this region attain a size larger than any other species of conifers (Norse 1990). The unique weather conditions of long wet and cool winters followed by unusually dry summers allows conifers to outcompete deciduous trees that require more moisture through the warm summer months for growth before loosing their leaves in the fall. Conifers, however, continue to grow during the long wet and cloudy period from November through May.

The dominant conifer species include Douglas-fir, Western redcedar, Western hemlock, and Sitka spruce. Other less abundant species include Sugar pine, Western white pine, noble fir, silver fir, grand fir, white fir, and Port Orford cedar. The most widespread species throughout the region is the Douglas-fir.

Because of the ideal climate for conifers, the primary land cover and land use throughout the Coastal PNMR - historically and currently - is forest and forest management. For example, 80% of the Oregon Coast Range is still forested (Ohmann et. al. 2007) (Figure 28). A summary of the overall extent of land uses in 2005 in the Oregon Coast Range is shown in Table 5.

Table 5. Extent of Land Uses in the Oregon Coast Range.

Land Use	Acres	Percentage
Forest	5,396,121	75.6
Mixed Forest/Ag	276,509	3.87
Intensive Ag	892,504	12.5
Low density residential/commercial	294,828	4.13
Urban	207,601	2.91
Sand/beach/water	69,843	.98

Table Source: Oregon Dept. of Forestry, 2005

Because of the relatively short distance from the crest of the Coast Range to the Pacific Ocean, much of the landscape is steep, uneven and generally poorly suited for farming. As a result, the best land for farming occurs along the river bottoms of the lower elevation portions of river valleys, which often means close to the coastline. Prior to settlement these areas were predominately natural prairie, woodlands, and wetlands. Figures 29 and 30 show an example of the changes in land use and land cover in the area around Tillamook Bay along the Oregon Coast from 1867 to 1985. Farther inland and at higher elevations forestry continues to be the dominant land use throughout the Coastal PNMR.

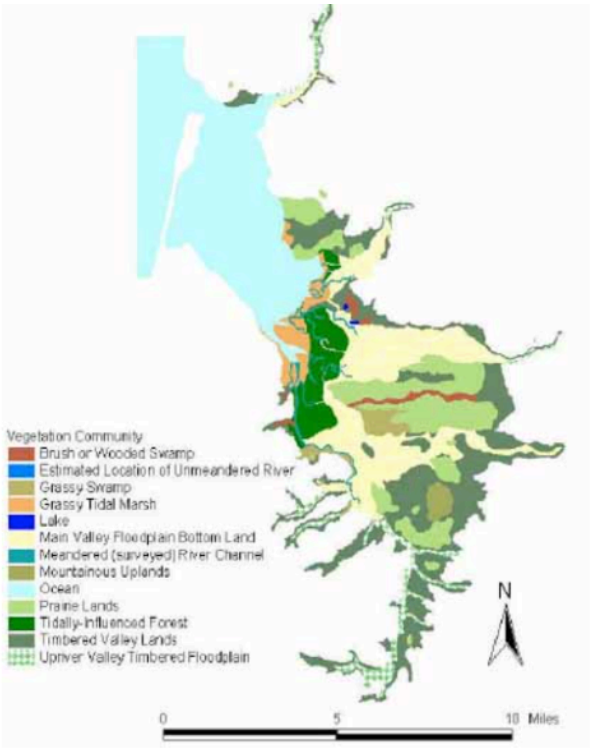


Figure 29. Land Use around Oregon's Tillamook Bay circa 1867 (Kilchis Watershed Analysis, Kieth Follansbee).

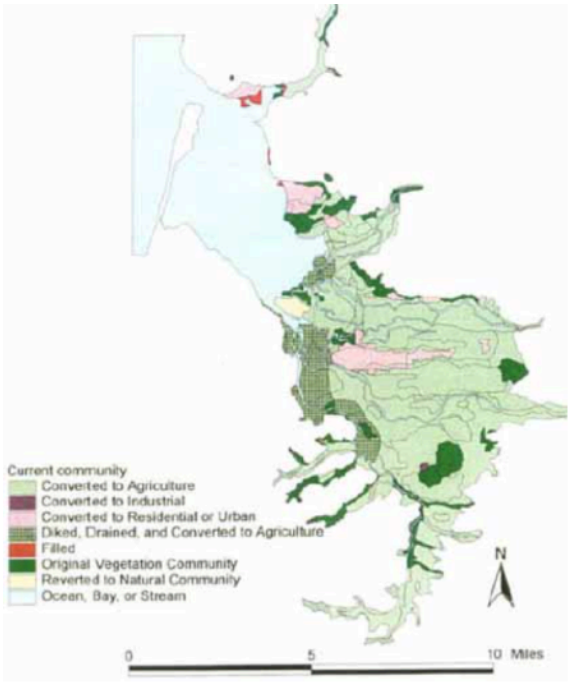


Figure 30. Land Use around Oregon's Tillamook Bay 1985 (Kilchis Watershed Analysis, Kieth Follansbee).

Ohmann et.al. (2007) created a vegetation map using the Gradient Nearest Neighbor (GNN) method for five vegetation classes in the Oregon Coast Range (Figure 31). Their results show the dominant vegetation cover in the Oregon Coast Range is Western hemlock forest (primarily composed of Douglas fir, Western hemlock, grand fir, and tanoak) covering 55% of the land area, followed by dry Western hemlock/mixed evergreen (19%), Sitka spruce (17%), foothill oak (7%) and high elevation true fir (2%).

Vegetation type and structure varied across five land ownership groups: State, Forest Service, BLM, forest industry, and nonindustrial private land (Ohmann et. al. 2007). Young to middle-aged forests (mean diameter <50cm) comprised the majority of all forests (71%) and occurred predominately on private lands (68%). Mature forests (mean diameter >50cm), but not considered old growth, occurred across 16% of the forest landscape with 68% of large trees and 89% of very large trees occurring on public lands. Old growth forest occurred on only 1% of the forest landscape and only on BLM or Forest Service lands. Age class simulations over the past 3,000 years suggests that old-growth forest coverage in the Oregon Coast Range varied between 25% and 75% of the landscape (Wimberly et.al. 2000).

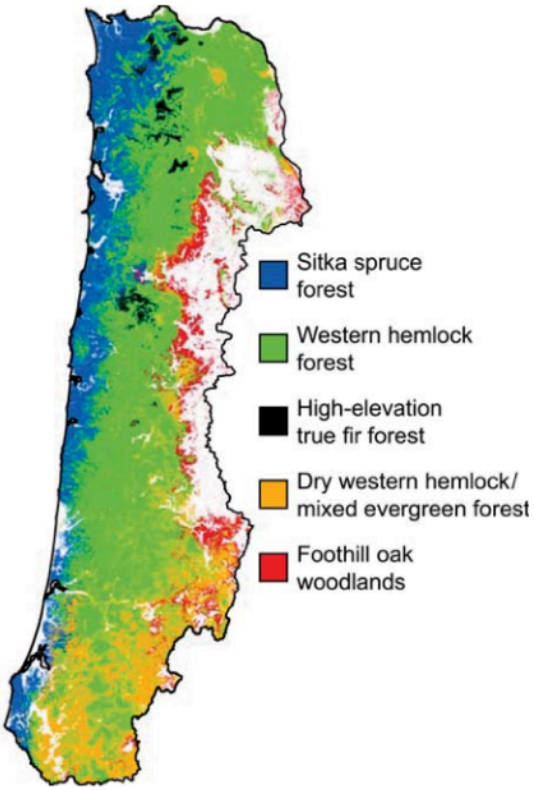


Figure 31. Current forest type distribution in Oregon Coast Range. (from: Ohmann et.al. 2007)

An environmental model based assessment of Washington's Olympic Peninsula, Henderson et.al. (2011) distinguished eight vegetation zones and their percent occurrence across the landscape (Table 6, Figure 32). While the vegetation classes are slightly different than those used for Oregon's Coast Range, the results show a similar distribution of forest cover with Western hemlock forest covering the greatest area (50.5% - Olympic Peninsula; 55% - Oregon Coast Range), Sitka spruce forest coming in as the second or third most extensive forest type (22.1% - Olympic Peninsula; 17% - Oregon Coast Range), and true firs and mountain hemlock occurring across regions of higher elevation.

Table 6. Extent of forest zones across Washington's Olympic Peninsula.

Vegetation zone	90-m pixels	Area	Mean elevation	Percentage of area
	Count	*Hectares*	*Meters*	*Percent*
Alpine	15,255	12,357	1,923.2	0.8
Subalpine parkland	67,304	54,518	1,513.6	3.7
Subalpine fir	17,557	14,222	1,542.9	1.0
Mountain hemlock	88,740	71,882	1,230.0	4.8
Pacific silver fir	305,348	247,341	738.3	16.6
Douglas-fir	9,247	7,490	743.6	0.5
Western hemlock	929,336	752,789	265.5	50.5
Sitka spruce	407,608	330,174	78.9	22.1
Sum/**mean**	1,840,395	1,490,773	**411.3**	100.0

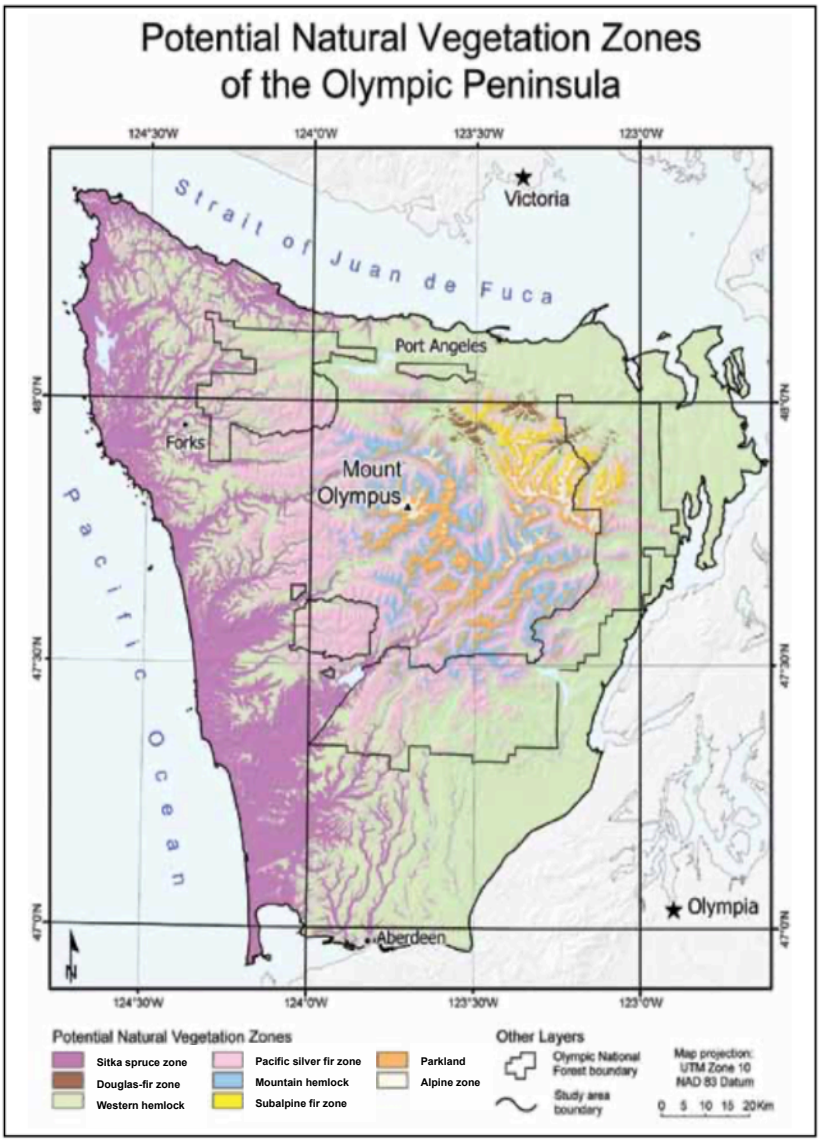


Figure 32. Map of forest vegetation zones across the Olympic Peninsula. (From Henderson et.al. 2011).

Conversion of land from more natural conditions to housing and development accelerated as population in Oregon and Washington rapidly increased. Changes in land use on non-Federal lands in both Western Oregon and Washington showed notable declines in wildland forest and agricultural lands along with large increases in low-density residential and urban uses from the mid-1970's to 2013/2014 (Table 7) . To reduce the loss of farmland, wildland forest, and wildland range to development, more strict land use laws were passed in Oregon (1984) and Washington (1994). After land use laws were enacted, conversion of private resource land to low-density or urban development declined by 54% and 6% in Oregon and Washington, respectively, compared to the period before land use laws (Lettman 2018).

Table 7. Change in land use on non-Federal lands in Western Oregon (1974-2014) and Western Washington (1976-2013) (From Lettman 2018).

Oregon: land use class	1974	2014	Net change 1974-2014	Net change 1974-2014
	Thousand Acres			*Percent*
Western Oregon				
Wildland forest	6,256	6,065	-191	-3.1
Mixed forest/agriculture	774	687	-87	-11.2
Intensive agriculture	1,938	1,754	-184	-9.5
Low-density residential	492	809	317	64.5
Urban	263	408	145	55.2
Washington: land use class	1976	2013	Net change 1976-2013	Net change 1976-2013
	Thousand Acres			*Percent*
Western Washington				
Wildland forest	5,932	5,421	-511	-8.6
Mixed forest/agriculture	333	225	-108	-32.4
Intensive agriculture	808	625	-182	-22.6
Low-density residential	863	1,406	543	63.0
Urban	331	584	253	76.6

The conversion of wildland forest lands to urban and low-density residential land has also impacted salmon bearing streams. From 1994-2013/2014 urban development along salmon bearing streams increased by 18.3% in Washington and 7.3% in Oregon (Table 8) (Lettman 2018).

Table 8. Percent of Land use change along salmon streams in Washington (1994-2013) and Oregon (1994-2014).

	Washington	Oregon
Land Use Category	*Percentage change of fish stream length within land use category*	
Wildland Forest	-1.5	-0.6
Low Density Residential	+13.7	+7.5
Urban	+18.3	+7.3

4.4 History of Human Settlement & Disturbance

4.4.1 European settlement history

The first Europeans to set foot in the Pacific Northwest arrived via the Pacific Ocean, so it was the coastal region of Oregon and Washington that European explores saw first. The first documented arrival was in 1543 by the Spanish explorer Bartolome Ferrelo (Oregon Blue Book). Sailing from the south he made it up Oregon's coastline as far as Cape Blanco before returning to Mexico. Other maritime explorers followed: Sir Francis Drake in 1579, Sebastian Vizcaino in 1603, then in 1774 Captain Juan Perez became the first European to explore the Pacific coast as far north as Washington, Vancouver Island, and southeast Alaska and claim the region for Spain. In 1778 Captain James Cook explored the coast of Oregon and Washington and continued as far north as the Bearing Sea, the furthest north of any European exploration to that time. The first American to explore the Pacific Northwest was Captain Robert Gray in 1788 and again in 1792 when on May 11, 1792 he was the first non-Native to enter and explore the Columbia River, which he named after the name of his ship. Also in 1792 the British explorer Captain George Vancouver explored the Oregon and Washinton coast and Vancouver Island for whom it is named. His expedition also entered the Columbia River in the fall of 1792.

The first Spanish expeditions to the Pacific Northwest were in search of a water route through North America and to claim the land for Spain (Reindhardt 2020). By the late 1770's, the economic value of the region's furs greatly heightened interest in the region from Great Britain and the United States. In 1790 Spain relinquished its claim to the region leaving the fur trade to the competing interests of Britain and the young country of America (Reindhardt 2020). Besides mapping and gathering first-hand knowledge of the vast area of North America west of the Mississippi River, the Lewis and Clarke expedition was also undertaken to further solidify America's claim to the territory over Britain. The American Jacob Astor established the first fur trading post (Fort Astoria) in the Northwest for the Pacific Fur Company in Astoria in 1811. However, in 1813 the British Northwest Company purchased the Pacific Fur Company and renamed the post Fort George. By 1821 the Hudson's Bay Company (HBC) had become the dominant fur trading company in the region with its headquarters at Fort Vancouver (Reindhardt 2020).

While the Native tribes acquired metal goods and guns from British and American fur traders for the furs they gathered, they also acquired diseases. Their lack of immunity nor knowledge of treatment for the diseases had devastating effects on their populations throughout the Northwest and North America (Boyd 2014). Even by the time Captain Gray entered the Columbia River in 1792 it is thought that the Native tribes had suffered significant losses from a smallpox epidemic in the early 1780's. This epidemic may have killed as many as 30-40% of the Native population in the region (Boyd 2014). More smallpox outbreaks occurred every ten to 20 years, as well as epidemics from other diseases including measles, malaria, yellow fever, typhoid fever, and cholera. Outbreaks of malaria along the Columbia River and into the Willamette Valley from 1830-1834 may have killed as many as 90% of the Native population. As a result, when Euro-Americans arrived in the Willamette Valley in the 1840's and 1850's the population of Native people was already greatly reduced from levels only twenty or thirty years early.

Settlers began arriving in significant numbers to the region in the 1850's, and population growth in the Pacific Northwest, especially in the area west of the Cascade Mountains, has continued unabated to this day. Figure 33 shows the population increase in Oregon from 1900 to 2030 (estimated) (Oregon Office of Economic Analysis 2019), and Table 9 shows the ten-year population changes in Washington from 1890 to 2010 (Washington Office of Financial Analysis 2012).

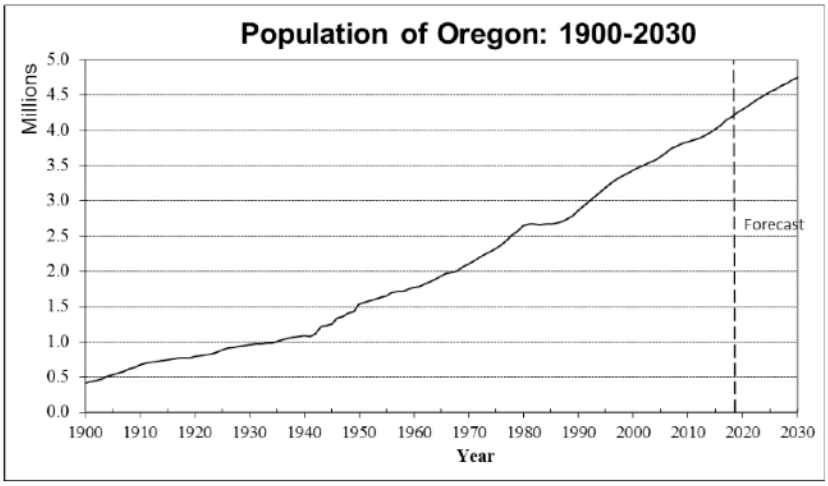


Figure 33. Oregon population growth from 1900-2030. (From Oregon Office of Economic Analysis, 2019)

Table 9. Washington population growth 1890-2010. (From Washington Office of Financial Analysis)

Year	Population
1890	357,232
1900	518,103
1910	1,141,990
1920	1,356,621
1930	1,563,396
1940	1,736,191
1950	2,378,963
1960	2,853,214
1970	3,413,250
1980	4,132,353
1990	4,866,663
2000	5,894,143
2010	6,724,540

Strong population growth has continued with increases from 2010 - 2020 ranging from 5 to 15% in counties of Western Oregon and Washington (2020 US Census Data). Based on the 2020 census, the population of Oregon and Washington has increased to 4,237,256 and 7,705,281, respectively. In both states the vast majority of the population (>70%) lives west of the Cascade Mountains and is concentrated in the Puget Lowlands and Willamette Valley regions.

4.4.2 Logging history

When Europeans first arrived to the Pacific Northwest, the land cover along the entire Coastal PNMR was dominated by extensive conifer forests. Early settlers were frustrated that there was little to no clear land for cultivation and considered the massive trees weeds that needed to be removed so the land could be farmed (Lucia 1975). The extensive forests, often described by disgruntled settlers as the Big Wood or Great Green Desert, was described in the late 1820's by David Douglas, the famous Scottish botanist, this way: *With respect to the appearance of the country, my expectations were fully realized, in its fertility and variety of aspect and of soil. The greater part, as far as the eye could reach, was covered with Pines of various species.*

By the 1850's however, lumbermen from eastern North America had arrived in the region. Rather than a green desert, they saw a landscape of trees of unimagined size and abundance with unlimited commercial potential. This was also a period of rapidly expanding population in the Pacific Northwest and California, and lumber for housing and ship building was in high demand. The "Big Wood" was about to change forever.

The history of logging the Pacific Northwest has been extensively documented and is widely available. The goal here is not to repeat what has already been written, but rather to provide a brief overview that summarizes the primary types of logging activity and forest management that has occurred across the region over the last 150 years, and discuss how forest practices have impacted the streams and aquatic resources of the region. The following is a general chronology of how logging in the region progressed from the first days of logging the massive trees with nothing but human and animal power to the highly mechanized operations of today.

1820's - 1880's:

The first lumber mill in the region was owned and built by the the Hudson's Bay Company (HBC) in 1828 along the banks of the Columbia River near HBC's headquarters at Fort Vancouver (Cox 1974). At this time logging had to be done by hand and with teams of oxen or horses. Because of the size of the trees and difficulty in moving them, only trees close to bays or large rivers were cut so they could be dragged to the water and then floated to a nearby mill. This meant that the volume of timber cut was low. Much of the lumber milled by HBC was shipped to Hawaii where missionaries needed the building material. Beginning around 1848 the gold rush in California greatly increased the demand for timber from the Pacific Northwest, and investors from San Francisco built numerous saw mills around Puget Sound. While the methods still relied on human and animal power, the timber industry increased significantly and became a major employer in the region.

Even with the crude logging methods, regulations were enacted to control logging activity. In the 1850's regulations passed that prohibited the cutting of timber from public land. Corruption within the industry and government, however, meant that these regulations were largely ignored. To further curb illegal logging in the region, the federal government in 1878 passed the Timber and Stone Act that allowed

residents of Washington, Oregon, California and Nevada to acquire 160 acres of timber or mineral land for $2.60 per acre. Though large companies were excluded from the act as it was meant to help farmers, big timber companies found ways to use it to acquire large tracks of land. In 1864 the federal government also began subsidizing the construction of railroads through the region, and the Northern Pacific Railroad was given 20-40 mile wide strips of land in alternate sections for every mile of track they built creating a checkerboard pattern of ownership. They purposely located the tracks through prime timberland to maximize the value of the land it acquired. By 1883 the Northern Pacific Railroad owned 7.7 million acres in the Washington Territory, which equaled 18% of the Washington Territory's area. Part of the legacy of this land grant is the checkerboard pattern of land ownership still seen in the region today.

Concerns over the impacts of logging to streams and fish during these early years of timber harvest were raised, but given the limited area and primitive methods used, large scale impacts to streams were yet to occur, and most impacts occurred near bays and along the lowest reaches of rivers where timber could be removed from the forest and transported to mills more easily. That changed with the mechanization of logging that began in the 1880's and the expanding reach of timber harvest throughout the region.

1880's - Present:

The 1880's and 90's saw a great expansion of harvest and improved methods available to cut and move the huge old-growth trees that covered the region. The big change began in 1881 with the invention of the steam donkey, a steam engine on skids with cables attached to a winch. The steam donkey could move the huge logs across the ground farther distances and much faster than teams of horses or oxen. Then in 1905 high-lead logging was developed where logs were suspended off the ground by cables attached to the top of a tall tree and yarded through the air rather than on the ground. This further expanded the reach and speed of what could be logged. The 1880's also saw the first use of narrow gauge railroads built into the forests to allow trees previously to difficult to access to be logged.

Only big logging companies could afford the large investments necessary to buy the expensive equipment and build railroads. This meant small operators either went out of business or sold their land to the big companies. In 1900 Frederich Weyerhaeuser, a Minnesota investor, purchased 900,000 acres from the Northern Pacific Railroad. More land purchases followed, and by 1903 Weyerhaeuser owned 1.3 million acres of land in Washington, 26 percent of all private land in the State. Other notable companies in Washington included the Puget Mill Company (owned by Pope and Talbot), and the St. Paul and Tacoma Lumber Company, owned by timber barons from the upper Midwest that moved west after the Midwest forests had been largely cut. By the 1890's and early 1900's, lumber mills were spread around Puget Sound (Figure 34), and by 1910 the timber industry was the largest employer in Washington: an estimated 63% of all wageworker jobs were related to the timber industry. The demand for lumber continued to expand throughout the early 1900's, and lumber from the Pacific Northwest was shipped across the country and all over the world.

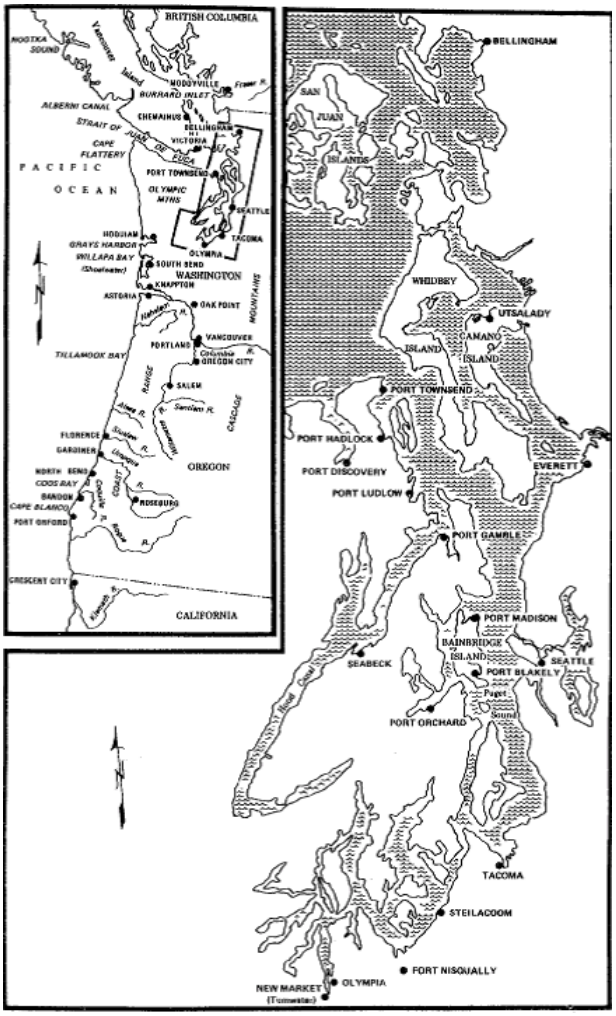


Figure 34. Lumber centers of the Pacific Northwest before 1900. Expanded area of Puget Sound. (From Cox 1974

Table 10. Change in board feet of timber cut from Washington's forests from 1880 to 2000. (From Center for the Study of the Pacific Northwest, Univ. of Washington.)

Year	Board Feet of Timber Cut
1880	160 million
1890	1 billion (b)
1900	1.9 b
1910	4.1 b
1920	5.2 b
1926	7.6 b
1940	4.5 b
1950	4.5 b
1960	4.6 b
1970	6.4 b
1980	5.7 b
1990	5.8 b
2000	4.1 b

The large expansion of the timber industry can be seen by the rapid increase in the amount of timber removed from Washington's forests from 1880 to 2000 (Table 10). Peak timber production occurred in 1926: 7.6 billion board feet cut compared to 160 million board feet in 1880. Timber production continued at record levels until 1930 when the Great Depression caused a dramatic drop in lumber prices and demand (2.2 billion board feet were cut in 1932). By 1933 half of all loggers and mill workers in Washington were unemployed. Timber harvest rebounded in the 1940's due to greater demand for lumber during and after WWII.

Timber harvest in Oregon lagged behind that in Washington until the early 1940's when in 1941 almost seven billion board feet were cut (Figure 35) (Andrews and Kutara 2005). Following the large drop in lumber production during the Great Depression in the 1930's, harvest increased rapidly equalling more than twice the harvest in Washington, with peak harvest occurring in 1952 at 9.8 billion board feet (Figure 35). With the exception of lower harvest during periodic national recessions, harvest continued

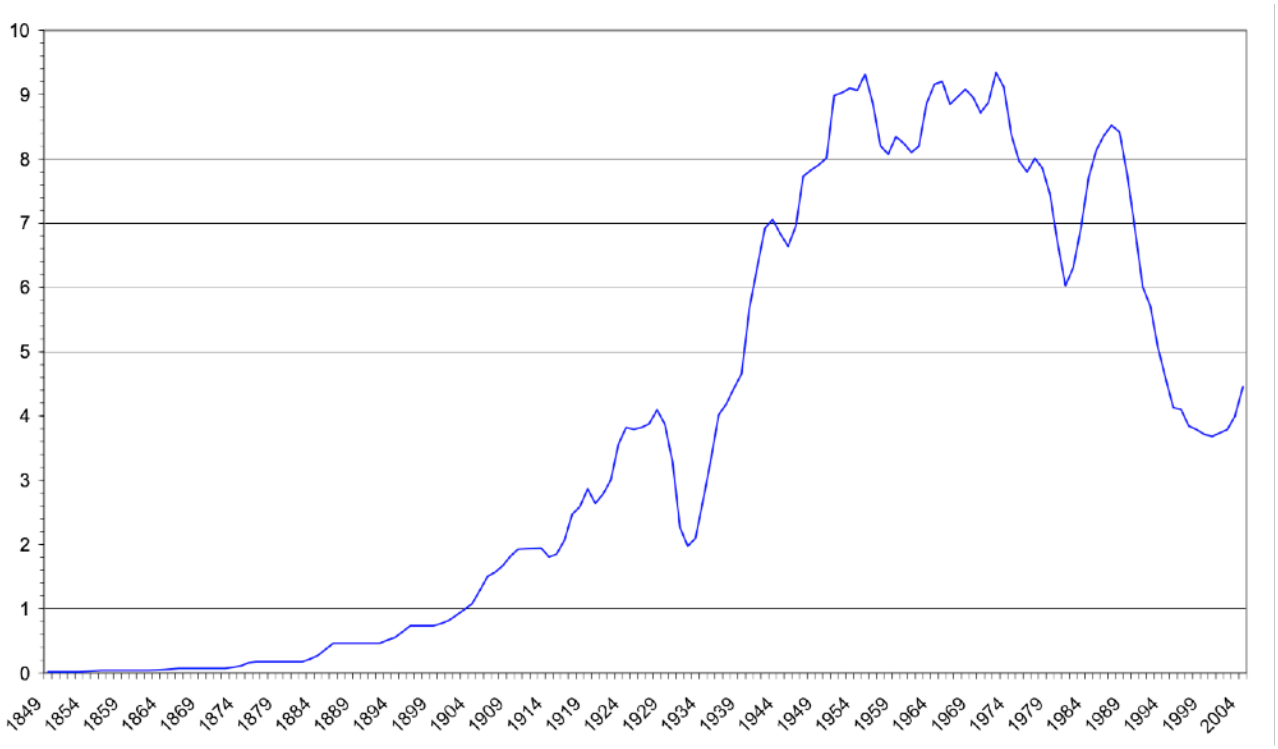


Figure 35. Timber harvest (billions board feet) in Oregon from 1849-2004. (From Andrews & Kutara 2005)

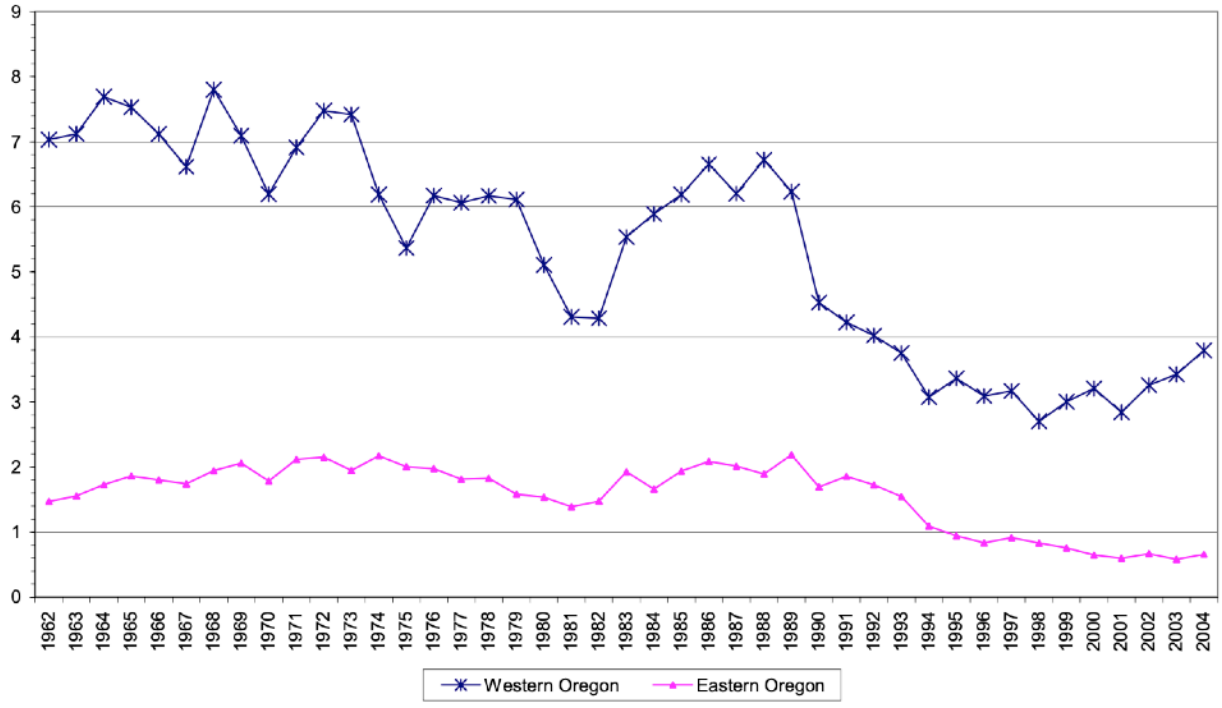


Figure 36. Timber harvest (in billions of board feet) from Western and Eastern Oregon from 1962-2004. (From Andrews & Kutara 2005)

at between eight and nine billion board feet annually until 1990 when the cut dropped to 6.2 billion board feet. From 1994 to the present, timber harvest has generally remained between 3.5 and 4.5 billion board feet (in 2020 3.6 billion board feet were cut). The total timber harvest in Western Oregon has consistently been two to three times greater than the harvest from Eastern Oregon (Figure 36). Since the late 1940's Oregon has been the largest producer of wood products in the country (Oregon Forest Resources Institute 2022).

Concerns about over harvest arose as early as 1890 (University of Washington, Center for the Study of the Pacific Northwest). In 1891, in an effort to protect watersheds from uncontrolled timber harvest and create a sustainable level of harvest into the future, the federal government passed the Forest Reserves Act that allowed the president to create forest preserves in western public lands. In 1897 President Grover Cleveland set aside millions of acres throughout the West, including a 1.5 million acre forest reserve on the Olympic Peninsula. The Forest Management Act of 1897 gave regulatory authority over the forest reserves to the Secretary of the Interior. Forest reserves later became the National Forests, and in 1905 Congress gave the Bureau of Forestry - which later became the US Forest Service - control over managing the reserves. Gifford Pinchot was appointed the first head of the Forest Service by President Theodore Roosevelt (University of Washington, Center for the Study of the Pacific Northwest). Before leaving office in 1909, and before Congress passed a bill preventing presidents from creating more forest reserves without congressional approval, President Roosevelt set aside nearly 11 million acres as National Forests in Washington state, almost 25% of the total land area of the state.

The competing interests of conservationists and the logging industry that began in the late 1800's only grew more intense as logging rates rapidly increased in the ensuing decades. Because logging was allowed in national forests, conservationists wanted land set aside that prevented all timber harvest. National Parks provided such protection, and in 1899 Mount Rainier National Park (236,381 acres) was established, though it excluded the timber rich lowland forests around the area. In 1902 President Theodore Roosevelt created Crater Lake National Park (183,224 acres), and in 1909 he set aside 615,000 acres of the Olympic National Forest to create the Mount Olympus National Monument. The Wilderness Act of 1964 also allowed land to be protected from all resource extraction or development.

While one of the goals of the Forest Service was to provide for "sustainable" timber harvest, Gifford Pinchot believed old-growth forests were "inherently wasteful" and promoted cutting them down to allow for more productive "managed" forests (University of Washington, Center for the Study of the Pacific Northwest). As a result timber harvest from public lands continued to increase throughout the 1900's reaching a peak in the 1960's-1970's when the Forest Service promoted clear-cut logging and allowed chemical treatment to control competing brush on national forests (Robbins 2015). Additionally, while Oregon passed the countries first forest practices act in 1941, it was largely drafted by the timber industry and had no effect on limiting timber harvest. Environmental regulations started to have an impact in the 1970's when DDT was banned in 1972, and 2-4-5-T was banned in 1979.

After the passage of the Endangered Species Act (ESA) in 1972, environmental groups began lobbying for the protection of species reliant on old-growth forests, especially for the Northern Spotted Owl. This

began a long drawn-out legal battle over forest management practices between environmental groups wanting more protection, especially for the small amount of remaining old-growth forests, and the logging industry and the rural communities with logging dependent economies. In 1990 the Northern Spotted Owl was listed as Threatened under the ESA, and in 1991 District Court Judge William Dwyer prohibited timber sales on federal lands in spotted owl habitat (Robbins 2015). As a result, in 1993 President Bill Clinton convened the Northwest Forest Summit to determine how to manage federal lands in Washington and Oregon. The outcome was the adoption in 1994 of the Northwest Forest Plan (NFP), after which Judge Dwyer lifted his ban on logging in national forests. The NFP allowed for a timber harvest of 1.1 billion board feet annually from national forests, almost a 90% drop from the pre-forest plan timber volume, and caused a large decline in timber industry employment in Washington and Oregon. Currently in Oregon about 74% of the annual timber harvest from the Oregon Coast Range comes from private industrial forest lands. As previously described in section 2.3 of this report, old-growth forests currently occupy only 1% of the Oregon Coast Range compared to historical estimates that range between 25-75% depending on fire and other natural disturbance factors (Wimberly et.al. 2000).

4.4.3 Land ownership and forest management

Throughout the Coastal PNMR, land management, including forest management, varies with who owns the land. Overall ownership falls into two broad categories: public land (state or federally owned and managed) or private land (owned and managed by corporations or private citizens). But even within these two categories the amount of timber harvest allowed and the rules guiding logging methods and restrictions very depending on land ownership classifications that fall into seven different categories each with different forest management objectives.

Most restrictive: Logging activity greatly reduced or completely prohibited

1. Federal Wilderness Areas

2. National Parks

Moderate restrictions: Logging allowed, but is managed under the limits and objectives of the Northwest Forest Plan

3. National Forest Service land (USFS)

4. Bureau of Land Management land (BLM)

Least restrictive: Logging managed under State Forest Practices rules

5. State forest land: State forests are managed for multiple uses including timber production, recreation, and fish and wildlife protection. As a result where and how logging is allowed varies between state forests.

6. Private industrial forests (land owned by large logging companies): The primary goal is timber production. Logging restrictions are regulated by the state's forest practices rules. State rules guiding logging practices on private lands differ between Oregon and Washington.

7. Private nonindustrial land (land owned by private citizens): Logging generally less intensive than on private industrial forest land. Where logging occurs it is regulated by state forest practice rules.

Differences in the type and extent of logging allowed, and the rules for protecting streams and rivers from logging impacts, depend on land ownership and their management objectives. Land ownership across the landscape, however, has evolved into a matrix of intermingled federal, state, and private land, often within a single watershed (Spies et.al. 2007). For example, private land in the Oregon Coast Range covers 63% of the land area and BLM land covers 15% (Figure 37), but 89% of BLM land falls within 1,000 meters of private lands. As a result, different forest stand conditions (e.g. ranging from <10% cover to old growth stands) varies across the landscape by ownership. Ohmann et.al. (2007) looked at the effect of disturbance and land ownership on forest stand condition across the Oregon Coast Range. While large, very large, and old-growth stands together make up only 17% of the Coast Range forest, 68%, 89%, and 95% of these classes

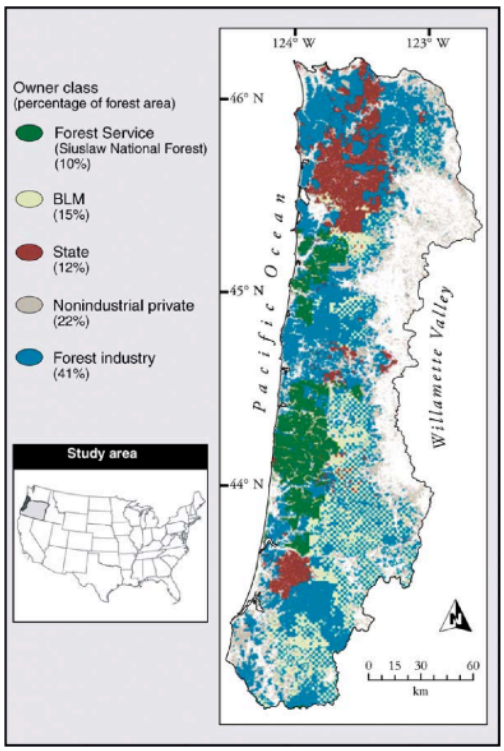


Figure 37. Land ownership across Oregon Coast Range showing mosaic distribution of federal, state, and private (From Spies et. al. 2007)

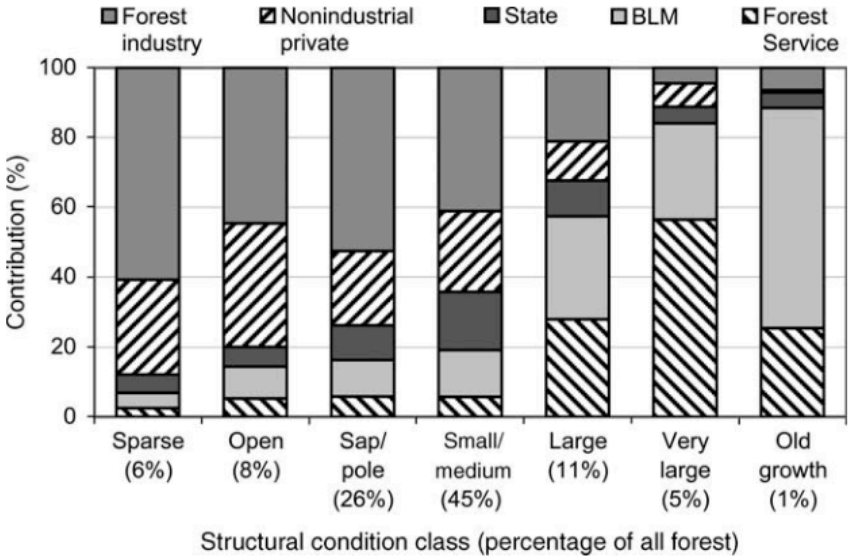


Figure 38. Distribution of structural condition classes among owner classes: sparse, <105 cover; open, 10-39% cover; sapling/ pole, >40% cover, 2.5-24.9 cm quadratic mean diameter (QMD); small/medium, >40% cover, 25-49.9 cm QMD; large, >405 cover, 50-74.9 cm QMD; very large, >75 cm QMD; old growth, old-growth habit index >0.75. (From Ohmann et.al. 2007))

occur on public land (state or federal) respectively, while the majority of sparse to small/medium stand conditions occur on private land (Figure 38). The majority of hardwood forests (oak woodlands, red alder, big leaf maple, etc) occur on nonindustrial private land, which is concentrated in lower elevation areas and valley bottoms of large rivers and streams (Ohmann et.al. 2007).

The forest practices acts in Oregon and Washington also differ in ways that could change how logging on state and private land effects streams and rivers in the two states. For example, stream riparian protection - riparian management zones (RMZ) - in Washington range from 90' to 200' from the edge of the bankfull width on both sides of the stream (Washington State Forest Practices Rules -Title 222 WAC). Final distance varies between five site classes determined by soil type, and stream width (< or > 10' stream width). Oregon's forest practices rules set riparian management zones of 50'-100' with a 20' no harvest zone on each side of fish-bearing streams with some harvest allowed outside the 20' no cut area (Oregon Forestry Department, Forest Practices Act 2021). Specific conifer retention requirements within each state's riparian management zones varies with stream size, distance from stream edge, and if they are fish bearing or non-fish bearing streams.

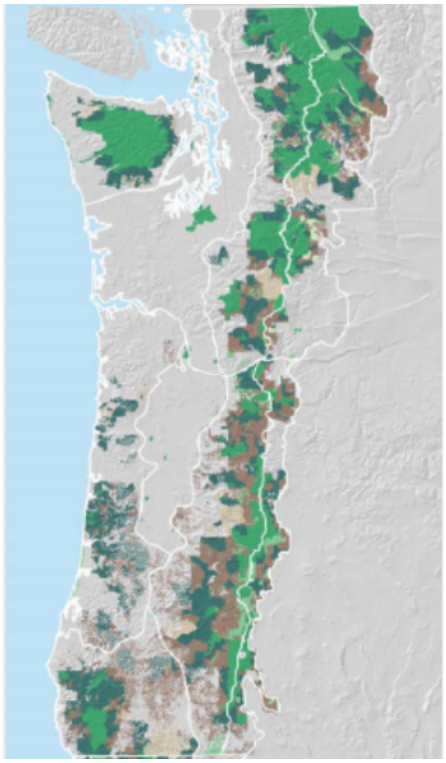
The federal lands of the US Forest Service (National Forests) and BLM in Western Oregon and Washington are managed under the guidance of the Northwest Forest Plan. Most federal land (60%) is set aside for fish and wildlife protections and restoration of late successional stage old-growth forests, and 16% is set aside for timber harvest and active forest management (Table 11). The NFP is reviewed and updated every five years. The distribution of management classes in the NFP are shown in Figure 39.

Table 11. Management objectives for the different land use classifications of the Northwest Forest Plan. (From Land Use Allocations: Northwest Forest Plan Regional Office website)

Land Classification	Acreage	Percent	Description
Congressionally reserved areas	7,320,600	30	Wilderness areas, wild & scenic rivers, national parks and monuments.
Late-successional reserves	7,430,800	30	Reserved for protection and restoration of late-successional old-growth (LSGO) ecosystems and habitat for northern spotted owl and marbled murrelet.
Managed late-successional areas	102,2000	<1	Restore and maintain LSGO stands where regular and frequent wildfires occur: allow for fire hazard treatment.
Administrative withdrawn areas	1,477,100	6	Areas in local forest district plans identified for recreation, visual set backs; does not include regular timber harvest.
Adaptive management areas	1,521,800	6	Some timber harvest allowed, but with ecological objectives.
Riparian reserves	2,627,500	11	Protective buffers along streams, lakes, and wetlands to provide good water quality and terrestrial species dispersal corridors.
Matrix	3,975,300	16	Federal lands where timber harvest and active forest management will occur.

Figure 39. Distribution of federal land and management classes in the Northwest Forest Plan for Western Oregon and Washington.





Riparian reserves on federal land require a streamside buffer on each side of fish-bearing streams of, "extending from the edges of the active stream channel to the top of the inner gorge, or to the outer edges of the 100-year floodplain, or to the outer edges of riparian vegetation, or to a distance equal to the height of two site-potential trees, or 300 feet slope distance (600 feet total, including both sides of the stream channel), whichever is greatest." The requirements for nonfish-bearing streams include: "the area on each side of the stream extending from the edges of the active stream channel to the top of the inner gorge, or to the outer edges of the 100-year floodplain, or to the outer edges of riparian vegetation, or to a distance equal to the height of one site-potential tree, or 150 feet slope distance (300 feet total, including both sides of the stream channel), whichever is greatest" (Northwest Forest Plan Standards & Guidelines). These riparian buffers are considerably larger and more restrictive to logging than either Washington's or Oregon's state requirements for private forest lands.

As a result of the checkerboard land ownership, and different logging requirements and forest management objectives based on land ownership, the current effects of forest land use on streams and aquatic life varies widely, often within the same watershed (Spies et.al. 2007). This makes developing coordinated and consistent plans for stream protection or restoration at a large scale, e.g. watershed or larger scale, difficult.

4.5 Stream Disturbance History

4.5.1 Natural Disturbance

Current stream conditions in the coastal PNMR are the result of a wide range of natural and anthropogenic disturbance factors. Fire, both natural and man-caused, has been a regular part of the disturbance regime throughout history. Estimates of the fire history over the last 9,000 years, based on charcoal from sediment cores at Little Lake in the central Coast Range of Oregon, show that natural stand replacement fires in the Oregon Coast Range occurred about every 100-230 years, with the range driven by long-term climate cycles (Long et. al. 1998). The natural fire cycle, however, has been altered through logging activities that change forest cover and fuel conditions, and aggressive suppression of all fires whether natural or man-caused (Means et.al. 1996). Even with aggressive fire suppression, there have been numerous large stand replacement fires over the past 150 years from both natural events and human activities, the impacts of which still effect forest condition and streams today. Four of the largest natural fires documented in the Oregon Coast range between 1848-1940 covered areas of 800,620; 311,350; 300,000; and 229,800 acres in size (Means et.al. 1996). In 1933 a major fire known as the Tillamook Burn started in the northern Coast Range from logging activity, and led to a series of fires over a twenty-year period that burned a total of 355,000 acres. In July of 2002, lightning storms in the Siskiyou Mountains started five fires in the Kalmiopsis Wilderness that eventually merged (named the Biscuit Fire), and burned an area of 500,000 acres, nearly all in the wilderness area.

Historical records of fire on the Olympic Peninsula and coast of Washington show an increase in fire frequency following European settlement, though native tribes were known to use fire throughout the coastal region to maintain habitat for game and food plants (Olympic Park fire history, National Park Service 2016). Two of the largest documented fires were the Dungeness Fire in 1890-1891 that burned 30,000 acres, and the Sol Duc Burn of 1907 that burned 12,800 acres. Over the last 40 years in the Olympic National Park, fires have burned 4,300 acres with 43% of ignitions and 63% of burned area started by natural causes, mostly lightning, and the remaining started by human activities primarily uncontrolled campfires. From 2002-2010 humane-caused fires adjacent to the park have burned 4,000 acres (Olympic fire history, National Park Service 2016).

4.5.2 Human Disturbance: Effects of logging and forest management on streams and aquatic life

Impacts to streams and aquatic life increased along with increased timber production. As early as the mid 1800's the environmental impacts from logging were a concern, and efforts were taken to minimize them starting with logging restrictions on public land in 1850, and the Forest Reserves Act in 1891, which led to the creation of National Forests and the US Forest Service. Despite efforts to better manage logging, the rate and amount of timber harvest continued to increase through the 1970's (see Logging History, section 3.2), with few restrictions or regulations in place to minimize the impacts of logging along streams.

Before the use of trains and logging trucks became widespread, transporting old-growth logs out of the forest to mills was a major challenge, and the cheapest and most efficient method was floating them down rivers to mills located along bays and river mouths. Throughout Washington and Oregon splash dams were used to facilitate moving large quantities of logs downstream to local saw mills. From the 1880's until banned in Oregon in 1957, splash dams were frequently used across Western Oregon. Miller (2010) mapped 232 splash dams and 213 log drive sites in Western Oregon (Figure 40). Splash dams were made of timber placed across the stream channel to create reservoirs, sometimes several kilometers long, where logs would be stored. The dams were then dynamited creating a flood of water and logs downstream to saw mills. To help the logs reach their downstream destinations, obstacles like large boulders and natural log jams were also dynamited (Miller 2010).



Figure 40. Confirmed splash dam locations in the Oregon Coast Range. (From Miller 2010).

The damage to stream habitat caused by splash dams from 50 to 130 years ago can still be measured in the affected streams today. Comparing splash-damed streams to similar streams with no history of splash dams, Miller (2010) found significantly more bedrock and fewer deep pools in the splash-damed rivers. She also measured significant differences in substrate and channel complexity.

The extensive logging history has resulted in a variety of landscape disturbances creating a complex range of environmental effects that vary depending on location (elevation and slope) and time following harvest (Murphy and Hall 1981; Newbold et.al. 1980). One widespread disturbance is the large network of logging roads that now extend throughout the forest landscape. By 1990 it was estimated that there were over 233,000 miles of logging roads covering an area of 2.4 million acres spread across northwest California, western Oregon, and western Washington (Norse 1990). Nearly all logging roads are unpaved and can become sources of fine sediment and alter runoff to streams (Wemple 1996). Cederholm et.al. (1981) found logging roads introduced 2.6-4.3 times more fine sediment to streams than the natural sedimentation rate in the basin, and resulted in reduced spawning habitat quality in the Clearwater River in western Washington. In addition, improperly constructed or damaged culverts at stream crossings can become fish migration barriers to rearing and spawning habitat.

The extensive use of clear-cuts and the removal of riparian vegetation has affected streams by altering the flow regime, nutrient inputs, and increasing sedimentation and stream temperatures (Poff et.al. 1997). The type and level of impact on the ecology and aquatic life appears to vary with stream type (size, slope, and elevation), extent of logging activity, including stream buffer size, and time since logging activity occurred (Keenan and Kimmins 1993). Brown and Krygier (1970) found that one year after clear-cut logging in a small watershed in the Oregon Coast Range, monthly maximum stream temperature rose by 14˚ F, and annual maximum temperature rose from 57˚ to 85˚ F, while no changes in temperature were observed in a nearby unlogged watershed. The effects of clear-cuts have also been shown to decline over time (Rosen et.al. 1996).

4.6 Current Ecological Condition in the Coastal PNMR

The Washington State Department of Ecology (WDOE) and the Oregon Department of Environmental Quality (ODEQ) have completed stream condition assessments throughout Washington and Oregon, including the Coastal PNMR, for habitat quality, water quality, and macroinvertebrate communities (Hubler et. al. 2009; Larsen et. al. 2019). These surveys sampled wadeable streams following a probabilistic sampling design, as well as targeted reference sites (sites with lowest levels of human disturbance), to determine regional stream conditions. While these surveys do not provide a historical account of stream conditions, they do provide a relatively recent comprehensive assessment of wadeable stream conditions for the Coastal PNMR and are summarized below. Selected data from these assessments were used in constructing the PNMR BCG. The selection was based on providing representative coverage of PNMR streams and full range of conditions along a gradient of least disturbed to highly altered streams. These surveys are described here to provide information on how current conditions have been assessed and reported (e.g. good, fair and poor). The good, fair and poor categories are based on reference conditions. A cross-reference to BCG levels has not been established, however, based on exploratory analyses, streams assessed as “good” generally correspond to BCG level 2 or 3 depending on location (e.g. high elevation streams generally score higher on BCG scale than low elevation streams due to level of human disturbance).

4.6.1 Washington

Washington DOE sampled 346 randomly selected wadeable sites and 75 reference sites from across the state (Larsen et.al. 2019). The survey however, did not sample sites on Federally owned lands. Besides aquatic macroinvertebrates, 14 chemical and eight habitat parameters were measured at each site. A sample site consisted of a stream reach 20 x the average bankfull width, and not less than 150m or greater than 2km. The results for each parameter were reported as a percent of stream kilometers falling into one of three condition classes: good, fair, or poor.

Each macroinvertebrate sample was a multi-habitat composite of eight, one-square foot, samples collected from eight randomly sampled pool/riffle transects using a D-frame kick net with a 500 µm mesh net. In the lab, a 500 organism random subsample was sorted from each sample, and each individual invertebrate was identified to the lowest practical taxonomic level, typically genus or species.

Table 12. Percent of stream kilometers in each of three B-IBI condition classes for four regions in Western Washington.

	B-IBI Condition classes: % of Stream Length		
	Good	Fair	Poor
Western Washington	42	26.3	31.7
Coastal Region	38.1	29.6	32.4
Lower Columbia Region	54	29.5	16.5
Puget Sound Region	34.2	20.9	44.9

From: Larsen et.al. 2019

Site results were compiled using Washington's Benthic Index of Biological Integrity (B-IBI) based on 10 diversity metrics describing different characteristics of the macroinvertebrate community (Larsen et. al. 2019). B-IBI results for Western Washington include three regions (Coastal, lower Columbia, and Puget Sound) that make up Washington's portion of the PNMR are shown in Table 12.

Of the three regions comprising Western Washington (Figure 41), the lower Columbia had the highest percent of stream miles in good biological condition (54%) and least in poor condition (16.5%), while Puget Sound had the lowest percent of stream miles in good biological condition (34.2%) and highest in poor condition (44.9%).

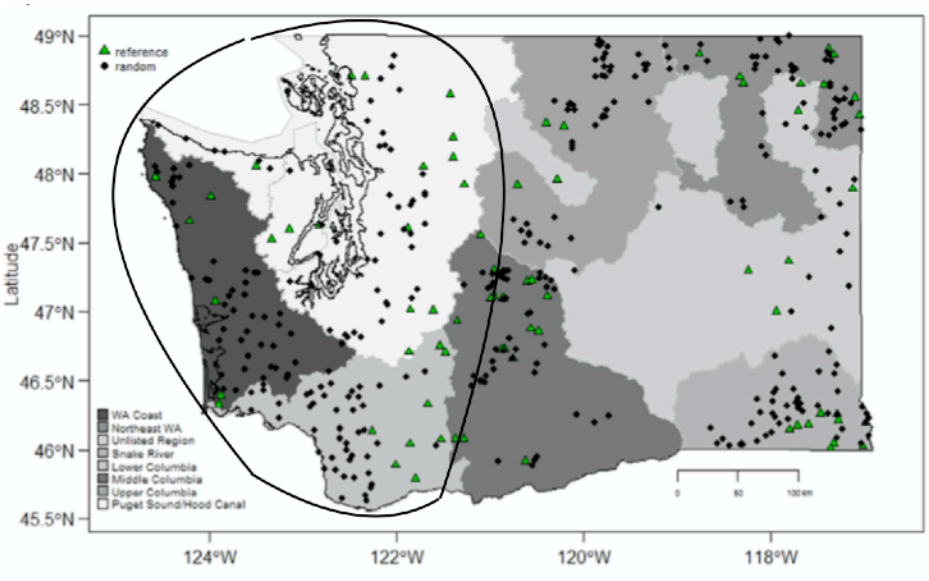


Figure 41. Monitoring regions in Washington. The three circled regions (Washington Coast; Puget Sound; and Lower Columbia) make up the PNMR in Washington (modified from Larsen et. al. 2019).

A relative risk analysis of 20 water quality and habitat parameters was performed to determine which factors posed the greatest risk to macroinvertebrates. This analysis found that % sand fines posed the greatest risk to the macroinvertebrate community. In addition, the percent of stream kilometers in poor condition due to high percent sand fines was the second highest (≈60% statewide) of the twenty assessed parameters, exceeded only by the percent kilometers in poor condition due to poor bed stability (log of relative bed stability) (≈80% of stream kilometers), which is also related to an increase in percent fines. Other parameters posing high risk to macroinvertebrates were canopy cover, streambed embeddedness, and high nitrogen and phosphorus levels. High water temperature had the sixth lowest relative risk value of the 20 assessed parameters, with high water temperature classified in poor condition occurring in ≈ 30% of stream miles statewide (Larsen et. al. 2019). This is in contrast to streams in Oregon, where high water temperature was found to be one of the highest stressors on the macroinvertebrate community (Hubler et. al. 2009).

4.6.2 Oregon

In 1996, NOAA Fisheries reviewed the status of salmon and steelhead populations throughout the Northwest and Columbia Basin. As of the year 2000, several evolutionarily significant units (ESUs) of steelhead and/or salmon within the PNMR had been listed as threatened (lower Columbia, upper Willamette, Oregon Coast, and Southern Oregon/Northern California) (Figure 42).

As a result of these listings, Oregon funded a program called “The Oregon Plan for Salmon and Watersheds” in order to track salmon and steelhead populations and identify key environmental factors for decline that threaten salmon survival or recovery. Part of this program funded ODEQ to undertake an ESU wide monitoring program within the Oregon Coast ESU, which includes most of the Coastal PNMR. The monitoring program developed by ODEQ sampled stream habitat, water chemistry, aquatic vertebrates, and aquatic macroinvertebrates at 129 randomly selected wadeable stream sites, 29 ‘least disturbed’ reference sites on wadeable streams, and water quality at 31 long-term water quality monitoring sites on large rivers within the Oregon Coast ESU (ODEQ 2005).

Each macroinvertebrate sample was a composite of eight, one-square foot kick samples collected from multiple riffle habitats within a selected stream reach. Each sample was preserved with 95% ethanol and processed in the lab by sub-sorting a minimum of 500 organisms that were identified to the lowest practical level, typically genus or species.

ODEQ analyzes macroinvertebrate communities using a predictive O/E RIVPACS type model called the PREDictive Assessment Tool for ORegon (PREDATOR). The PREDATOR tool is a multivariate predictive model used to assess the integrity of an aquatic macroinvertebrate assemblage. Predictive modeling estimates the expected occurrence of macroinvertebrate taxa at a sample location. This is accomplished by developing a list of macroinvertebrates that commonly occur at least disturbed, or reference, locations that have similar natural characteristics to the sample locations. The list of species generated from the reference locations is known as the “Expected” taxa list or “E”. This list is compared to the captured macroinvertebrates, or “Observed” taxa (“O”), at an assessment site. The predictive model

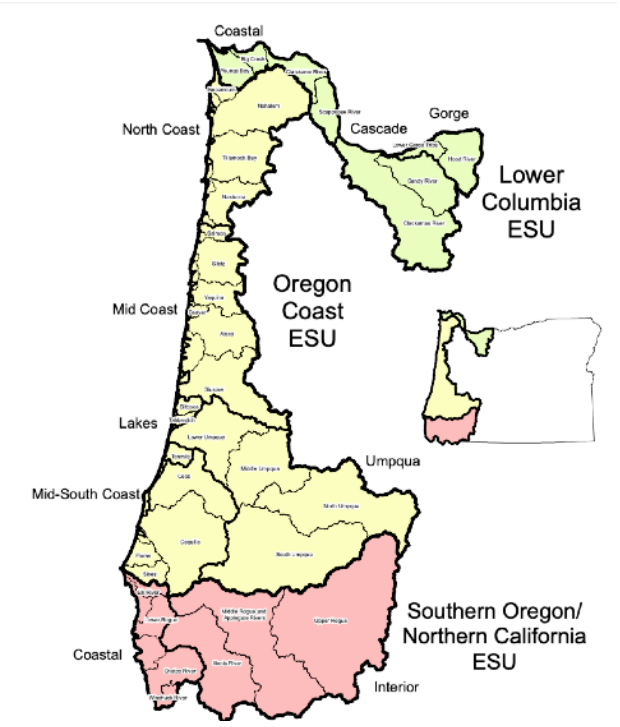


Figure 42. ESUs in Oregon with salmon or steelhead populations listed as threatened by NOAA Fisheries.

output is the observed to expected (O/E) taxa ratio. Scores less than 1.0 have fewer taxa at a site than were predicted by the model. Scores greater than 1.0 show more reference taxa were collected than predicted to occur at a site. For a detailed description of ODEQ's PREDATOR models, see Hubler (2008). For the Oregon Coast ESU, PREDATOR scores less than the 25th percentile of reference site scores (<0.9) indicate impaired macroinvertebrate communities. Figure 43 shows the percent of stream miles with impaired macroinvertebrate communities within the entire Oregon Coast ESU, four sub-regions within the ESU, and within five land use categories.

The results show that within the ESU and its four sub-regions, the percent of stream miles in poor condition was not significantly different than reference streams. A comparison of the different land ownership categories, however, shows that Federal forest lands had the fewest impaired stream miles (16%), while agricultural lands had the highest (63%). A number factors can affect stream condition based on ownership. Agricultural lands, for example, are predominately found at lower elevations and along valley floors with lower gradients. These areas have the longest history of human disturbance and streams there will be more sensitive to habitat impacts that raise water temperatures and increase erosion and fine sediment deposition in stream channels. On the other hand, Federal and State lands typically occur at higher elevations and streams there have higher gradients, and thus are somewhat buffered against high water temperature and fine sediment accumulation. In addition, riparian conditions and runoff from agricultural areas are often a problem as agricultural best management

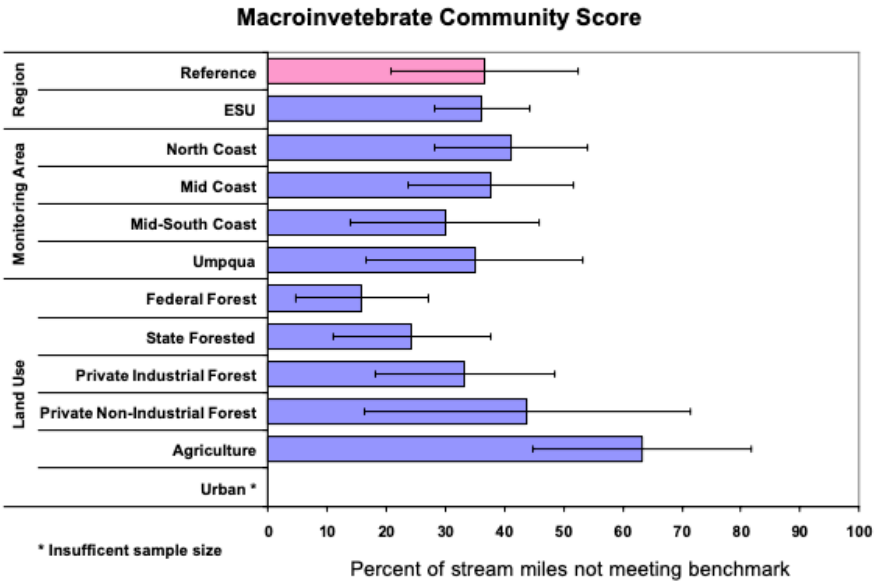


Figure 43. The percentage of stream miles below the benchmark for the macroinvertebrate index across reference streams, the ESU, four monitoring units, and six land use categories. (Error bars represent the 95% confidence interval). (From ODEQ 2005)

practices are largely voluntary and often not implemented, while logging activities on forest lands must follow well defined Federal or State forest practice regulations.

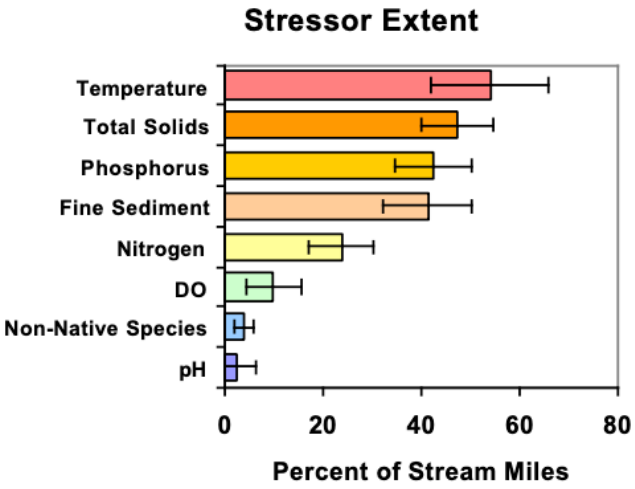
Assessing the eight environmental parameters that effect the condition of macroinvertebrate communities found that water temperature had the greatest extent of stream miles (53%) in poor condition (above State temperature standards). Total solids had the second highest percent of stream miles exceeding benchmarks, followed by phosphorus, fine sediment, nitrogen, dissolved oxygen, non-native species, and pH (Figure 44). When the relative risk to macroinvertebrates (and fish and aquatic vertebrates) was calculated and considered along with the 'stressors' extent,' high water temperature and high total solids were found to be the highest risk factors to aquatic biota (Table 13).

Water quality from long-term monitoring sites on large rivers was assessed using Oregon's Water Quality Index (OWQI). The OWQI analyzes a defined set of water quality variables and produces a score describing general water quality. The water quality variables included in the OWQI are temperature, dissolved oxygen (DO) (percent saturation and concentration), biochemical oxygen demand (BOD), pH, total solids, ammonia and nitrate nitrogens, total phosphorus, and bacteria (E. coli). OWQI scores range from 10 (worst case) to 100 (ideal water quality). Because ambient monitoring sites were not randomly selected, results are reported as a "percent of monitored sites" rather than percent of stream miles.

Ambient monitoring data showed 42% of the 31 large river monitoring sites had good to excellent water quality based on the OWQI, while 19% of sites had poor to very poor water quality (Figure 45). A ten-year trend analysis from 1992-2002 showed no sites sites with declining trends, 39% with improving trends, and 61% with no trend (ODEQ 2005).

Table 13. Qualitative relative rank of water quality stressors incorporating extent and relative risk for the macroinvertebrate and vertebrate assemblages. (From ODEQ 2005)

Relative Rank of Risk	Macroinvertebrates	Vertebrates
Higher Risk	Total Solids	Temperature
	Temperature	Total Solids
	Fine Sediment	Phosphorus
	Phosphorus	Fine Sediment
	Nitrgen	DO
	DO	pH
▼	Non-Native Species	Nitrogen
Lower Risk	pH *(insufficient data)*	Non-Native Species



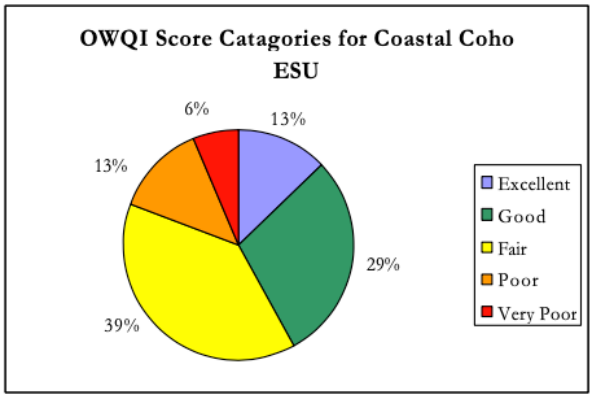


Figure 45. Water quality results from large river ambient monitoring sites. (From ODEQ 2005)

When parameters were assessed for their relative contribution to the OWQI score, and ranked according to their impact on water quality, total solids was found to cause the greatest stress, followed by BOD, nitrogen, phosphorus, bacteria, temperature, DO and pH (Table 14). The impact of total solids is consistent with the water quality results at probabilistic sites and inline with the results from Washington State that found the percent of fine sediment posed the greatest risk to macroinvertebrates in their assessment. It should be noted that the dominant sources of these parameters are non-point sources of pollution (primarily agriculture and forestry), which are largely unregulated and rely on the voluntary application of best management practices to control their discharge to streams and rivers.

Table 14. Relative risk of stressors on water quality in larger rivers. (From ODEQ 2005)

Stressor	Relative Rank
Total Solids	Most Important
BOD	↓
Nitrogen	
Phosphorus	
Bacteria	
Temperature	
DO	
pH	Least Important

4.6.3 Fisheries

The fact that numerous stocks of salmon and steelhead throughout the West Coast, including the PNMR, have been federally listed as threatened indicates the severity of the decline in anadromous fish stocks that has occurred since Europeans first arrived in the region (Figure 46). Historical data for the Coastal PNMR, however, is limited. A review by NOAA Fisheries for the Oregon Coastal Coho ESU provides some historical information and relevant data (NOAA Fisheries 2016). They estimated that prior to European development runs of Coho salmon in Oregon's coastal rivers were in the range of one to two million fish, compared to 60 to 300 thousand today. Oregon coast Coho salmon where also the most numerous species harvested by commercial and sport fisheries during the 1950s through 1970s. Figure 47 compares historical (1890-1956) Coho abundance to recent years (1958-2015). Figure 48 shows a more detailed graph of adult Coho spawning abundance for wild and hatchery origin fish in the Oregon Coast Coho ESU from 1990-2016 (Sounhein et. al. 2016). Other salmon and steelhead stocks have also suffered significant declines (NOAA Fisheries 2016), one result of which is a significant change in stream productivity from the loss of marine derived nutrients left in freshwater when spawning salmon die.



Figure 46. West Coast ESUs with listed salmon and/ or steelhead stocks. (From NOA Fisheries website).

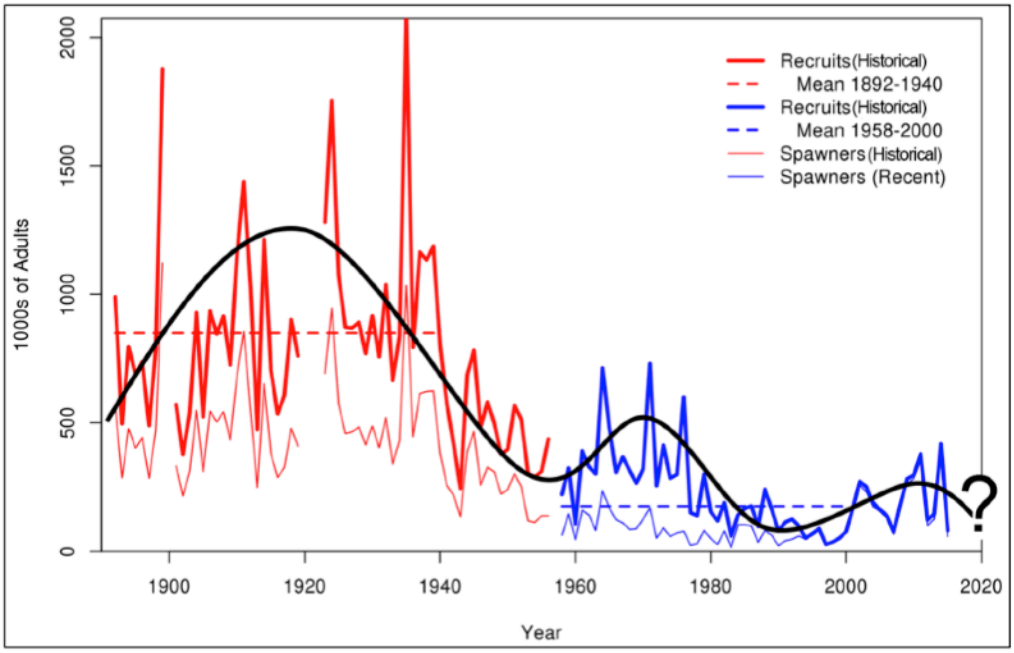


Figure 47. Comparison of historical (1892–1956) and recent (1958–2015) estimates of spawner abundance and pre-harvest recruits. Horizontal dotted lines are the geometric mean recruits for 1892–1940 and 1960–2009. (From NOAA Fisheries 2016: Analysis based on data from Cleaver 1951, Mullen 1981a, and Mullen 1981b; recent data from Wainwright et al. 2008 and ODFW 2016. Dark line is one interpretation of the long-term trend.)

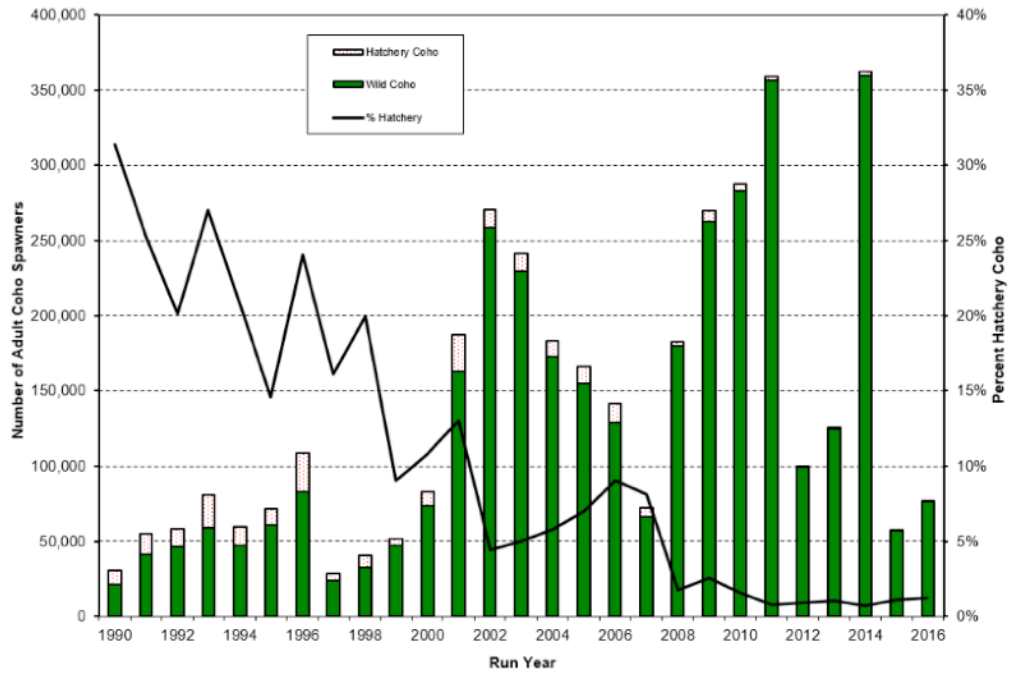


Figure 48. Adult Coho spawner abundance for wild and hatchery origin fish from 1990 through 2016 run years. (From Sounhein et. al. 2016).

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Attachment I

Comparison of Stream Classes and Regions within BCG Model

Comparison Across Regions & Stream Classes in BCG Model

The previous in-depth descriptions of the three regions within the PNMR (Puget Lowlands, Willamette Valley and Coastal Region) show both similarities and differences in their physical habitat, geology, geologic history, land use, and human development patterns. When evaluating macroinvertebrate communities across the entire PNMR however, it became clear that a site's elevation and stream gradient were overarching factors in determining the expectations for macroinvertebrate communities. In addition, the type of disturbance and how that disturbance affects stream habitat, and ultimately the responses expected in macroinvertebrate communities, is also influenced by the stream site's elevation and gradient.

For these reasons, as previously described, the BCG model has been developed for three subclasses of wadeable streams across the region: 1) low elevation (< 750meters), low gradient (< 1% slope) streams; 2) high elevation (> 750 meters), low gradient (< 1% slope) streams; and 3) high elevation (> 750 meters), high gradient (> 1% slope) streams.

The following information describes the major characteristics (similarities and differences) across these three stream subclasses and regions within the PNMR. It is grouped into five sections: 1) physical characteristics (climate and soils/topography/geology); 2) settlement and land cover patterns; 3) hydrology; 4) water quality; and 5) biology.

1 Physical characteristics

Some of the underlying differences between the regions in the PNMR stem from differences in the geologic forces that created them. The Puget Lowlands and Olympic Peninsula, for example, were largely shaped by glaciation during the last ice age, as well as volcanic activity. Ice sheets, however, did not extend as far south as the Willamette Valley or Oregon Coast Range. While the effects of the last ice age affected the regions differently, the overall land form across the PNMR is relatively similar. This has resulted in both differences and similarities in the physical characteristics across the PNMR as described below.

Soils, topography, and geology

- Similarities:
    - The PNMR is bounded along its eastern border by the Cascade Mountain range and the Coast Range along its western border. The Cascade Mountain range significantly influences weather patterns - snow and rainfall - in a similar way throughout the PNMR. In addition, volcanic activity (both past and present) has also affected the landscape, geology, and streams in the PNMR much the same way.

- All three stream subclasses (low gradient, valley bottom streams with predominantly fine, soft substrates to higher gradient, higher elevation rocky-substrate streams) are well represented across the PNMR.

- Differences:
  - During the last ice age, ice sheets covered most of the Puget Lowlands and the Olympic Peninsula, but did not extend into the Willamette Valley or Western Oregon.
  - Because of the ice sheets in Western Washington, glacial till (sediment deposited directly by the ice) and outwash (sediment deposited by meltwater in front of the glacier) make up a large portion of the soils found at or near the surface, resulting in more varied and complex substrate across stream subclasses in the Puget Lowlands and Olympic Peninsula compared to the Willamette Valley and Oregon Coast Range.
  - Soils in the Willamette Valley were created largely by historic floods (Missoula or Bretts Floods) that laid down thick sediment deposits.

Climate

- Similarities:
  - The entire PNMR is known for warm, dry summers, and mild wet winters.
  - Most of the moisture, with the exception of the high Cascades, is deposited as rain with only a small percentage (~5% of total) appearing as snow.
- Differences:
  - The presence of Puget Sound and the Olympic Mountains in Western Washington creates a diverse topography that results in highly variable local-scale climates. For example, a portion of the Olympic Peninsula is in a rain shadow created by the Olympic Mountains and receives only 15 inches of rain annually, while other locations close by receive over a 100 inches per year.
  - Because streams in Western Washington are more strongly influenced by the marine environment due to Puget Sound, they experience cooler and wetter summers than in the Willamette Valley or most of Western Oregon.
  - Because of the greater maritime effect and more northern latitude, summer maximum air temperatures and maximum stream temperatures are less extreme in Western Washington compared to Western Oregon.

The above differences in underlying physical conditions create a landscape with an overall greater variety of stream conditions (greater channel complexity, substrate types, vegetation, etc) in Western Washington streams compared to Western Oregon streams. As a result, macroinvertebrate data show a slightly greater range of species diversity in Western Washington streams compared to Western Oregon.

2 Settlement patterns and land cover change

Human settlement patterns have been similar across the PNMR with Native tribes arriving about 13,000 years ago and European settlers arriving from the mid-1700s to early 1800s. Though the general settlement histories are similar, some differences evolved in the pattern and timing of settlement.

Settlement patterns and land cover change

- Similarities:
  - Prior to European settlement, both ecoregions had been home to large and diverse populations of Native tribes.
  - After European settlement, streams across the PNMR experienced similar patterns of disturbance from agricultural development, lumber extraction, and urban and industrial development due to similar drivers. For example:
    - Fur trappers were among the first European settlers to arrive, and they decimated beaver populations across the PNMR by the late 1830s.
    - Extensive logging occurred, resulting in changes to riparian areas (loss of shading and wood and leaf inputs) and increased sediment inputs into streams.
    - Overfishing of salmonids decimated once-abundant anadromous native fish populations, resulting in a loss of marine-derived nutrients to streams.
    - Large tracts of wetlands have been lost over time; many were drained before the 19th century and converted to pastureland, cropland, and forests.
    - Urban/rural residential development and industrial activity have become principal land use activities across the PNMR with the human population concentrated in Western Washington and Western Oregon (both are home to approximately seventy percent of each state's entire population).
  - Lowland streams (low elevation/low gradient) have been severely degraded over time, while the upland streams (with minimal agriculture and urban development) have experienced lower levels of disturbance primarily from logging and forest management activities.
  - The majority of land with greater environmental protection (e.g. Federal Forests and Wilderness Areas) occur at higher elevations across the PNMR. As a result, the greatest proportion of reference sites are located on mid-elevation to upland streams (high elevation/high gradient sites).

- Differences
  - European settlement occurred slightly earlier in Western Oregon and the Willamette Valley compared to Western Washington and Puget Lowlands.
  - Because of earlier settlement in Western Oregon and the Willamette Valley, extensive agriculture started earlier than in the Puget Lowlands or Western Washington.

- Historic vegetation patterns differed across the interior portions of the PNMR (Puget Lowlands vs Willamette Valley). Pre-European vegetation in the Puget Lowlands included a mix of coniferous forest, swamp or bog-like wetland communities, as well as prairies mixed with open stands of oak. In the Willamette Valley, vegetation in the valley floor consisted of oak savanna, grass savanna, and wetlands with mixed timber along stream riparian areas. At higher elevations, upland prairie and coniferous forests dominated the vegetation.
- Because the most valuable land for agriculture and urban development occurs at low elevation and low gradient regions, low elevation/low gradient streams have experienced greater and earlier disturbance from human activity compared to high elevation/high gradient streams.

3 Hydrology

The characteristics of streamflow, and the factors that affect it, are quite similar across the PNMR. Throughout the region, streamflow is primarily driven by wet winters and snow pack at high elevations and the resulting runoff from snowmelt out of the Cascade Mountains. Similarities and differences across the region and stream classes are described below.

- Similarities:
  - Stream hydrology across the PNMR is influenced by precipitation patterns; most watersheds are predominantly rain-dominated, with low flows typically occurring during the summer dry season and peak flows occurring during the wetter winter months. Streams draining the Cascades Mountains however, are primarily affected by snowpack with peak flows occurring during spring snowmelt and low flows in the late summer and early fall.
  - Historically (pre-European settlement), low to mid-order streams had ample shade, cold water, clean spawning gravels, abundant pools formed by woody debris and beaver ponds, and abundant food. Higher order streams included side channels, wetlands, and sloughs that provided ample refuge habitat for developing juvenile fish as well as migration corridors unblocked by dams, dikes, or irrigation channels.
  - Since major European settlement began in the early to mid-1800s, the hydrology has been heavily altered across the PNMR due to similar drivers, such as:
    - The near-complete elimination of beaver throughout the PNMR by the late 1830s
    - Human activities (e.g., logging, agriculture, urbanization, industrialization), resulting in:
      - Channelization to protect property along the rivers, which resulted in a significant loss of channel complexity and areal extent of water
      - Construction of dams, dikes and revetments to protect property, generate power, etc.
      - Water withdrawals for irrigation, industry, water supply, etc.

        - Loss of wetlands, including beaver-impounded wetlands, which altered recharge patterns.
    - Due to these hydrologic alterations, at least two distinct patterns have appeared:
        - Lower water levels during the dry seasons
        - Flashier, more damaging flows during the wet season

- Differences
    - Low elevation/low gradient streams are more heavily impacted by irrigation and water withdrawals than higher elevation/high gradient streams.
    - Higher elevation/high gradient streams have been more impacted by logging. The presence of large clearcuts produces flashier flows that can result in debris torrents during large rain and snowmelt events.
    - Some regional differences within the PNMR also exist. For example, streams in the Puget Lowlands generally have stronger groundwater influence (the porous nature of the glacial till and glacial outwash in the Puget Lowlands creates the potential for strong connectivity between streams and groundwater) compared to other low elevation/low gradient streams.

4 Water quality

As described below, similar influences have altered water quality across the PNMR, while human disturbance has had different impacts depending on stream elevation and gradient.

- Similarities:
    - Following European settlement, water quality declined significantly over time due to human-induced changes such as soil erosion, instream habitat disturbance, reduced streamflows, dams, human and animal waste, and toxic runoff of pesticides and herbicides.
    - Water quality today has improved significantly from critical conditions that existed in the early to mid-1900s, but many water quality problems still exist from both point source and non-point source pollutants. The greatest proportion of water quality problems today are caused by non-point source pollution.
    - Legacy impacts that persist in streams today include toxic substances (like PCBs and DDT) from their widespread use up to the 1970s, and increased fine sediment and loss of spawning gravels as a result extensive past logging that has exposed soils on steep-sided hills and reduced buffers in riparian areas. Widespread agricultural practices, especially cattle grazing in riparian zones, have also caused extensive soil erosion and other water quality impacts in stream channels (Agouridis et.al. 2005).

- Differences:
  - Human disturbance affecting water quality tends to be most pronounced in low elevation/low gradient streams. This is particularly true for agricultural and urban related pollution.
  - Due to climate and some landscape differences within the PNMR, high water temperature is a more prevalent and greater water quality problem in the Willamette Valley compared to other regions. Warmer water in the Willamette can be explained partly by natural differences like lower latitude, warmer ambient weather conditions, less marine weather influence, and often lower stream gradients (and thus slower moving water) in the Willamette Valley relative to other regions on the PNMR. However, human related impacts, such as extensive agriculture with resulting loss of riparian vegetation, water withdrawal for irrigation, and stream channel simplification, plus extensive urban development, all contribute to warm stream temperatures and temperature related stress to aquatic life in the Willamette Valley and throughout the PNMR.

5 Biology

One of the dominant natural factors influencing the biology of streams in the PNMR is the widespread presence of anadromous fish runs. Their runs were - and still are - significantly important to the economy of the region, and as a result, have been a major focus of fish and wildlife management agencies. Throughout the PNMR the size of these runs have dramatically declined from historic levels. Because of the economic importance of anadromous fish to the region, long-term data and information is often available documenting the size and distribution of anadromous fish runs over time. Such extensive historical data do not exist for macroinvertebrates. Similarities and differences in macroinvertebrates and anadromous fish across the PNMR and stream classes are described below.

- Similarities:
  - Historic stream macroinvertebrate data do not exist for streams throughout the PNMR.
  - While we can infer that there have been losses in macroinvertebrate diversity over time due to habitat alterations, water quality degradation, and changes in food sources, we have no documentation about what specific taxa have been lost, though the taxa currently found in beaver ponds, on wood debris, and other slow-water habitat provide some indication of what was once more common and abundant.
  - Relatively good records exist for anadromous fish runs due to their economic importance. These records show that anadromous fish populations have experienced dramatic declines over time throughout the PNMR. This, in turn, has resulted in a loss of marine-derived

nutrients to freshwater streams affecting all levels of the food chain (aquatic and terrestrial) including macroinvertebrates.

- Differences:
  - Macroinvertebrate communities in low elevation/low gradient streams have generally been more severely impacted from human disturbance than in high elevation/high gradient streams.
  - The Willamette River is unique in the region as it has a natural waterfall (the Willamette Falls) located close to its mouth with the Columbia River that historically controlled anadromous fish movement patterns in the basin. While the addition of a fish ladder has increased access of anadromous fish to the basin throughout the year, the large-scale reduction of anadromous fish runs across the PNMR has still resulted in a significant decline of anadromous fish to the basin from historic levels and the resulting effects of that on the rest of the stream ecosystem, including macroinvertebrates.
  - Because of differences in land use, human disturbance, stream habitat, and climate between high elevation/high gradient streams and low elevation/low gradient streams, more reference sites (low levels of human disturbance) exist and have been sampled in moderate to high elevation areas.
  - Because a greater range in habitat diversity and greater extent of land with lower levels of human disturbance occurs at higher elevations, more streams with macroinvertebrate communities classified as belonging to BCG Level 2 by the BCG model occur at high elevation/high gradient sites compared to low elevation/low gradient sites.
  - In addition, because low gradient/low elevation streams in the PNMR are heavily affected by agriculture, with cities and urban development filling in the area not being farmed, modeling low gradient/low elevation streams has been difficult as very few high quality streams with low levels of human disturbance exist for them.
  - High quality (low disturbance), low gradient/low elevation streams, while rare throughout the PNMR, are less rare in the Puget Lowlands ecoregion, and many of these sites have been sampled. The data from these sites were used in creating the BCG model so that the model is still effective for this class of streams, though some adjustments to the model were made to better account for low gradient stream conditions and expected macroinvertebrate communities.

Table 1, summarizes the major environmental changes that have occurred on streams in the PNMR from the mid-1800's to present and how these changes may have impacted aquatic life. A better understanding of these changes can help inform stream restoration work in the future.

Table 1. Summary of the key environmental changes from the mid-1800s to present and how those changes can affect stream habitat and biology and possible restoration actions based on our understanding of historical conditions.

Key Environmental Changes from 1850's to present	Watershed Effects	Biological Results	Possible Restoration Actions
Rapid and near complete removal of beaver	Significant shift in hydrology and stream habitat due to: - Less water storage. - More rapid runoff. - Lower summer streamflows. - Warmer water temperatures. - Increased erosion & higher amount of fine sediment deposition. - Loss of lentic habitat for aquatic and terrestrial species.	- Significant loss of biological diversity. - Loss of lentic and depositional aquatic invertebrate species. - Increase in sediment tolerant and temperature tolerant invertebrate species. - Increase in erosional habitat & rheophilic invertebrate species. - Loss of fish habitat, especially for juvenile salmon and trout. - Loss of habitat for waterfowl and other terrestrial plants and animals that depend on wetlands and diverse aquatic habitats.	- Reintroduce beaver. - Protect and restore wetlands.
Channel modification and simplification	- Loss of stream habitat diversity (fewer side channels & alcoves). - Increase in fast water habitat. - More rapid stream runoff - Lower summer streamflows. - Warmer summer temperature - Increased erosion & sedimentation. - Wider (& straighter) stream channels. - Higher substrate embeddedness.	- Shift in aquatic invertebrates from diverse lentic and lotic species to primarily lotic. - Increase in erosional habitat and rheophilic invertebrate species. - Increase in sediment tolerant and temperature tolerant invertebrate species. - Decline in winter and summer habitat for juvenile salmonids. - General decline in habitat quality for salmonids due to increased sedimentation and warmer temperatures. - Decline in terrestrial animal and plant diversity.	- Remove riprap and streambank armoring. - Reconnect stream channels with floodplains. - Reestablish natural channel morphology. - Stream restoration (e.g. add in-channel structures like large logs where appropriate) to increase habitat diversity and channel complexity.
Dam construction	- Shift in hydrology of stream systems. - Higher summer stream flow below dams. - Change in natural annual temperature regime. - Lower maximum streamflows during storms and spring runoff (little to no flooding). - Block fish migration. - Loss of movement of substrate from watershed above dams to replenish substrate below dams. - Increased substrate embeddedness - Channel simplification.	- Major reduction in available spawning and rearing habitat for anadromous salmonids. - Lower aquatic invertebrate diversity due to less variable streamflows and unchanging habitat. - Altered life cycle timing (and lower abundance) for some fish and invertebrate species due to modified temperature regime.	- Retrofit dams to improve fish passage. - Manage dam releases to better reflect natural hydrograph and temperature regimes. - REMOVE DAMS
Agriculture and forestry expansion	- Increased sedimentation - Loss of riparian vegetation and habitat. - Decline in water quality from toxic chemicals (herbicides and pesticides). - Decline in in-stream water quantity for irrigation. - Lower streamflows, especially in the summer. - Warmer summer water temperature. - Excess nutrient loading from fertilizers and animal waste. - Loss of wetlands.	- Decline in invertebrate abundance (especially for sensitive species) from water quality, quantity, and habitat impacts. - Decline of native fish abundance from habitat and water quality impacts. - Increase in non-native tolerant fish species (carp, bass, etc.)	- Maintain and increase riparian areas, and replant with native trees and vegetation. - Reduce or eliminate the use of chemicals and fertilizers (promote organic farming). - Monitor water use for irrigation, and use best available methods to reduce amount of water needed for irrigation. - Remove non-native fish species where possible. - Improve stream habitat.

Table 1 cont.

Key Environmental Changes from 1850's to present	Watershed Effects	Biological Results	Possible Restoration Actions
Urban development and population growth	- Loss of wetlands. - Flashy stream hydrology due to high percent of impervious area. - Decline in water quality from impervious surface runoff and lawn and garden chemical use. - Loss of riparian vegetation and habitat. - Warmer summer temperature.	- Decline in invertebrate abundance (especially for sensitive species) from water quality and habitat impacts. - Native fish abundance decline from habitat and water quality impacts. - Increase in non-native tolerant fish species (carp, bass, etc.)	- Reduce or eliminate lawn and garden chemical use. - Create bio-swales to capture street and urban runoff before entering streams. - Create green spaces along stream corridors. - Protect riparian areas from development. - Improve stream habitat.
Climate warming	- Warmer water temperatures - Lower minimum streamflows	- Shift to more temperature tolerant species. - Decline in abundance and diversity of cold adapted species.	- Increase riparian vegetation for more stream shading. - Improve stream habitat and restore wetlands to improve low summer streamflows and provide more cool water. - Reintroduce beavers to watershed. - Improve irrigation methods and reduce amount of water used for irrigation. - Reduce human water consumption in urban areas.

Attachment II

Rational for Natural Condition Assessment

1 Synthesis

1.1 The BCG Disturbance Gradient

It's important to remember that the purpose of developing this broad historical understanding of natural and human influences on streams in the PNMR, as well as the historical changes to aquatic life, is to more precisely understand and define BCG Level 1 characteristics for both biotic condition (Y axis) and stressor (human disturbance, X axis) gradients. In other words, it defines the upper most left-hand corner of the conceptual graph in Figure 1 and provides important context for where current conditions of biota and human disturbance exist relative to our best understanding of "natural condition and function."

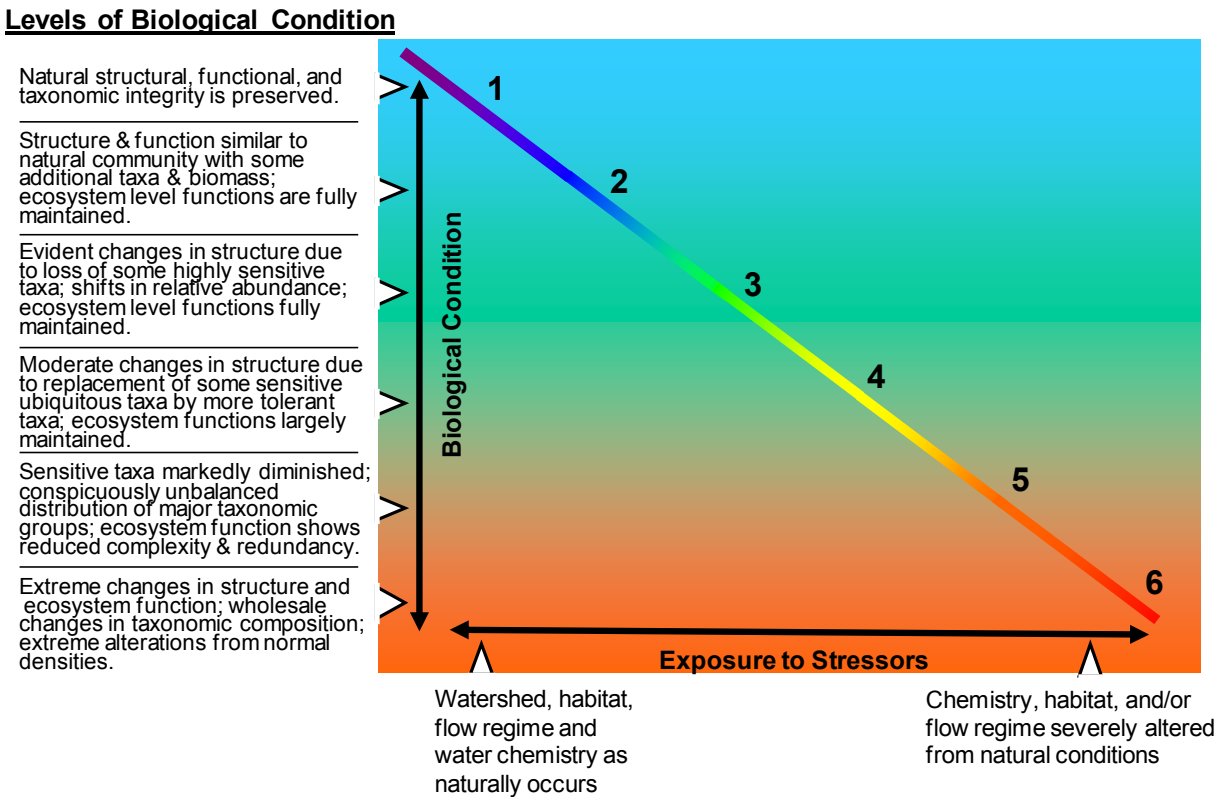


Figure 1. Conceptual diagram of the Biological Condition Gradient (BCG).

Based on our understanding of historical land use and habitat conditions we can identify the following key components that provided for the "natural" biotic communities that existed prior to extensive EuroAmerican settlement and development.

1. Extensive beaver populations created a complex matrix of pond and flowing water habitats throughout the PNMR. This created a diverse mixture of habitat types for all aquatic organisms including macroinvertebrates.
2. Riparian vegetation along stream channels was intact and complex with a mix of large mature trees (conifers and deciduous) providing shade and allochthonous food sources, and an understory of native shrubs and grasses.
3. Stream banks were generally intact and supported by vegetation resulting in minimal soil erosion.
4. Stream channels included a complex mixture of slow and fast water habitat often created by large wood in the stream channel that acted to create pools and trap sediment and organic material.
5. Stream hydrology was unaffected by irrigation withdrawals and extensive human use, or by dams that changed natural annual flow patterns. Therefore, greater summer streamflows existed due to lack of human water uses and natural vegetation structure throughout the watersheds.
6. High water quality unaffected by chemical contaminants, domestic animal waste, human waste, or urban and agricultural runoff. Lower overall water temperatures.
7. Large runs of anadromous salmonids provided significant inputs of marine derived nutrients to most streams throughout the watersheds in both regions. Willamette Falls on the Willamette River restricted anadromous fish access only to winter and spring seasons.

An overview of habitat conditions and aquatic life characteristics in streams that may represent BCG Level 1 conditions is provided in Table 1. This table serves as a template for narratively describing and, as data and information enable, develop semi-quantitative or quantitative measures to inform defining BCG level 1 conditions when such conditions no longer exist. This template is applicable to other aquatic systems and can be used to prioritize the type of historic records, data, and photos that would be most useful in defining BCG level 1 conditions. For the PNMR, a quantitative biodiversity index has been proposed to describe BCG Level 1 macroinvertebrate communities (see the narrative "biodiversity" characteristic included at the bottom of Table 1). This index is being further developed and reviewed for use in the Pacific Northwest Maritime region (Wisseman 2020) and is described in Appendix C.

2 Lessons learned

2.1 Why bother, since in most cases pre-settlement conditions are not attainable?

This was the first time we delved into historic/BCG level 1 conditions at this level for a BCG project. The intent of going through this exercise was not to set BCG level 1 (pre-European settlement conditions) as the bar streams should attain. Rather we believe that this exercise is able to provide several important insights for stream management:

- Provide a shared narrative starting point for assessing incremental changes from BCG level 1 to BCG level 6
- More clearly understand and communicate how biological communities have been altered and possible restoration actions that can be taken. (see summary in Table 1)
- Serve as documentation that can be used as a defense against baseline shifts (i.e., public perception of what "high" quality stream conditions look like can change over time)
- Inform restoration and protection goals for waterbodies with potential not only to improve but to maintain those improvements (see potential restoration actions in Table 18; also, to read about an example of a restoration project currently underway in the Willamette Valley that ties in nicely with these strategies, see - https://www.nature.org/en-us/about-us/where-we-work/united-states/oregon/stories-in-oregon/reconnecting-the-willamette-river-/?src3=e.gp.local.Mar2019.Oregon)

This information can also be very valuable for public outreach. This became apparent during a talk given by Rick Hafele on the historic conditions in the Willamette Valley to 50 or so members of the public in Portland. Most members of the general public had little to no understanding of present-day conditions, let alone how they had changed over time. By providing this historical perspective, it created greater concern and interest among the attendees to protect what we have left.

2.2 The process

This effort was an iterative and collaborative process with the timeline explained below:

- **Workshop #1** (January 2017) - Held brainstorm session to develop (rough) BCG level 1 narratives (which the panelists kept in mind when assessing samples). Discussed ecological and landscape drivers of benthic community health.
- **Between workshops** (February 2017-March 2018) - Rick & Rob built on these narratives and researched historic information on the Willamette Valley and Puget Lowlands, respectively; compared draft outlines then wrote up their findings independently; their write-ups were later combined into a draft appendix for the BCG report. During this time, Rick, Rob & Bob also worked on a conceptual/narrative definition of BCG level 1.

- **Workshop #2** (March 2018) - Rick & Rob presented on their findings and proposed a draft BCG level 1 definition, which included a 'key components' table. Bob came up with the idea of the BCG Level 1 Biodiversity Index.
- **Post-Workshop # 2** (April 2018-current) – Rick completed a similar historical condition overview for the Coastal regions of Washington and Oregon to expand the project to the entire Pacific Northwest Maritime Region. Bob developed the BCG Level 1 Biodiversity Index, (independently of Rick and Rob's work). The index can be calculated with a free R tool

After having gone through this process, we feel it can be broken into the following tasks:

Task 1: Identify Information and Expertise Needs

- Put together expert team (subgroup of expert panel – Rick Hafele, Rob Plotnikoff and Bob Wisseman).
- Construct conceptual narrative and framework for organizing the information.
- BCG level 1 narrative (see main report, Section 4) - started with more general statement, refined with research and dialogue among experts.

Task 2: Information Gathering

- Literature searches, talking with experts, evaluating the best of what is left in existing datasets.
- Revisit and develop framework and narrative.
- Summarize findings in write-ups that follow the agreed-upon framework.

Task 3: Synthesis

- Bring together the write-ups; edit for consistency; identify similarities and differences across the ecoregions; identify data gaps; update narrative and to extent possible quantify.

One recommendation that we have for others who are considering embarking on a similar undertaking is to reach consensus on an outline early on. The structure of this document may work well for others as well, or at least could serve as a good starting point. Agreeing to the structure and content early on is especially important if multiple people are working on the write-up, and if the study area covers more than one ecoregion or class.

Table 1. The fundamental characteristics of BCG level 1, conceptual expectations for BCG level 1 ("as naturally occurs") in perennial, freshwater wadeable (watershed area of 1 to 100 mi2) streams in the Pacific Northwest Maritime Region. This table was developed through an iterative process (refined with research and dialogue among the other BCG panelists).

Fundamental Characteristics	Description
Stream channel	Channel connected to hyporheos and flood plain including wetlands, beaver ponds, etc.; diverse habitats present (e.g. braided channels, side channels, debris jams, mixture of steps and pools consistent with stream gradient); wood debris typically present and may be abundant; quality of habitat and refugia persists during periods of both low and high streamflows.
Riparian & watershed	Riparian zones support intact community of overstory, understory, and ground cover plants (including a mixture of mature conifer and hardwood trees with a diverse age structure in forested watersheds); upper watershed vegetation intact, supporting delivery of water of high chemical and thermal quality to lower reaches.
Hyfrologic regime	Hydrologic regime natural, without alteration from dams and/or irrigation withdrawals or return flow; cool-cold water common from springs, groundwater accretion, and/or natural runoff; perennial surface or subsurface flow. Re-charge in the watershed sustains flow, especially during years of extreme drought. Perennial surface water in some portion of watersheds maintain endemic taxa that serve as recolonization sources sustaining high biodiversity at select locations. These locations promote resiliency in steam reaches that are periodically de-watered.
Disturbance regime and resilience	Natural seasonal range of high and low streamflows present, which enhances and maintains channel and habitat complexity. Natural sediment transport based on local geology, oils and stream gradient. High resilience (ability to recover from disturbance) to natural and anthropogenic watershed stressors (Flotemersch et.al. 2016). Watershed endemic taxa and promotes connectivity for purpose of recolonization.
Ecosystem function	Watershed supports full range of ecological processes and functions essential to maintaining high biodiversity provided by a minimally disturbed ecosystem. Food web, nutrient and energy flow linkages between aquatic and terrestrial environments fully supported.
Biodiveristy	Benthic macroinvertebrate community typically with high taxa richness (low richness may occur under some natural conditions; e.g. high altitude, cold spring systems, etc.), including many micro-habitat specialist taxa and taxa sensitive to human disturbance. Habitat complexity results in diversity of both rheophilic and lotic-depositional taxa. Non-native invasive taxa not present.

Attachment III

The Biodiversity Index

The goal of the biodiversity index is not to describe or predict what historic (pre-settlement) macroinvertebrate communities were like, but rather to create an index that can help identify streams and/or stream reaches that currently exist that provide overall high habitat complexity with minimal human disturbance and have outstanding biodiversity of macroinvertebrates. Wisseman (2020) describes the purpose of the index as:
"Through this BCG Level 1 Biodiversity Index, we attempt to define measurable biological indicators, or signals, for a benthic macroinvertebrate community comparable to BCG level 1, an exemplary community that has not been impacted by human disturbance."

Table 1 shows the first draft of the BCG Level 1 biodiversity index metrics, and Table 2 shows the final site ranking scores for streams in the Pacific Northwest Maritime Region. For more details and explanation of the BCG Level 1 Biodiversity Index see the paper by Wisseman (2020).

One of the limitations of this biodiversity index is that it is, "… limited to moderate to high gradient streams [>1.0%] with coarse mineral substrates and extensive erosional habitat" (Wisseman 2020). As a result it cannot be used to identify high quality, low gradient streams with mostly fine sediment present. The reason is that we currently lack sufficient data to characterize the aquatic invertebrate community from high quality, low gradient streams in the PNMR (including Puget Lowlands and Willamette Valley ecoregions) in order to determine appropriate community metrics and scoring ranges.

The BCG Level 1 biodiversity index provides a tool for describing the aquatic invertebrate community found in mid-elevation and mid to high-gradient streams with minimal human disturbance (BCG Level 1). However, because of the lack of historical biological data and the current absence of low gradient/low elevation streams without significant human disturbance, we cannot effectively create a similar index or an in-depth description of macroinvertebrate communities that represent "natural conditions" (BCG Level 1) for low gradient/low elevation streams. Instead, based on our understanding of the historical "natural" conditions and how they have changed, we can begin to develop an understanding of how aquatic macroinvertebrate communities have also changed, and can make inferences about what BCG Level 1 macroinvertebrate communities where like in conditions unaltered by extensive human use and development. For example, most pre-settlement streams would have a complex mix of fast and slow water with abundant wood debris, abundant slow-water habitat such as beaver ponds, sloughs and wetlands, plentiful cold water throughout the year, and water without chemical contaminants or excess nutrients. As such one would expect to see macroinvertebrate communities to include many of the characteristics described below.

1. Well represented diversity of cold-water species, particularly cold-adapted species of pollution sensitive mayflies, stoneflies, and caddisflies (EPT taxa), which are known to disappear when streams become warmer and/or have degraded water quality. Macroinvertebrate data in Oregon, collected along with stream temperature data, have been used by the Oregon Department of Environmental Quality (ODEQ) to determine the temperature tolerance ranges for most species of macroinvertebrates in the state (Huff et.al. 2008). This list could be used to determine which species were likely more abundant under historical natural conditions when stream temperatures, based on historical habitat and land cover information, were on average colder throughout the year.

2. Taxa preferring depositional habitats (pools and ponds with slow water and fine sediment) would also be well represented. A sediment-macroinvertebrate model, also developed by ODEQ, provides a list of taxa that have a high tolerance to fine sediment (Hubler et.al. 2016). This includes species of some aquatic beetles, Diptera species especially of the families Chironomidae and Tipulidae, Odonata, and species of freshwater crustaceans.
3. High water quality would also result in an abundance of sensitive non-insect taxa like freshwater clams and mussels. Note that there is an active conservation effort to protect and recover freshwater mussels in the Pacific Northwest. Information about these efforts can be found at: www.Xerces.org
4. Last, the variety of stream habitats available would result in an increased presence of micro-habitat specialist species rather than the current dominance of habitat generalist species that can survive across a wide habitat range.

Table 1. BCG Level 1 Biodiversity Index (draft, version 1) - community composition metrics and scoring thresholds.

Metric	Scoring criteria (points)			
	0	1	2	3
Total taxa richness	<40	40-49	50-59	≥60
EPT taxa richness	<20	20-24	25-29	≥30
BCG attribute 1i, 1m & 2 taxa	<2	2-5	6-8	≥9
Shannon-weaver diversity ($log_e x$)	<2.75	2.75-2.99	3.0-3.24	≥3.25
Long-lived taxa richness	<6	6-8	9-11	≥12
Ephemerellidae taxa richness	0-2	3	4	≥5
Heptageniidae taxa richness	0-2	3	4	≥5
Nemouridae taxa richness	0-2	3	4	≥5
Perlidae taxa richness	0	1	2	≥3
***Rhyacophila* taxa richness**	0-2	3-4	5	≥6
Predator taxa richness	<10	10-12	13-15	≥16
Noteworthy taxa richness	Add an additional score point for each noteworthy taxa present			

Table 2. BCG Level 1 Biodiversity Index (draft, version 1) – overall scores and ratings, as well as

Rating	Score	Description
High	≥ 30	Exemplary biodiversity and high habitat complexity and resilience probable. Acquire additional information and data on the site and watershed that is readily available. Alert stakeholders, including government and non-government organization conservation agencies.
Medium	21-29	Moderate habitat complexity/resilience and biodiversity indicated. Further evaluation of the site and watershed is recommended.
Low	<20	Unexceptional biodiversity indicated. Mostly widespread and common taxa present.